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MODELING RADIATION AND DUST DYNAMICS IN PROTOPLANETARY DISKS

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REFEREES

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to Amma and Appa, for everything.

Abstract

Radiation and dust dynamics control the large-scale transport and evolution of protoplanetary, or planet-forming disks, from microscopic to macroscopic scales. Protoplanetary disk thermodynamics sets the temperature structure that governs radiation transport, opacity, observational emission and absorption signatures, as well as the chemical inventory relevant for planet formation and various dynamical processes. Dust is a key ingredient to this story. Through both chemistry and its role as the dominant carrier of opacity, dust is not only deeply intertwined with radiation, but its dynamics—including drift, growth, and settling—shape the structure and composition of solids in the disk available for planet formation.

This thesis explores the modeling of radiation and dust in protoplanetary disks, with detailed studies of the physics of stellar irradiation affecting the disk's temperature structure and geometry, and numerical developments in modeling dust dynamics. Using radiation-hydrostatic and hydrodynamic simulations with frequency-dependent stellar irradiation, we explore starlight-driven shadowing and the formation of staircase geometry in flared, passively heated disks. In addition, we present the comprehensive numerical modeling of dust as pressureless fluids in the `PLUTO` code.

Our results show that shadows can form at intermediate disk radii in optically thick, slowly cooling disks. Thermal waves form and propagate inward in hydrostatic models, making them susceptible to the thermal wave instability. In contrast, in the hydrodynamic picture, we find steady shadows that form and converge to a long-lived staircase structure for several thermal diffusion timescales.

Motivated by the fact that dust dynamics coupled with radiation is important for accurate studies of the thermal structure of disks, we focus on developing the fluid dust module for `PLUTO`. The single- and multispecies dust–gas drag and diffusion solvers we present can be easily coupled to complex physics, including not only radiation but also processes such as dust growth via coagulation, dust charge, magnetic fields, etc. We test the results and convergence properties of the solvers using detailed problems in all geometries, in global and local shearing-box simulations that directly compare to analytical solutions, linear theory, or previous implementations in the literature.

Overall, our results provide a detailed picture of the temperature structure and geometry of protoplanetary disks, and the numerical efforts of modeling dust dynamics that are fundamental to our understanding of the initial conditions for planet formation.

Zusammenfassung

Strahlung und Staubdynamik sind entscheidend für die globale die Entwicklung and den Transport in protoplanetarer, also planetarischer, Scheiben – von der mikroskopischen bis zur ma- kroskopischen Skala. Die Thermodynamik der protoplanetare Scheibe beeinflusst die Temperaturstruktur und diese kontrolliert den Strahlungstransport, die Opazität, die beobachtbaren Emissions- und Absorptionssignaturen, verschiedene dynamische Prozesse und die für die Planetenentstehung relevante chemische Zusammensetzung. Staub ist ein zentraler Baustein in der thermischen Interaktion. Sowohl durch seine chemischen Eigenschaften als auch durch seine Rolle als dominanter Träger der Opazität ist Staub nicht nur eng mit der Strahlung verflochten, sondern auch die Struktur und Zusammensetzung des festen Materials in der Scheibe, das für die Planetenentstehung zur Verfügung steht ist von der Staub Dynamik geprägt, welche Drift, Transport, Wachstum und Sedimentierung einschließt.

Diese Dissertation befasst sich mit der Modellierung von Strahlung und Staub in protoplanetaren Scheiben. Diese Arbeit vereint Studien der detailgetreuen Physik des Einfluss der Sterneinstrahlung auf die Temperaturstruktur und Geometrie der Scheibe und die Entwicklungen numerischer Modelle der Staubdynamik. Mithilfe von strahlungshydrostatischen und hydrodynamischen Simulationen mit frequenzabhängiger Sternstrahlung untersuchen wir die durch Sternlicht verursachte Schatten und die Entstehung einer Stufengeometrie in passiv erwärmten, sich aufweitenden Scheiben. Darüber hinaus untersuchen wir aufführliche, numerische Modelle von Staub im `PLUTO`-Code als Flüssigkeiten ohne eigene Druckkomponente.

Unsere Ergebnisse zeigen, dass sich auf mittlere Distanz in optisch dichten, langsam abkühlenden Scheiben Schatten bilden können. In den hydrostatischen Modellen bilden sich durch diese Schatten thermisch bedingte Wellenstrukturen, die sich nach innen ausbreiten, wodurch diese Modelle anfällig für die sogenannte thermische Welleninstabilität sind. Im Gegensatz dazu finden wir im hydrodynamischen Szenario stationäre Schatten, die sich über mehrere thermische Diffusionszeitskalen hinweg aufbauen und in eine langlebigen Stufenstruktur konvergieren.

Um nun auch die thermischen Struktur von Scheiben mit mit Strahlung gekoppelter Staubdynamik zu verstehen, konzentrieren wir uns auf die Entwicklung des Fluidstaubmoduls für `PLUTO`. Die von uns vorgestellten Solver für den Strömungsdrift und die Diffusion von Ein- und Mehrkomponenten-Staub-Gas-Systemen lassen sich leicht mit komplexen physikalischen Prozessen kombinieren, darunter nicht nur Strahlung, sondern auch zusätzliche Prozesse wie Staubwachstum durch Koagulation, Staubladung, Magnetfelder usw. Wir testen die Ergebnisse und Konvergenzeigenschaften der Solver anhand wohldefinierter Probleme in globalen und lokalen Shearing-Box-Simulationen, die direkt mit analytischen Lösungen, der linearen Theorie oder früheren Implementierungen in der Literatur vergleichbar sind, für alle Geometrien.

Zusammenfassend liefern unsere Ergebnisse ein detailliertes Bild der Temperaturstruktur und daraus folgenden Geometrie protoplanetarer Scheiben und des numerischen Fortschritt bei der Modellierung der Staubdynamik, die für unser Verständnis der Anfangsbedingungen der Planetenentstehung von grundlegender Bedeutung sind.

Tamaso mā jyotirgamaya
"Lead me from darkness to light"

— Brihadaranyaka Upanishad 1.3.28

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Table 2 Initial conditions for the `DUSTYWAVE` test, corresponding to eigenvectors for perturbations in the dust density, gas and dust velocities for $St \in [10^{-3}, 100]$. Perturbations in gas density are always assumed to be $\delta\hat{\rho}_g/\rho_g^0 = 1$. These values are used in Eq. (50). 72

Table 3 Initial conditions for the `DUSTYWAVE` test, corresponding to eigenvectors for perturbations in the dust density, gas and dust velocities for $St \in [0.001, 1, 10]$. Perturbations in gas density are always assumed to be $\delta\hat{\rho}_g/\rho_g^0 = 1$. These values are used in Eq. (50). 78

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Acronyms

CAIs	Calcium–Aluminium-rich Inclusions
JWST	James Webb Space Telescope
VLT	Very Large Telescope
PPDs	protoplanetary disks
GMCs	giant molecular clouds
YSO	young stellar object
HST	Hubble Space Telescope
ALMA	Atacama Large Millimeter Array
ISM	interstellar medium
SED	spectral energy distribution
MRI	magnetorotational instability
MHD	magnetohydrodynamic
RWI	Rossby wave instability
VSI	vertical shear instability
COS	convective overstability
ZVI	zombie vortex instability
IR	infrared
UV	ultraviolet
HD	hydrodynamic
MRN	Mathis–Rumpl–Nordsieck
SI	streaming instability
SPHERE	Spectro-Polarimetric High-contrast Exoplanet REsearch
DSHARP	Disk Substructures at High Angular Resolution Project
eDisk	Early Planet Formation in Embedded Disks
ARKS	ALMA Debris Discs Survey
AGE-PRO	ALMA Survey of Gas Evolution in PROtoplanetary Disks
MINDS	MIRI Mid-infrared Disk Survey
GPI	Gemini Planet Imager
HiCIAO	The High-Contrast Coronographic Imager for Adaptive Optics
ODE	ordinary differential equation
2D	two dimensional
FLD	flux-limited diffusion

PDE	partial differential equation
CFL	Courant–Friedrichs–Levy
ELT	Extremely Large Telescope

Part I

INTRODUCTION

1

Prologue

"How unique is life on our planet?" is a question that has not only occupied the minds of astronomers and philosophers, but generations of humankind. Observing the night sky led to independent discoveries of different astronomical objects—stars, planets, the Milky Way—from all parts of the world. Along with the often-quoted Greek thinkers came scholars from the Indian and Arabic civilizations who built catalogs and derived mathematical formulae for how and when these celestial objects appeared in the night sky, as well as their nature. In the history of Indian astronomy, among the body of work by *Aryabhata* (476–550 CE) and *Brahmagupta* (598-668 CE) were sophisticated systems to model planetary motions. *Brahmagupta* was reported to have used corrections due to *parallax* in lunar eclipse calculations, recognizing the effect of the observer's location on Earth. This concept remains foundational in contemporary astronomy where *parallax* measurements are used to determine distances and properties of celestial bodies with modern-day telescopes.

With telescopes, our understanding of planets and the solar neighborhood has grown exponentially, leading to the discovery of the nature of terrestrial planets, giants and their moons, reinforcing the Copernican heliocentric theory. Spacecraft missions since the late 1950s have put constraints on the geological make-up of these bodies, with examples including volcanoes on Io, oceans in Europa, the atmosphere of Titan with *Cassini–Huygens* and the surface of Mars with the *Curiosity* and *Perseverance* rovers. The primordial composition of the solar system is evidenced by meteorites that have helped constrain the timescales of the solar system's evolution. Calcium–Aluminium-rich Inclusions (CAIs) in chondritic meteorites date back to 4.56 billion years ago (Amelin et al., 2002). Sample returns from asteroids such as *Ryugu* and *Bennu* (Lauretta et al., 2019; Watanabe et al., 2019) provide records of chemical origins of the solar system by constraining complex organic chemistry before the formation of the Earth.

Discoveries in the solar system and its neighborhood made us think about the question posed in the beginning of this prologue more, and ask whether there exist other worlds similar to ours. Perhaps one of the most significant reminders of our place in the universe that is iconic in pop-culture is the image of the Earth (as a tiny speck in Fig. 1) taken by the camera on Voyager I in 1990, colloquially known as the "Pale Blue Dot", from 6 billion kilometers away ($\sim$ 40.2 astronomical units or au), outside the orbit of Neptune. This picture is known to have sparked artistic creations, environmental activism, and, more importantly, inspired many young people. For me, this image is the first thing that appears to mind when asked about what drew me to astronomy.

The discovery of the first exoplanet *51 Pegasi b* around a sun-like star in 1995 by Michel Mayor and Didier Queloz (Mayor and Queloz, 1995) marked a dramatic shift in history, where planets around other stars no longer remained elusive. This was recognized by the 2019 Nobel Prize in Physics as a new era for planetary sciences. Today, we report more than 6000 confirmed exoplanets with different detection methods like radial velocity, transits, micro-lensing, etc., and we are able to start to decipher the compositions of their at-



Figure 1: Pale Blue Dot. Image credit: NASA's Scientific Visualization Studio NASA/JPL-Caltech.

mospheres with space- and ground- based telescopes such as the JWST and the Very Large Telescope (VLT). Figure 2 demonstrates this demographic of the different exoplanets in a plot of estimated planet masses against their semi-major axes from the NASA Exoplanet Archive grouped by their detection methods.

We have also traced back the origins of these solar systems by finding younger planetary systems, just a few million years old, in their forming environments, their "protoplanetary disks". protoplanetary disks (PPDs) are rotating disks composed of gas and dust, around a central forming star, that form as a natural consequence of the conservation of angular momentum during the gravitational collapse of a molecular cloud. Four confirmed "proto"-planets *PDS 70 b & c* (Haffert et al., 2019; Keppler et al., 2018) and *WISPIT 2 b & c* (Lawlor et al., 2026; van Capelleveen et al., 2025) have been identified in their disks, with the latter two having been discovered only in the past year, which also occupy the top right part of the scattered plot in Fig. 2. This provides the ground for us to test our theories of planet formation and whether the solar system is *unique*.

The process of planet formation cannot be disentangled from star formation, and modeling protoplanetary disks entails combining the effects of magnetic fields, radiation, dust dynamics, and the interstellar environment coupled to the dense gas. Within this environment, planet formation occurs through the growth of solid material—with dust grains processed from micron sizes to several kilometer-sized planetary cores, spanning about 14 orders of magnitude in size (and 42 orders in mass, since mass $\propto$ size^3).

This thesis is a humble part of the larger body of work that is looking into the physics of protoplanetary disks and their dynamics. It focuses on radiation and dust processes and how to model them, and how they affect the overall density, temperature structure, and dynamics of protoplanetary disks.

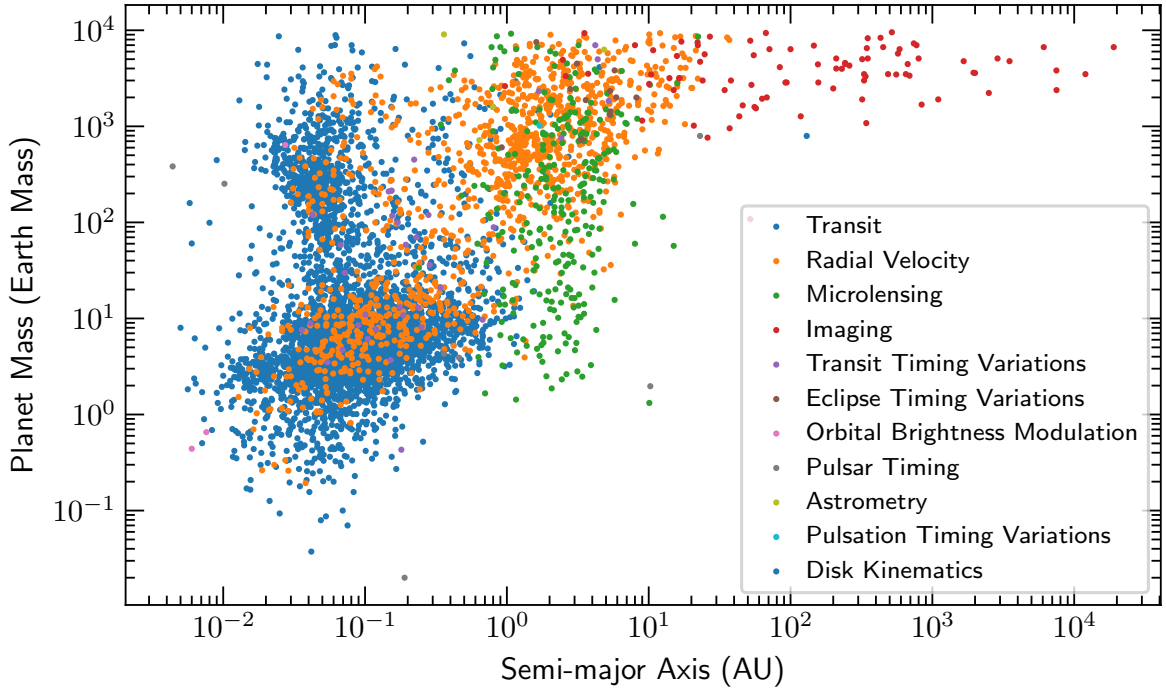


Figure 2: Planet mass in Earth masses against the semi-major axis in astronomical units (au) of the discovered planets as of August 3, 2026. Data taken from the NASA Exoplanet Archive, which is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program.

2

The planet forming environment

This chapter provides a brief introduction to star and planet formation. We start by describing a hierarchical view of star formation and, in conjunction, planet formation, and describe the classification of YSOs. We discuss how protoplanetary disks are observed, constraints on disk properties from these observations, and the origins of the observed substructures. Later, we provide an overview on angular momentum transport and gas disk evolution, and following that, we discuss dust growth from small sizes to planetesimals and finally, planet formation.

2.1 A hierarchical picture of star and planet formation

Planetary systems form alongside their host stars, and are fundamentally interlinked to them. Gravitational collapse of giant molecular clouds (GMCs) at parsec scales triggers the formation of pre-stellar molecular clouds. In this thesis, we focus on low-mass star formation as we are interested in understanding the initial conditions of planet formation around solar-type stars. The distinction from the high-mass star formation theory lies somewhat broadly at 8 solar masses, or $M_\odot$, and below this value protostars are thought to have luminosities that are largely dominated by accretion (Armitage, 2017; McKee and Ostriker, 2007).

2.1.1 *Birth of a low-mass star*

Low-mass stars form cores of the order of a few thermal Jeans masses ($\sim M_\odot$) from the fragmenting overdense molecular clouds. The collapse of these "Larson" cores (Larson, 1969) follows a two-step sequential process: first, it proceeds isothermally, with the gravitational binding balanced by efficient radiative cooling and maintaining thermal equilibrium at $\sim$ 10–30 K. As the density of the core and the optical depth increase, this process transitions into an adiabatic phase, heating it up and leading to the formation of the "first" hydrostatic core. This core adiabatically contracts until H_2 dissociation (an endothermic process that begins at $\sim$ 2000K), after which heat from the gas is extracted rapidly, resulting in the polytropic index γ (defined from $P \propto \rho^\gamma$ where P and ρ are the gas pressure and density) falling from 1.4 to about 1.1. This culminates in the formation of a second core, i.e., the birth of the "protostar". Since the pioneer work of Larson, many numerical studies have tried to study the radiative properties of these shocks. Modeling this multi-scale problem with radiation hydrodynamic simulations covering timescales much longer than the birth of the protostar is a computationally expensive effort and an active field of research in itself (Bhandare et al., 2025, 2020; Teyssier and Commerçon, 2019).

A fundamental problem in star formation is the angular momentum budget: a star has far less angular momentum and magnetic flux than the dense gas in the interstellar medium (ISM). Angular momentum can be efficiently removed by magnetic fields when core contraction is sub-Alfvenic and ion-neutral coupling is good (when this process is too efficient, it

results in magnetic braking, preventing disk formation). Once either of these two conditions breaks down, gas starts to infall with a nearly conserved specific angular momentum j, forming a rotationally supporting star–disk system, or a young stellar object (YSO), with a disk of radius $R \propto j^2/GM_\star$, where G is the gravitational constant and $M_\star$ is the mass of the star (McKee and Ostriker, 2007). The sequence of low-mass star and planet formation is indicated by the schematic in Figure 3 from Öberg and Bergin (2021).

2.1.2 *Classification of YSOs*

The evolution of the star–disk system is better understood by describing the standardized classification. Observational estimates for the duration of the gas disk phase are typically a few Myr (Haisch et al., 2001). These YSOs were first observationally classified based on their spectral energy distributions (SEDs) in the infrared by Lada and Wilking (1984) and Lada (1987). YSOs exhibit two characteristic features in their SEDs:

1. An infrared excess on top of the stellar black-body continuum arising from the thermal emission of the hot dust in the disk.
2. A ultraviolet (UV) excess due to accretion hotspots on the stellar surface, coming from the accretion of the disk onto the host star via magnetospheric accretion columns.

Figure 4[1] illustrates the different formation stages of a YSO. The classification system is based on the slope of the SED in the infrared between the K and N bands (so $2.2 - 19$ μm), i.e., the infrared spectral index, which can be defined as (following Armitage, 2017):

$$\alpha_{IR} = \frac{d \log(\nu_{KN} F_{\nu_{KN}})}{d \log \nu_{KN}} \tag{1}$$

where ν_{KN} is the frequency of the observed radiation and $F_{\nu_{KN}}$ is the spectral flux density in the IR. Based on this, the four classes of YSOs are:

- **Class 0:** Deeply embedded sources with no detectable emission in the mid- to near IR and the SED peaks in the millimeter end of the spectrum ($\sim$ 100μm) coming from the cold envelope. The reported mass accretion rates for Class 0 objects are about $\dot{M} \sim 10^{-4} M_\odot/yr$.
- **Class I:** For these objects, the SEDs start rising to the mid-IR with $\alpha_{IR} > 0$. This reflects contributions from both the circumstellar disk and the remaining infalling envelope, with signatures of strong bipolar outflows, and the mass accretion rates are estimated to be about $\dot{M} \sim 10^{-6}$–$10^{-5} M_\odot/yr$. Both Class 0 and Class I objects exhibit episodic accretion events, such as FU Orionis events, with bursts causing variability of several magnitudes over short timescales.
- **Class II:** These sources, comprising the classical "T Tauri" stars, show a falling SED in the mid-IR with $-1.5 < \alpha_{IR} < 0$. These are optically visible sources with a well-developed circumstellar disk and ongoing magnetospheric accretion, with mass accretion rates of the order of $\dot{M} \sim 10^{-7} - 10^{-6} M_\odot/yr$. A large portion of the results presented in this thesis that pertain to radiation hydrodynamic models are primarily concerned with the Class II phase of evolution.

1 https://home.ifa.hawaii.edu/users/jpw/classes/star_formation/lectures/protostars.pdf

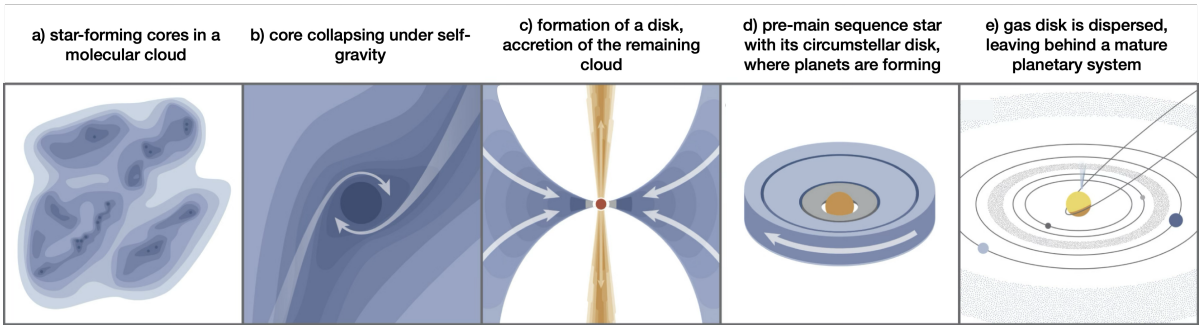


Figure 3: The different formation stages of star and planet formation. Figure is taken from Öberg and Bergin (2021)

- **Class III:** These sources are pre-main sequence stars with almost no excess in the near-IR. By this stage, the disk is largely dissipated, and the contribution to the SED is almost purely from the stellar photosphere. In many cases, sources retain a debris disk with planetesimals moving from the "protoplanetary" stages to formed planetary systems.

While the first observations of protoplanetary disks came indirectly through spectral energy distributions, modern high-resolution instruments on the JWST and VLT and more, such as *NIRSpec*, *CRIRES*, and *X-Shooter*, have revolutionized the spectral information available for these objects. Perhaps more remarkably, direct imaging of YSOs in recent times has been an extremely challenging but fruitful endeavor, unveiling rich morphological detail and transforming our understanding of planet formation.

2.2 DIRECT IMAGING OF PROTOPLANETARY DISKS

The composition of protoplanetary disks is directly linked to that of their parent molecular clouds (Mathis et al., 1977). This means that about 99% of their mass is composed of gas, and the remaining 1% is in the refractories, i.e., solids. The gas is mainly composed of hydrogen (about 70% in both atomic and molecular phases), about 28% of helium, and the remaining 2% is composed of heavier elements. At typical disk temperatures and densities, observing hydrogen and helium is hardly possible; hence, observations of dust particles and molecular tracers are what the community relies upon. For the dust, scattered light and thermal emission are sensitive to grain sizes $a \sim \lambda/2\pi$ for wavelength λ. Small sub-micron-sized grains are probed in the optical and near-IR, and millimeter (mm) and centimeter (cm)-wavelength observations are sensitive to larger solids, the so-called "pebbles".

From the first observations of young stellar objects in the optical with the HST to the galleries of disk observations in the infrared and the sub-millimeter continuum with the Atacama Large Millimeter Array (ALMA), and the Spectro-Polarimetric High-contrast Exoplanet REsearch (SPHERE) instrument on the VLT, JWST, we have obtained unprecedented structural detail within these disks. Figure 5 shows one of the first observations of the disk HH 30 in the optical with HST and the counterpart from recent JWST observations, respectively, showing the striking difference in how much the science has transformed.

Disks have now been imaged across all morphological classes, as discussed previously in Sect. 2.1.2, through various ALMA large programs. Disk Substructures at High Angular Resolution Project (DSHARP) surveyed Class II disks in the millimeter continuum (Andrews et al., 2018), Early Planet Formation in Embedded Disks (eDisk) captured the envelopes and

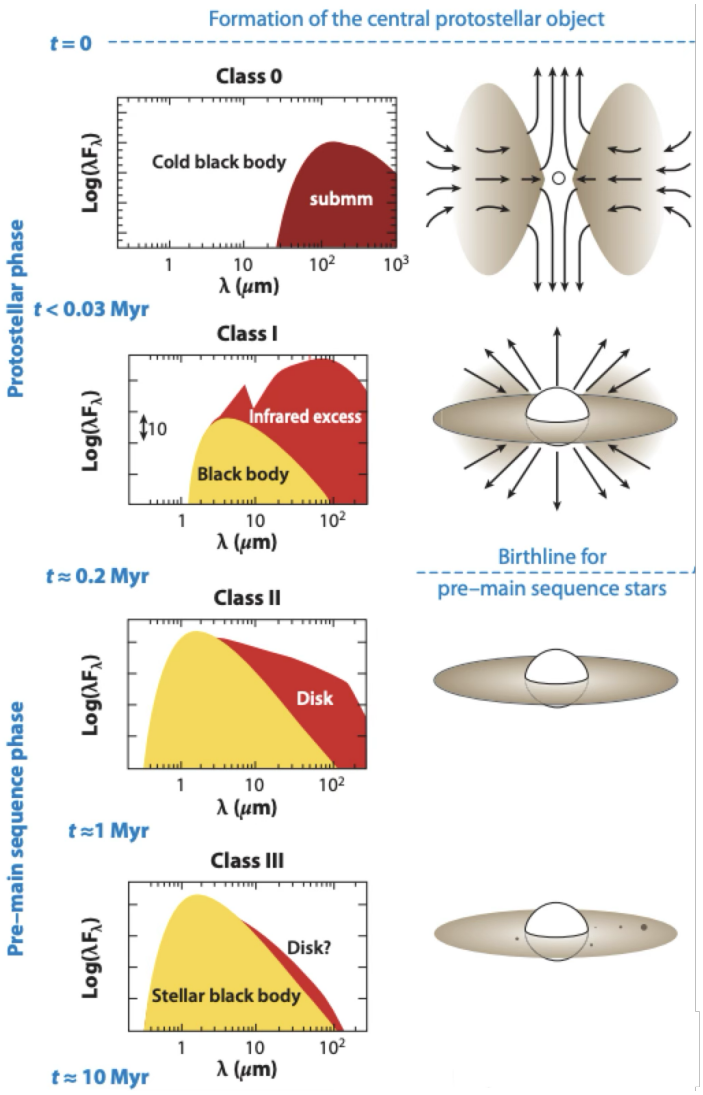


Figure 4: An illustration depicting the classification of young stellar objects. The figure is taken from the star formation course by Jonathan Williams. Originally from Dauphas and Chaussidon (2011).

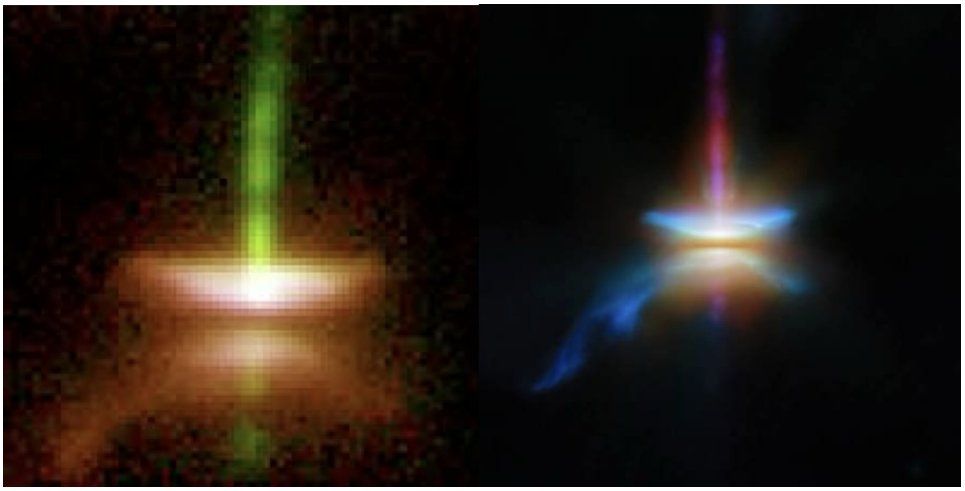


Figure 5: HST and JWST images of HH 30 (Burrows et al., 1996; Tazaki et al., 2025)

the environments of the younger Class 0/I stages (Ohashi et al., 2023), and more recently ALMA Debris Discs Survey (ARKS) has studied the properties of debris disks (Marino et al., 2026). Beyond large programs, targeted high-resolution observations of individual sources have yielded equally important results, characterizing disk morphology with the discovery of substructures, kinematics, and chemistry.

Different regions of the disk have been observed with different tracers, allowing us to begin building more complete theories of the disk's vertical and radial structure. Beyond structure, this has also let us test the different physical processes operating in those regions. Figure 6 highlights this clearly, illustrating the point with emission regions (right panel) and the structure and physical processes (left panel). Molecular lines have been helpful in tracing different vertical layers of the disk. With optically thick-to-thin molecular tracers, one can measure the gas density structure with great accuracy Law et al., 2021; Paneque-Carreño et al., 2023. In the dust, the sub-millimeter continuum traces the large-grain population in the midplane, which is expected to undergo growth, conducive to planet formation. Scattered light traces the disk surface, providing insights into the upper layers. With the SPHERE instrument at the VLT, we have probed spirals, shadows, and warps in the near-IR using polarized light scattered off small dust grains at the disk surface—shadows being of particular relevance to this thesis. The inner regions of the disk at sub-AU scales are now being revealed in the infrared with interferometry using the *MATISSE* and *GRAVITY* instruments on the VLT.

2.3 CONSTRAINTS FROM DISK OBSERVATIONS

Here, we discuss a few important, though non-exhaustive, constraints on different disk properties from observations, focusing on aspects such as radial and vertical structure, gas and dust masses, sizes, turbulent properties, etc.

- **Dust mass:** Measurements of dust masses are typically made from continuum emission assuming it is optically thin: the intensity can be written as $I_\nu = B_\nu(T_{dust})\Sigma_d\kappa_{abs}$ using the dust temperature and opacity (see early works such as Beckwith et al., 1990; Hildebrand, 1983). This method is subject to uncertainties related to grain sizes and composition, which has led to the "mass budget problem," where the derived disk masses are

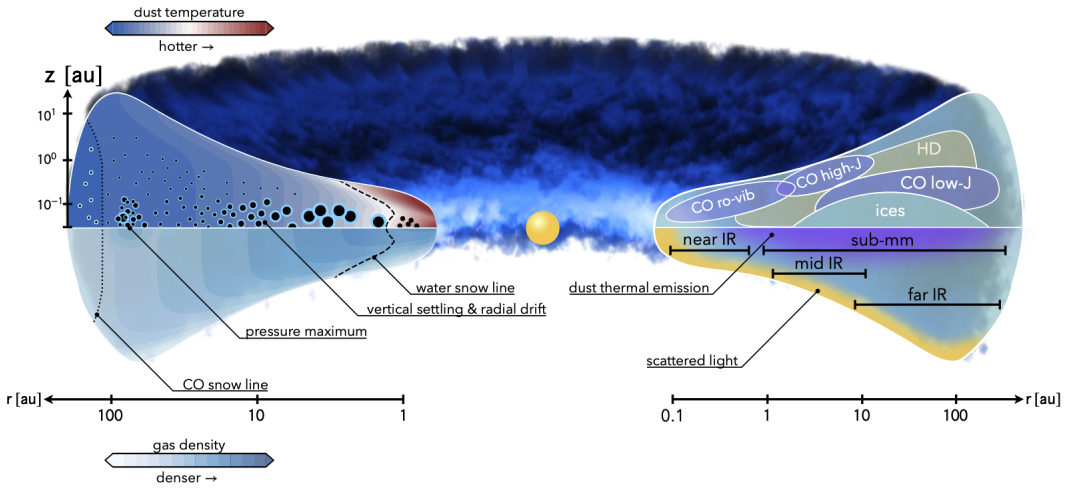


Figure 6: A schematic that highlights the different emission regions of tracers used to observe disks. Correspondingly, the relevant dynamical processes are also mapped (left). Figure is taken from Miotello et al. (2023).

insufficient to form cores of known planetary systems. However, the discovery of large samples of optically thick substructures in disks (Andrews et al., 2018; Huang et al., 2018; Long et al., 2018) has made it clear that dust mass estimates using this method are severely underestimated, especially when scattering is taken into account. Across different star-forming regions like Lupus, Taurus, Chamaeleon I and Upper Scorpius, dust masses differ by two orders of magnitude from young to old disks (Tychoniec et al., 2018), but measuring the true dust mass without systemic optical depth effects and uncertainties from grain sizes remains an open challenge.

- **Gas mass:** Estimating gas disk masses is challenging since H_2 gas lacks a permanent dipole moment and cannot be used to observe the fraction of gas in protoplanetary disks (Bergin and Williams, 2017). Indirect tracers such as CO isotopologues ^{13}CO, $C^{18}O$, which requires the relative abundance of CO with respect to H_2, "χ_{CO}" (usually a value of $\approx 10^{-4}$ is used) to measure gas disk masses (Deng et al., 2023; Miotello et al., 2016; Williams and Best, 2014). Further, CO-based gas disk masses are found to be lower than those derived from HD emission, suggesting unaccounted chemical effects. Recent work has shown that combining CO-isotopologues with N_2H^+ can break this degeneracy and yield masses more consistent with those from the HD emission (Anderson et al., 2019). Additionally, Yoshida et al. (2022) proposed a new method of constraining the gas surface density using the pressure-broadened wing emission of spectral lines.

- **Disk size:** For disk sizes, there is no single agreed-upon definition. Studies use radii that contain fixed fractions of the total fluxes, like 68% or 90% (Long et al., 2022; Tripathi et al., 2017). Gas sizes are measured from optically thick CO, and dust sizes are inferred via the millimeter continuum. Because tracers respond differently to optical depth, it is not straightforward to compare gas and dust disk sizes (Facchini et al., 2017). However, Trapman et al. (2019) suggest that gas-to-dust size ratios above 4 can be confidently attributed to radial drift of dust. Recently, the ALMA Survey of Gas

Evolution in PROtoplanetary Disks (AGE-PRO) large program (that looked at K and M stars) reported median gas disk sizes of 74 au and 110 au in Lupus and Upper Scorpius star-forming regions, and size ratios spanning 1-5.5 (Trapman et al., 2025; Vioque et al., 2025).

- **Dust settling and vertical structure:** The vertical distribution of dust grains reflects a balance between gravitational settling and turbulent mixing by the gas, with a degree of coupling set by the grain's Stokes number St, i.e., the ratio of the aerodynamic stopping time of the dust particle with respect to the orbital time (Dubrulle et al., 1995). This can be probed by dust observations in multiple wavelengths, with infrared scattered light tracing small, poorly coupled grains lifted above the midplane to the millimeter continuum, which probes settled, larger dust grains. Settling is confirmed by edge-on disk surveys like Villenave et al. (2020). With ALMA, settled and puffed-up ring structures in inclined disks (Doi and Kataoka, 2023; Pinte et al., 2016) can be used to measure dust scale heights. This can also provide constraints on disk turbulence, as the dust scale height values provide a way to constrain α_z/St where α_z is the turbulent diffusion parameter in the vertical direction.

- **Rotation curves and gas pressure structure:** High-resolution molecular line observations enable measurements of the gas rotational curves in protoplanetary disks, allowing for constraining kinematic signatures of substructures or embedded planets (Izquierdo et al., 2023; Pinte et al., 2023). Assuming centrifugal balance, the azimuthal velocity of the gas is given by

$$v_\phi(R,z) = \sqrt{\frac{GM_\star R^2}{(R^2+z^2)^{3/2}} + R\frac{\partial \Phi_{gas}}{\partial R} + \frac{R}{\rho_{gas}}\frac{\partial P_{gas}}{\partial R}} \tag{2}$$

  where $M_\star$ is the stellar mass, ρ_{gas} and P_{gas} are the gas density and pressure, and Φ_{gas} is the gravitational potential sourced by the disk's own mass (i.e., self-gravity), satisfying the Poisson equation $\nabla^2\Phi_{gas} = 4\pi G\rho_{gas}$. The three terms are contributions from the stellar gravity, the disk's self-gravity (which becomes significant for massive disks Kratter and Lodato, 2016), and finally the pressure gradient. Deviations from Keplerian rotation—where the velocity profile is set purely by the stellar gravity and $v_K = \sqrt{GM_\star/R}$—can now be measured with high spatial and spectral resolution observations from ALMA (Teague et al., 2018). The correlation between the continuum rings and the gas pressure maxima derived from the velocity profiles supports the idea of dust trapping (Pinilla et al., 2012; Whipple, 1972). The gas pressure profiles themselves, and the rotation curves of many disks, have been inferred from the large sample of high-resolution observations from the exoALMA large program (Stadler et al., 2025).

- **Turbulent properties of the disk:** Turbulent motions in the protoplanetary disk span a broad range of physical scales, from large, energy-injecting motions to small eddies (Lyra and Umurhan, 2019). As discussed above, turbulent properties can be inferred from dust scale height observations of disks. Overall, the results from this method indicate low levels of turbulence, with a turbulent diffusivity parameter $\alpha_z \sim 10^{-4}$. In the radial direction, using dust ring widths, (Dullemond et al., 2018; Villenave et al., 2025) estimated the radial turbulent parameter α_r, and the typical values are higher than α_z, with α_r on the order of 10^{-3} but with larger scatter.

More direct methods have been proposed in the literature. Non-thermal broadening of molecular lines has been used to find subsonic turbulence in a few disks (Flaherty et al., 2020; Paneque-Carreño et al., 2024), pointing towards values of turbulent $\alpha_t \lesssim 10^{-3}$ to 10^{-2} in the outer disk (Flaherty et al., 2018; Flaherty et al., 2020; Teague et al., 2018). These measurements come with caveats: unresolved structure being misinterpreted as line broadening, sensitive assumptions about the disk radial and vertical structure, and observational degeneracies, making it very difficult to identify the physical origins of turbulence (Hardiman et al., 2026; Rosotti, 2023). Coherent, large-scale deviations from Keplerian motion in kinematic measurements have been proposed as evidence for turbulence-driving instabilities (Barraza-Alfaro et al., 2021; Hall et al., 2020). In particular, for gravitationally unstable disks, this has been done using the "wiggle" kinematic signature from the spiral structure (Paneque-Carreño et al., 2021).

- **Disk temperature:** Gas temperatures can be constrained by direct methods, through line fluxes or channel maps, or using indirect chemical tracers like snowlines (Najita et al., 2007). For the inner disk (< 1 au), near- to mid-IR lines of CO, OH, and water can be traced, yielding temperatures ranging between $500 - 1700$ K, although there could be degeneracies due to unresolved emitting regions (Salyk et al., 2011). The intermediate disk regions ($3 - 10$ au) are largely unconstrained, limited by the sensitivity of ALMA and the absence of observing facilities in the far-IR. With the resolved channel maps of the CO isotopologues calibrated against the CO snowline ($\sim 17 - 30$ K), the clearest temperature constraints are for the outer parts of the disk, although it is difficult to isolate the radial and vertical temperature structure with just one tracer (Schwarz et al., 2016). Vertical emission surfaces for different tracers (sampling different heights) for edge-on or inclined disks can help locate CO snowlines, reconstruct a 2D map, and constrain turbulence measurements (Law et al., 2021; Paneque-Carreño et al., 2023; Pinte et al., 2018).

- **Magnetic fields:** The morphology of magnetic fields can be probed with polarized dust emission from magnetically aligned grains (in protostellar envelopes); however, it is difficult to do this in protoplanetary disks. Where we expect to see the toroidal magnetic field, polarization observations in disks are often consistent with dust self-scattering or gas-grain alignment. This is attributed to weak fields and high gas densities, with notable exceptions such as HD 142527 (Kataoka et al., 2016). Line polarization (the Goldreich–Kylafis effect) has also been used, although the interpretation is not accurate, as it shows degeneracies with disk geometry. The magnetic field strength can be estimated using the Zeeman effect, although this is difficult to probe due to the lack of suitable paramagnetic molecules (with CN emission being the only viable tracer) and other line-of-sight constraints (Lankhaar and Teague, 2023). Recently, Teague et al. (2025) provided the first robust measurement of the magnetic field strength in the TW Hya system ($\sim$ 10mG at $60 - 120$ au), notably stronger than previous estimates from wind-driven accretion modeling.

Several other important disk properties and observational advances have also been made, including an estimation of mass accretion rates, which have recently been transformed using instruments such as *X-Shooter* and *MUSE* (Alcalá et al., 2017; Manara et al., 2015, 2020), and observations of winds and outflows (Pascucci et al., 2023). With the JWST program MIRI Mid-infrared Disk Survey (MINDS), we understand the chemical inventory of the inner disk much

more, revealing compositional differences between solar-type stars and M dwarfs (Tabone et al., 2023), and the water content in the inner disk (Perotti et al., 2023; Temmink et al., 2025). Variable cold water emission in disks with dust traps resolved in the ALMA continuum further suggests that these traps are leaky, allowing small grains to pass through (Arulanantham et al., 2025; Banzatti et al., 2025). Shock tracers in the disk (SO emission) point to localized planet- or infall-driven shocks (Zagaria et al., 2025). About 50% of stars form in multiple systems, and binarity and multiplicity have been confirmed in large samples of YSOs (Tobin et al., 2022). Disk evolution is further shaped by external factors, including observations of stellar encounters (Dai et al., 2015), external photoevaporation (Winter and Haworth, 2022), and late-stage infall detected in Class I and even Class II sources (Garufi et al., 2022; Gupta et al., 2024; Pineda et al., 2023). For the sake of brevity, we do not discuss these topics further here, and note that there are many more aspects of disk observations that we do not summarize in this section.

2.4 SUBSTRUCTURES

Observations of disks across multiple wavelengths have shown that substructures are ubiquitous in large samples of disks, though this picture is not complete, as there is a bias toward high-angular-resolution observations directed toward large and bright sources. Disk substructures can only be characterized when the emission is spatially well-resolved and observed with a high signal-to-noise ratio. Gaps, rings, spirals, crescents, and asymmetries are common types of structures found in observations, and they appear across different observational primers such as continuum emission, scattered light, molecular tracers, and more.

At the time of the Protostars and Planets VII meeting in 2023, Bae et al. (2023) showed that in their sample of 355 objects with confirmed "disk" detections, substructures were found in 73 disks. There is also a correlation of the fraction of disks with substructures and the stellar mass, with a higher occurrence rate for high-mass disks (Andrews et al., 2013; Pascucci et al., 2016). However, it is extremely important to note the observational biases underlying these statistics. For disks observed with low angular resolution in their sample, the occurrence rate of substructures was about 2%, and for those observed with high effective angular resolution, 95% of them had substructures.

Figure 7 shows a sample of these disks with substructures observed with high-contrast imaging using instruments SPHERE, Gemini Planet Imager (GPI) and The High-Contrast Coronographic Imager for Adaptive Optics (HiCIAO) from Benisty et al. (2023) (and specific references therein), based on the type of structure identified in the optical and near-IR. Galleries such as this one are now available across multi-wavelength studies, and the recent ALMA survey paper by Kurtovic et al. (2026) provides a large, informative collection of continuum substructures in the mm.

2.4.1 *Dust traps and their origins*

Two main questions that arise when understanding these substructures are: What physical processes are responsible for substructure formation, and can we observationally constrain them? Furthermore, what are their implications for dust growth and planet formation? Understanding this requires a grasp of how dust particles move in a protoplanetary disk: Gas orbits at sub-Keplerian speeds due to pressure support (as seen in Eq. (2)), and hence, dust

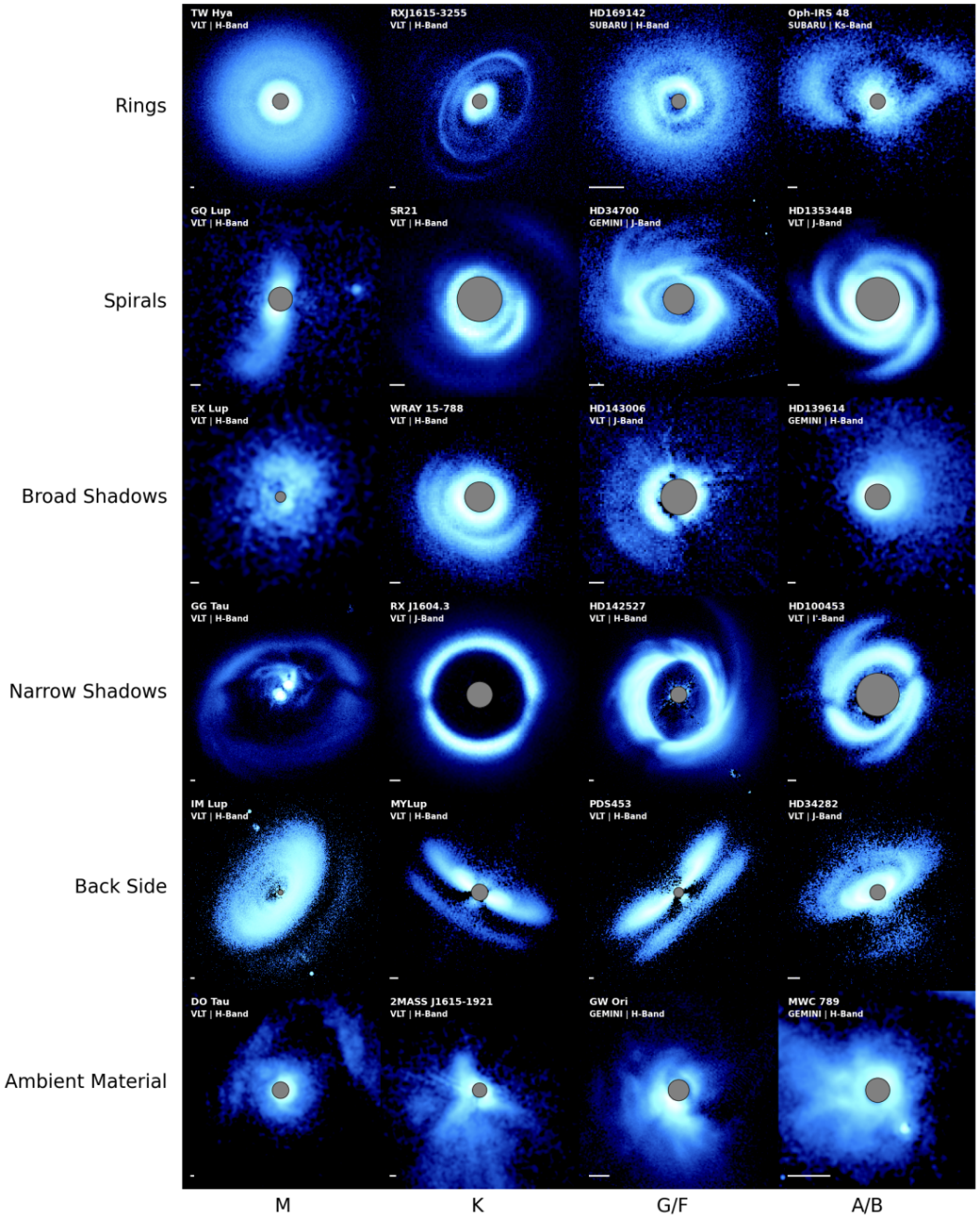


Figure 7: A gallery of different types of substructures observed with optical and near-IR observations taken from Benisty et al. (2023). For more details, see references therein.

grains (lacking the pressure component) experience a headwind that removes angular momentum and causes them to drift inwards. This process, also known as "radial drift", is strongest when the particles are marginally coupled to the gas (St ~ 1). We have also seen that the vertical distribution of dust is controlled by settling and mixing (see Sect. 2.3). However, this radial drift of dust can be stopped by pressure traps in the disk. Radial dust trapping has been discussed for many years, dating back to Weidenschilling (1977) and Whipple (1972). The direction and the speed of drifting dust depend on the local gas pressure gradient. In protoplanetary disks, the pressure profile decreases monotonically outwards, hence the dust drifts inwards. If there is a local pressure maximum, or a "pressure bump", dust particles converge towards this region, acting as a "dust trap" (Pinilla et al., 2012). We now know that substructures are regions where the local pressure gradient of the gas can be inverted, acting as dust traps with implications for dust growth and planetesimal formation (Johansen et al., 2007; Youdin and Goodman, 2005a).

Various dynamical processes in the disk can create substructures. These range from companions such as planets and binaries, hydrodynamical and magnetohydrodynamical processes, tidal interactions, shadowing, etc (Bae et al., 2023).

Dynamical perturbers: Planets drive spirals and clear out gaps (Goldreich and Tremaine, 1980; Rafikov, 2002), creating symmetric ring and gap structures in the continuum (Pinilla et al., 2012; Rice et al., 2006; Ziampras et al., 2023). These pressure bumps can also form vortices azimuthally (Li et al., 2000; Lovelace et al., 1999) and velocity kinks in the channel maps (Pinte et al., 2018). Because we expect planets to be forming in protoplanetary disks (and have imaged four of them), planets can be the leading explanation for most observed substructures. However, this does not mean that all substructures are caused by them, nor that every ring corresponds to a planet. Larger companions, like binary stars, can clear out cavities in their circumbinary disk (Kurtovic et al., 2022; Sudarshan et al., 2022) along with azimuthal asymmetries at the cavity edges. Tidal interactions, such as stellar flybys, can warp the disk, launch spirals within it, and sometimes break the disk, causing structure. Externally driven processes such as infall can produce spiral density waves and ring–gap structures (Kuznetsova et al., 2022), and photoevaporation can create cavities and, in some cases, annular gaps in the disk. For more information on photoevaporation, the reader is referred to Alexander et al. (2014).

Magnetic fields: Although it is difficult to constrain the magnetic field strength in a protoplanetary disk, theoretical studies have placed constraints on how they could affect the disk geometry at different regions. In the ideal-magnetohydrodynamic (MHD) scenario, the presence of a weak magnetic field can drive the (magnetorotational instability (MRI) Balbus and Hawley, 1991). This generates turbulence and can provide the required angular momentum transport in the inner disk (Flock et al., 2011). For the MRI, Heinemann and Papaloizou (2009) suggest that spiral substructures can be generated when the waves swing from leading to trailing. Zonal flows are also thought to create ring structures in ideal MHD simulations (Johansen et al., 2009). Pressure bumps can form at the inner/outer edge of the "dead zone" (Gammie, 1996) between the MRI active and the stable regions (with MRI stabilized by non-ideal MHD effects). Gaps can form at this transition, with a density contrast large enough to be unstable to the Rossby wave instability (RWI), generating vortices (Flock et al., 2017a, 2015; Lyra et al., 2015). Other non-ideal effects like the Hall effect and ambipolar diffusion

can also drive zonal flows (Béthune et al., 2016), and formation of rings (Riols and Lesur, 2019), driving structure. In both ideal and non-ideal MHD scenarios, magnetized winds can be launched, and can lead to structure (Suzuki et al., 2016).

Hydrodynamics: Hydrodynamic instabilities that operate in the dead zone can also generate substructures. Examples include the previously mentioned RWI, vertical shear instability (VSI), convective overstability (COS), and zombie vortex instability (ZVI).

Studies have shown that several regions can be susceptible to RWI. Namely, the edges of the MRI-dead zone, where there is a significant transition in viscosity and accretion rates, the edges of planet-driven gaps, and the region subject to mass addition during infall onto the disk. RWI can also occur between corrugated modes induced by the VSI (see below). The instability arises from a local minimum in the radial profile of vortensity described by the "Lovelace criterion", a quantity that is a function of the density, temperature, and orbital frequency of the gas (we refer the reader to the definition and equations within Li et al., 2000; Lovelace et al., 1999) and see also recent work by Chang and Youdin (2024). Thus, strong perturbations in density and/or temperature causing large shear in the velocity lead to a Kelvin-Helmholtz-like instability, creating anticyclonic vortices. Asymmetries observed in disk observations are often attributed to these RWI-induced vortices arising at the edge of a dynamical transition zone. Additional structures, such as spirals, could also be launched after the vortices form due to the RWI (Paardekooper et al., 2010).

The VSI is an instability originating from vertical variations in the disk's angular velocity (or a "vertical shear" Nelson et al., 2013; Urpin and Brandenburg, 1998). The vertical shear condition is met when there is a radial temperature gradient, which is generally likely in protoplanetary disks. As a result, VSI is an excellent candidate for driving turbulence in the disk that matches current observational constraints ($\alpha \sim 10^{-4}$). However, VSI additionally requires a short cooling timescale (rapid cooling of the gas Lin and Youdin, 2015) that is expected to be present in the outer regions of disks (Pfeil and Klahr, 2019). The VSI drives vertical gas flows, and numerical simulations (Flock et al., 2017b, 2020) show nearly-axisymmetric corrugation modes with radially alternating upward and downward gas motions. These vertical motions can cause dust particles to diffuse strongly, and mm-sized particles could be lifted up to vertical heights $z/R \approx 0.1$. These turbulently mixed dust particles manifest as bright and dark rings in the continuum (Flock et al., 2017b). The VSI motions can cause perturbations in the vortensity, generating vortices via the RWI (Flores-Rivera et al., 2020; Richard et al., 2016).

The COS is another hydrodynamic instability driven by radial buoyancy, where epicyclic perturbations are amplified exponentially in time, growing into sustained oscillatory motions in the disk. It operates for intermediate cooling timescales (when the disk cools at a rate comparable to the orbital time) and has been suggested to drive small and large-scale vortices in the saturated state (Lyra, 2014; Raettig et al., 2021). The ZVI (Barranco and Marcus, 2005; Barranco et al., 2018) is another candidate for forming vortices that require slow cooling, where baroclinic critical layers (i.e., where constant pressure and density surfaces are misaligned) can be excited before being damped. For an in-depth discussion on these instabilities, we refer the reader to (Lesur et al., 2023b). Figure 8 taken from (Bae et al., 2023) shows an illustrative picture of the above-mentioned processes collectively.

Radiation: Another important effect that has not been discussed above is that of radiation. Radiation hydrostatic and dynamic simulations have shown that shadows can create sub-

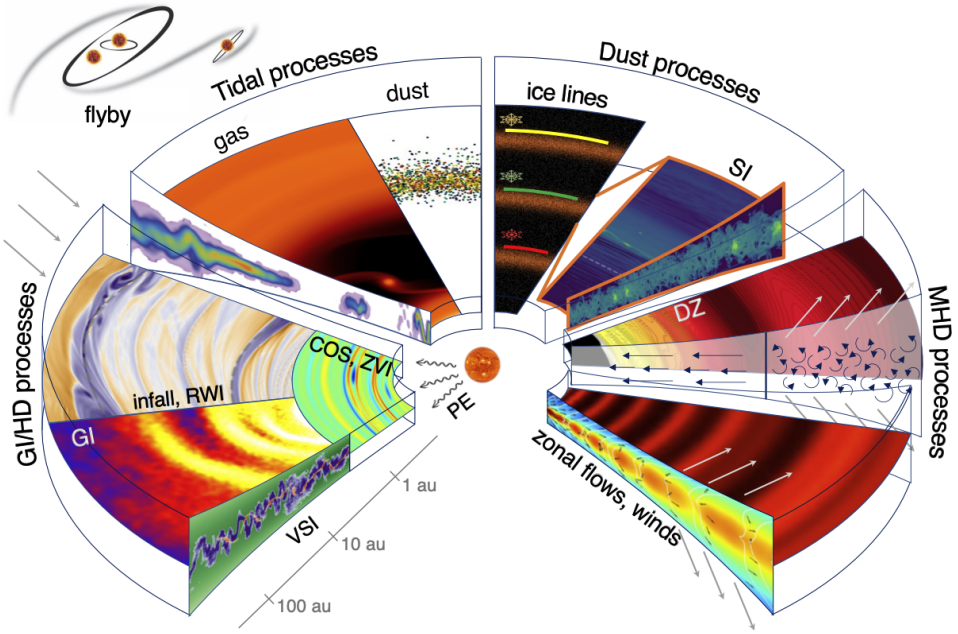


Figure 8: An illustration showing the different substructure forming mechanisms in the protoplanetary disk, taken from Bae et al. (2023, and references within)

structures in the disk. Misaligned inner disks can cast narrow and wide shadows in the outer disk that can drive spiral waves, observable in scattered-light observations (Montesinos et al., 2016; Zhang and Zhu, 2024) and also drive gap- and ring-like structures, and in some cases, vortices in the mm-continuum (Su and Bai, 2024; Ziampras et al., 2025a). Recently, it was shown that in externally irradiated disks, ring–gap pairs can form due to non-axisymmetric heating (Ziampras et al., 2026b). The inner rim of the disk can cast a strong shadow in the outer disk, as studied in Dullemond and Dominik (2004b) and Dullemond and Monnier (2010), and outbursts in the inner disk can form bright and dark shadows that show up as ring-gap structures in scattered light (Schneider et al., 2018).

Studies have further proposed that scale height perturbations in the disk, as a result of starlight hitting the surface of the disk unevenly, can result in growth and inward propagation of thermal waves (Dullemond, 2000; Watanabe and Lin, 2008; Wu and Lithwick, 2021). These thermal waves correspond to variable bright and dark ring–gap pairs (a staircase-like geometry) that were thought to produce observable features in the scattered light and ALMA emission (Ueda et al., 2021). This mechanism is part of the main results section of this thesis, where we drop simplified assumptions used in previous literature, perform detailed 2D radiation-hydrostatic and dynamic modeling, and discuss the implications of substructure formation.

2.5 MODELING ANGULAR MOMENTUM TRANSPORT

The hydro- and magnetohydrodynamic instabilities that we discussed in the previous section, coupled with radiation transport, do not merely generate substructures observed in gas and dust; they are primary candidates for driving turbulence in the disk, generating accretion stresses that redistribute angular momentum throughout the disk. To connect these mecha-

nisms to the global disk evolution, it is necessary to use a model to describe the resulting angular momentum transport and consequently, the mass accretion rates.

Historically, this has been parameterized using the α-disk model of Shakura and Sunyaev (1973), where the radial turbulent stress tensor (particularly, the $R\phi$ component Balbus and Papaloizou, 1999, that may include the hydrodynamic Reynolds stress, the gravitational stress, and the Maxwell stresses depending on the driving mechanism) can be written as $T_{R\phi} \equiv \alpha_{SS} P$, corresponding to a kinematic viscosity $\nu = (2/3)\alpha_{SS} c_s H$ with the dimensionless viscosity parameter α_{SS}, and the gas scale height H of the gas and sound speed c_s. Since the molecular viscosity of the gas in a disk is very low, this effective viscosity α_{SS} is thought to arise from the instability mechanisms discussed previously (e.g., MRI, VSI). With this prescription, the mass accretion rate due to radial viscous evolution can be written as,

$$\dot{M}_{acc}^{rad} \approx \frac{4\pi}{\nu_K} \frac{\partial(\alpha_{SS} \Sigma c_s^2 R^2)}{\partial R} \tag{3}$$

with the classical behavior of inner-disk accretion and viscous spreading of the outer disk (Lynden-Bell and Pringle, 1974).

In recent years, angular momentum transport via magnetic winds has emerged as an important, if not dominant, mechanism in the weakly ionized MRI-stable outer disk, according to many studies (Bai and Stone, 2013; Gressel, 2017; Suzuki et al., 2016). Rather than a radial angular momentum transport mechanism, wind-driven accretion removes angular momentum vertically, with gas escaping along open magnetic field lines that extend from the disk surface to larger distances. Consequently, this can be parameterized into a vertical stress α_{wind}, with a corresponding mass accretion rate $\dot{M}_{acc}^{vert} \approx 2\pi \alpha_{wind} \Sigma H c_s$. The viscous disk evolution equation, thus, comprises both contributions and can be written following (Bai, 2026; Tabone et al., 2022),

$$\frac{\partial \Sigma}{\partial t} = \frac{2}{R} \frac{\partial}{\partial R} \left[\frac{1}{\Omega R} \frac{\partial}{\partial R} \left(\alpha_{SS} \Sigma c_s^2 R^2 \right) \right] + \frac{1}{R} \frac{\partial}{\partial R} \left(\frac{\alpha_{wind} \Sigma c_s^2}{\Omega} \right) - \frac{\alpha_{wind} \Sigma c_s^2}{\Omega R^2 [2(\lambda - 1)]} \tag{4}$$

where λ is the magnetic lever arm parameter that sets the efficiency of wind-driven accretion, $\Omega = v_K/R$ is the Keplerian angular velocity, and $\alpha = \alpha_{SS} + \alpha_{wind}$.

It is important to emphasize that protoplanetary disks are unlikely to be described using constant values for α_{SS} and α_{wind}; in reality, they are radial and vertical profiles that vary depending on the region and over time. This model provides a predictive and useful framework, rather than identifying the underlying physical mechanisms that set these parameters. The values of α that determine gas accretion also dictate the local pressure structure felt by the dust grains, controlling both their radial velocity and the turbulent stirring that hinders further collisional growth (see Sect. 2.6.1). Moreover, as discussed above in Sec. 2.4, deviations from the smooth disk evolution picture, whether by local pressure maxima from zonal flows, or substructures due to MHD or hydrodynamic (HD) instabilities, can cause dust particles to be locally trapped. Understanding dust evolution and growth, and ultimately planet formation, therefore needs modeling dust on top of such a gas dynamical model. In the following section, we discuss how solid material, controlled by these gas properties, grows and is transported in the disk.

2.6 DUST EVOLUTION, PLANETESIMALS AND PLANETS

Understanding dust evolution is critical as the small fraction of dust in protoplanetary disks makes up pebbles, planetesimals, and planetary cores.

2.6.1 *Dust growth*

Dust grains begin from sub-micron interstellar monomers that coagulate via collisions, with relative velocities set by Brownian motion and by drift, settling, and turbulence (Ormel and Cuzzi, 2007; Weidenschilling, 1977). This growth is stalled by the fragmentation barrier, i.e., once collision speeds exceed 1–10 m/s. Then, dust grains prefer to fragment into smaller grains rather than stick together (Blum and Wurm, 2008; Güttler et al., 2010). This fragmentation barrier depends on gas properties like the local turbulent strength and surface density, and on material properties like the grain monomer density, setting the maximum grain size in the inner disk (Birnstiel et al., 2012). Further out, drift velocities between differently sized grains can trigger fragmentation before the turbulence-driven limit is reached, where fragmentation is controlled by drift. In the absence of fragmentation, the growing particles might not keep up with the radial drift, causing them to spiral inwards before reaching larger sizes, producing a drift-limited barrier in the outer disk (Brauer et al., 2008). Drift velocities peak for particles with Stokes numbers $\mathrm{St} \sim 1$—corresponding to meter-sized bodies for typical disk conditions—and this "meter-size" drift barrier means that these particles are swept away before they can grow past these sizes. At intermediate relative velocities, particles could also bounce off one another without growing or breaking apart, creating a "bouncing barrier" before fragmentation is relevant. Together, these barriers cap dust growth at typical sizes of about a mm or cm (Zsom et al., 2010). The resulting size distribution then depends on which of the mechanisms dominate at particular locations, ranging from the classical Mathis–Rumpl–Nordsieck (MRN)-like distribution $n(a) \propto a^{-3.5}$ that describes how the number density of grains scales with size a, to steeper distributions as turbulence goes down (Mathis et al., 1977).

Given these limits imposed by bouncing, fragmentation, and radial drift, forming planetesimals through growth alone might seem nearly impossible. However, several pathways have been proposed that could allow particles to overcome these barriers. Among them, "lucky" particles are proposed, which manage to stochastically avoid growth barriers (Windmark et al., 2012); however, this is still likely limited by the drift barrier (Birnstiel, 2024). Okuzumi et al. (2012) propose "fluffy growth", where the porosity of the particles can help them grow via enhanced drag. However, there are more favored pathways for planetesimal formation that we will discuss below.

2.6.2 *Forming planetesimals*

The most widely accepted theory of planetesimal formation invokes the gravitational collapse of pebble clouds (Safronov, 1972). This requires the local dust density to be much higher than the Hill density $\rho_{\mathrm{Hill}} \simeq {}^{9}M_{\star}/4\pi r^{3}$. The streaming instability (SI), provided the right conditions, is a key mechanism close to the midplane that can increase dust concentrations to provide the densities required for gravitational collapse.

SI belongs under the broader umbrella of resonant-drag instabilities (Squire and Hopkins, 2018) and arises from dust–gas drag. Because dust orbits at the local Keplerian speed but the gas lags behind due to additional pressure support, drag between the dust and gas causes the former to drift inward, pulling the gas with it in the azimuthal direction. A local dust overdensity enhances the azimuthal drag on the gas, which in turn reduces the dust speed and allows more particles to catch up, becoming overdense, in a self-sustaining runaway growth. While its robustness in the presence of external turbulence (Bi et al., 2026; Eriksson et al., 2026) and with multiple grain sizes (Paardekooper and Aly, 2025) is an active topic of research, current models suggest that strong clumping is possible for local dust-to-gas ratios above a few percent and Stokes numbers above $\mathrm{St} \sim 0.01$ (however, see Li and Youdin, 2021; Lim et al., 2024, for more recent work). Once the collapse of these strong clumps is initiated by SI, populations of planetesimals can be formed. There has also been work that suggests testing the SI scenario in the Solar System asteroid belt (Morbidelli et al., 2009), or in the Kuiper belt (Nesvorný et al., 2019); see also recent relevant work by Li and Chiang (2025).

To summarize planetesimal formation in three steps (Birnstiel, 2024): i) dust grows via collisions into sizes that are partially decoupled from the gas, often called "pebbles", ii) pebbles are concentrated by pressure bumps, zonal flows, snowlines, etc. to dust-to-gas ratios sufficient for the SI to operate, and iii) once local dust densities surpass the Hill density, direct collapse of pebble clouds produces planetesimals, bypassing the meter-size barrier altogether.

2.6.3 *Planet formation*

Once planetesimals are formed, planet formation can proceed via core accretion. There is a second, "gravitational fragmentation" scenario, but we focus mainly on the former, as it is thought to be the more dominant picture to form both terrestrial planets and the cores of gas giants.

Beyond the planetesimal stage, growth toward terrestrial planets is initially via direct collisions. For large enough planetesimal masses, this is aided by gravitational focusing, which increases the effective cross section of a particular body by a factor of $1 + v_{\mathrm{esc}}^2/\sigma^2$ where v_{esc} is its surface escape velocity and σ is the local velocity dispersion. An additional growth pathway is through pebble accretion, where aerodynamically coupled solids are captured as the gas drag slows down these "pebbles" to be effectively captured by the body via gravitational encounters (Lambrechts and Johansen, 2012; Ormel and Klahr, 2010). Dynamical friction keeps the velocity of the bigger body low (σ is small), and this in turn lets the body keep growing efficiently, driving a phase of runaway growth. Once the body becomes massive enough to stir up other planetesimals in its velocity, the runaway growth is halted, and the body proceeds to grow faster than its neighbors, transitioning into an oligarchic growth phase (Kokubo and Ida, 2002). This continues until the isolation mass is reached, at which point growth slows down even further. Now the relatively isolated planetary embryos further grow via collisions (Chambers and Wetherill, 1998). Their orbits are further perturbed by the influence of giant planets and resonances. The final assembly of terrestrial planets is expected on timescales on the order of 100 Myr (Raymond et al., 2005).

Giant planet cores are also thought to assemble through the same planetesimal and, more efficiently, pebble accretion scenarios (Ormel and Klahr, 2010; Pollack et al., 1996), as pebble accretion can dramatically reduce the timescales of core growth compared to planetesimal accretion (Chambers, 2014; Lambrechts and Johansen, 2012). As the core grows, it starts to

accumulate a massive envelope of surrounding gas and dust. Once the envelope becomes comparable to the core mass– the critical core mass (of the order of $\sim 10\ M_\oplus$) is reached, hydrostatic support fails (Ikoma et al., 2000), and the gas is accreted in a runaway process. This runaway growth phase is ultimately shut off when the local gas, or the disk, starts to disperse.

Gas giants are also thought to have formed via gravitational instability (Kratter and Lodato, 2016) if the disk is sufficiently massive ($M_{\rm disk} \gtrsim 0.1 M_\star$) to satisfy the Toomre criterion (Toomre, 1964) $Q = c_s \Omega_K / \pi G \Sigma \lesssim 1$. Along with this condition, sustained fragmentation can occur only when the disk's cooling timescale is very short (Gammie, 2001), restricting this mechanism to the early, and more massive phase of disk evolution.

Once planets have formed, they continue to dynamically interact with the disk, producing spirals and gaps; in return, disk torques act on the planet, resulting in planet migration (Kley and Nelson, 2012; Ziampras et al., 2024). The planet can also accrete the nearby dust and gas, forming a circumplanetary disk (Krapp et al., 2024b). The final composition of planets and the location they end up in requires modeling these complex gas flows and dynamical interactions.

2.7 OUTLINE OF THIS THESIS

Following the above introduction of the planet formation process from the early star formation stages to the growth and assembly of planets, we focus on discussing the questions important to this thesis. The physics results in the project mostly pertain to the Class II stage of protoplanetary disk evolution, and the numerical results can be applied throughout the different stages. Throughout this introduction, we talked about the importance of radiation, its role in controlling the microphysics, and opacities to macrophysical dynamical instabilities in the disk that strongly depend on the disk's radiative cooling timescales. Further, the importance of solids, the dynamics and growth of dust, and their role in planet formation has been a central point of discussion. Having this in mind, we can broadly characterize the results of this dissertation into two main questions:

Can stellar irradiation impact the thermal structure of the disk, generating shadows and substructures?

and

How can we accurately model dust dynamics in modern hydrodynamic codes?

To address this, we use the following outline for the rest of the thesis:

In Chapter 3, we present theoretical and numerical preliminaries relevant to the results section of the thesis. This includes the basics of irradiated disks and radiation transport. On the numerical side, we briefly discuss finite-volume methods, the Riemann problem, and timestepping. In Chapter 4, we present the results for the radiation hydrodynamic models of the disk for the project studying the temperature structure. In Chapter 5, we present the results for the dust fluid module of `PLUTO` (both the single and the multi-species versions) In Chapter 6, we present ideas on future work, outlook, and conclusions.

3

Theoretical and numerical concepts

In this section, we discuss a few theoretical and numerical concepts relevant for the forthcoming result chapters. We start by focusing on the physics of starlight-heated disks and radiation, and later move towards the numerical concepts required to understand how a code like PLUTO is used to solve hydrodynamic equations.

3.1 Static equations for an irradiated disk

Following Chiang and Goldreich (1997) and Dullemond (2000), for a geometrically thin, vertically isothermal disk that is heated by a central star of radius $R_\star$ and effective temperature $T_\star$, we can derive the static equations for temperature or the aspect ratio h of the disk under vertical hydrostatic equilibrium. The notation in the equations below follows Dullemond (2000). The incident stellar flux at radius R on the disk surface, given a flaring angle α (the angle between the incident stellar light and the disk surface), can be written as,

$$F_i = \alpha \left(\frac{R_\star}{R}\right)^2 \sigma T_\star^4. \tag{5}$$

The disk with temperature $T_{\rm disk}$ emits thermal radiation from its surface, with the flux $F_e = \sigma T_{\rm disk}^4$. Assuming energy balance, and equating $F_i = F_e$, we get,

$$T_{\rm disk} = \alpha^{1/4} \left(\frac{R_\star}{R}\right)^2 \sigma T_\star^4 \tag{6}$$

The flaring angle α can be defined in terms of the disk surface. For a height H_s above the midplane, a stellar ray grazing the surface at a radius R travels on a line with slope H_s/R, and the disk surface itself has a slope $\partial H_s/\partial R$. The angle between the incoming ray and the surface α is the difference between these two slopes,

$$\alpha \approx \left(\frac{\partial H_s}{\partial R} - \frac{H_s}{R}\right) = R\frac{\partial}{\partial R}\frac{H_s}{R}. \tag{7}$$

Under the vertically isothermal approximation, the disk temperature remains constant with height z. The vertical hydrostatic equilibrium for a rotating disk then yields a Gaussian density profile,

$$\rho(z) \approx \frac{\Sigma}{\sqrt{2\pi}H} \exp\left(-\frac{z^2}{2H^2}\right) \tag{8}$$

where H is the gas pressure scale height given by $H = \sqrt{R_{\rm gas} T_{\rm disk} R^3/\mu G M_\star}$ where R is the gas constant, and μ is the mean molecular weight. Typically, the ratio of the surface height to the

gas scale height can be written as the parameter $\chi = H_s/H$ (in Chiang and Goldreich, 1997, they use a value of $\chi \approx 4$). Substituting χ and α into Eq. 5, we obtain,

$$T_{\rm disk} = \sqrt{\frac{\mu G M_\star H^2}{R_{\rm gas} R^3}} \tag{9}$$

It is easier to work with dimensionless quantities $r \equiv R/R_\star$, $h(r) \equiv H/R$, we can arrive at,

$$r\frac{\partial h}{\partial r} = \frac{C}{r^2 h^8}, \tag{10}$$

where the constant

$$C = \frac{1}{\chi}\left(\frac{\mu G M_\star}{R_{\rm gas} T_\star R_\star}\right)^4 \tag{11}$$

yields the ordinary differential equation (ODE) that describes the radial evolution of the pressure scale height H in the disk. This first-order ODE has a family of solutions, and we write the result from Dullemond (2000)

$$h(r) = \left(\frac{7C}{2r^2} + K\right)^{-1/7}. \tag{12}$$

Here, K is an arbitrary integration constant. Thus, $K = 0$, yields the power-law solution (Chiang and Goldreich, 1997)

$$h(r) = \left(\frac{2}{7C}\right)^{1/7} r^{2/7}. \tag{13}$$

For negative K, the solutions have a singularity at radius $r = \sqrt{-7C/(2K)}$. When K is positive, the solutions approach a constant value at $r = \sqrt{7C/(2K)}$. As $r \to 0$, all remaining solutions tend to the above power law. As $r \to \infty$, the solutions either diverge from the power law at a certain radius to $h \to \infty$ or approach a flat (conical) profile ($h(r) \to$ const.). During their study, Dullemond (2000) discuss that when the integration was done in the direction of causality (from the inside out, as the disk is heated from inwards, i.e., the inner regions of the disk cast shadows over the outer disk), the solutions strongly deviated from each other, hinting towards a "physical" instability of the time-dependent equations. Under the above assumptions, this is indeed true, and was called the "self-shadowing" or the "thermal wave instability" (Ueda et al., 2021; Watanabe and Lin, 2008; Wu and Lithwick, 2021). In Chapter 4, we relax the assumptions of vertical isothermality and hydrostatics to solve two dimensional (2D) radiation hydrodynamic simulations with self-consistent modeling of the vertical and in-plane diffusion of radiation, along with a frequency-dependent prescription for the stellar irradiation.

3.2 MODELING RADIATION

Following the previous section, here we discuss the physics of radiation and the way we model radiation coupled to the hydrodynamic gas evolution. The full time-dependent, frequency-dependent radiative transfer equation is given by

$$\frac{1}{c}\frac{\partial I_\nu}{\partial t} + \mathbf{n}\cdot\nabla I_\nu = \eta_\nu - \chi_\nu I_\nu \tag{14}$$

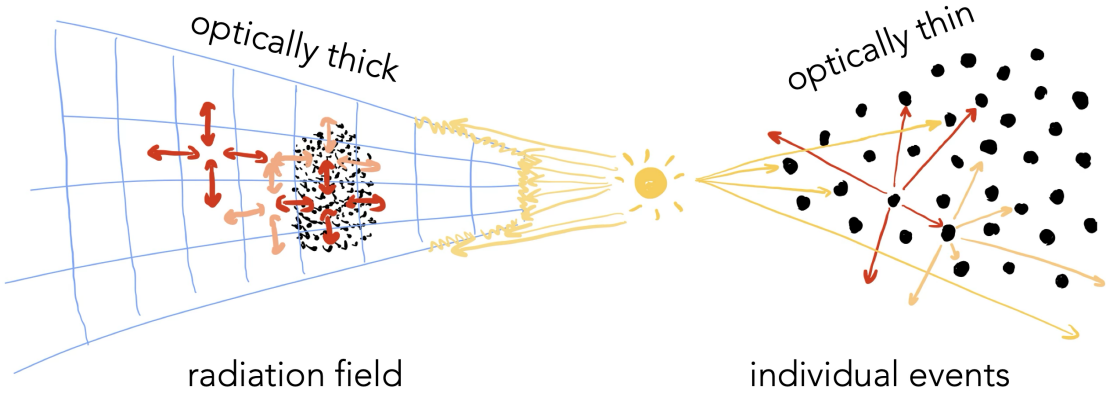


Figure 9: A schematic showing the limiting regimes for radiation: the optically thick "diffusion" limit (left), where the full radiation field must be treated, or the optically thin "free-streaming" limit (right), where rays can be traced as individual events. Credit: Alex Ziampras.

describing the change in the specific intensity I_ν due to emission η_ν and absorption $\chi_\nu I_\nu$ at a given frequency ν. Here, η_ν and χ_ν are emission and absorption coefficients, respectively; c is the speed of light, and $\mathbf{n}$ is the unit direction vector. Moreover, I_ν is a function of solid angle (η_ν and χ_ν can depend on direction too). Hence, this equation is not only a function of space and time, but also angle and frequency (Mihalas and Mihalas, 1984). This poses a difficult problem not only to accurately solve for the radiation field, but also to store the output—both can be computationally prohibitive. Moreover, it is essential that a good radiation transport scheme can capture the limiting regimes of radiation: when the mean free path is much larger than the typical length scale of the gas, matter is "optically thin" ($\tau << 1$) to radiation. On the other hand, when typical length scales are larger, then matter is opaque or "optically thick" ($\tau >> 1$) to radiation, where τ is the optical depth (see Fig. 9 for an illustration of these limiting regimes).

Monte Carlo, or particle-based methods like RADMC-3D (Dullemond et al., 2012) sample photon packets and follow their trajectory as they propagate through the medium. However, it becomes computationally very expensive to couple this with hydrodynamics, due to extremely slow convergence in the optically thick regions. Alternatively, a ray description can be used. This requires integrating intensities across straight lines: rays originate at each radiation source, and end at the boundaries where the radiation is completely absorbed. Ray-tracing-based methods can be used to calculate heating, cooling, and ionization (Abel and Wandelt, 2002). Another variation to this prescription is the "short characteristics" method, where rays propagate locally between active points onto nearby points (Bailey et al., 2024). In recent years, a discrete ordinate method has been used, which takes into account both the direction and frequency dependence of radiation along with the time-dependent transport (Jiang, 2021, 2022; Ma et al., 2026). This resembles the short characteristics approach, as it discretizes the radiation field across different directions, but also resembles fluid-based moment methods (which we use and discuss below) in that it solves for the time-dependent evolution of radiation. However, the discrete ordinates method can be computationally expensive and often requires simpler assumptions such as few directions and isotropic scattering along with short evolution times. Moment methods such as flux-limited diffusion (FLD) and M1 are

therefore very useful for radiation hydrodynamic simulations, and the former approach is particularly straightforward for simulating objects over long timescales and for generalizing to more complex physics (Flock et al., 2013).

In a moment method, instead of calculating radiative intensity, the moments of the radiation transport equation are used. The first moment is proportional to the radiative energy density E_{rad}, the second to the radiation flux $\boldsymbol{F}_{rad}$, etc., following

$$E_{rad} = \frac{1}{c}\int I_\nu \, d\Omega \quad ; \quad \boldsymbol{F}_{rad} = \int \mathbf{n} I_\nu \, d\Omega \tag{15}$$

and so on, where Ω is the solid angle. Then the time-dependent equations for E_{rad} and F_{rad} can be written in the following forms:

$$\frac{\partial E_{rad}}{\partial t} + \nabla \cdot \boldsymbol{F}_{rad} = \text{source terms} \tag{16}$$

and

$$\frac{\partial \boldsymbol{F}_{rad}}{\partial t} + \nabla \cdot \boldsymbol{P}_{rad} = \text{source terms.} \tag{17}$$

where $\boldsymbol{P}_{rad}$ is the radiation pressure tensor. Expanding on this, higher-order moments can also be utilized (see e.g., Variable Eddington Tensor approaches like Menon et al., 2022) that solve for $\boldsymbol{F}_{rad}$ and E_{rad} along with an approximation for the Eddington tensor $\boldsymbol{f}_e = \boldsymbol{P}_{rad}/E_{rad}$. Going back, the first moment is used for the diffusion, i.e., the FLD approximation. Eq. (16) is used along with a closure relation between E_{rad} and $\boldsymbol{F}_{rad}$ given by Fick's law $\boldsymbol{F}_{rad} \propto -\lambda \nabla E_{rad}$. Here, λ is a flux limiter (Levermore and Pomraning, 1981) that smoothly handles the transition between optically thick and free-streaming limits by modifying the effective diffusion. For the complete system of equations with FLD, we refer the reader to Sect. 4.3.1 of Chapter 4. Further, in combination with Eq. (16), Eq. (17) is also solved for the M1 approach (Melon Fuksman et al., 2021), along with a closure relation for $\boldsymbol{P}_{rad}$, and for non-relativistic flows, the introduction of a parameterized "reduced" speed of light approximation. The detailed formulation of the M1 equations can be found in Melon Fuksman et al. (2021) and Melon Fuksman and Klahr (2022).

We solve this set of coupled partial differential equations (PDEs) numerically using a code such as PLUTO. In the next section, we briefly describe how a code like PLUTO solves equations numerically.

3.3 THE FINITE-VOLUME METHOD IN PLUTO

Instead of solving a continuous problem that is described by a set of PDEs, a numerical method solves for a discrete, finite set of points or values. First, the domain of the PDEs is discretized into a finite set of points or volumes, forming a numerical *grid*. Then, the PDEs are sampled on the grid points. For a finite-difference approach (used in codes such as FARGO3D, Benítez-Llambay et al., 2019), discrete point values are defined at the grid points. In finite-volume codes, on the other hand, such as PLUTO (Mignone et al., 2007), *averages* over finite volumes are evolved in time.

PLUTO is designed to solve a system of conservation laws; the general form of such an equation can be written as

$$\frac{\partial \boldsymbol{U}}{\partial t} = -\nabla \cdot \boldsymbol{T}(\boldsymbol{U}) + \boldsymbol{S}(\boldsymbol{U}) \tag{18}$$

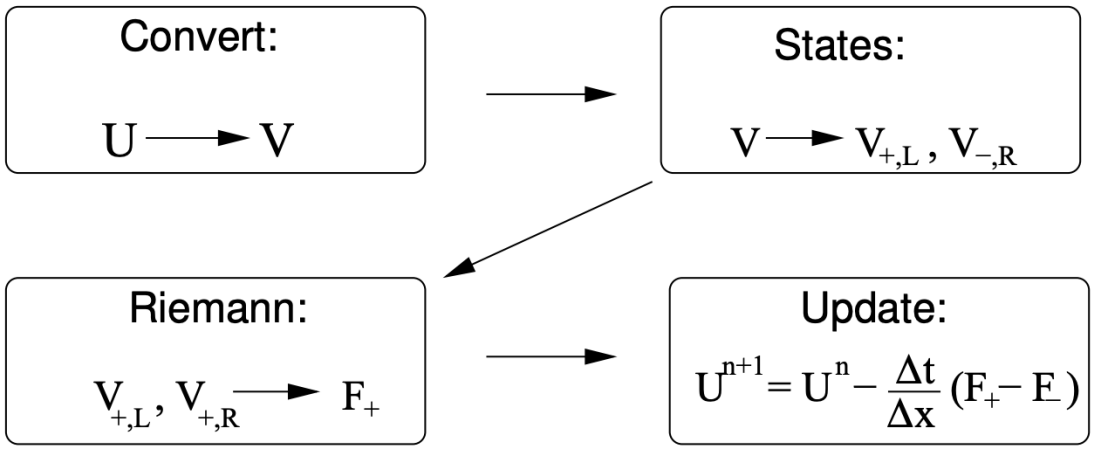


Figure 10: Numerical integration scheme used by the PLUTO code. The figure is taken from the Mignone et al. (2007) paper.

Here, $\boldsymbol{U}$ is called the "state" vector of conservative quantities, $\boldsymbol{T}(\boldsymbol{U})$ is a rank-2 tensor that contains the fluxes corresponding to each term in $\boldsymbol{U}$, and $\boldsymbol{S}(\boldsymbol{U})$ denotes the source terms. An arbitrary number of advection equations can be added to Eq. (18). While the conservative variables $\boldsymbol{U}$ are the ones being evolved in time, fluxes are more easily computed over primitive variables $\boldsymbol{V}$. The numerical integration of Eq. (18) follows: an interpolation routine, the solution of a Riemann problem across cell interfaces, and then a final time-evolution stage (Mignone et al., 2007). This is described by the schematic of the solution strategy used in PLUTO in Fig. 10.

During the interpolation step, cell-averaged primitive quantities are used to construct the left and right states at each interface, which are used by the Riemann problem. This is achieved using a *reconstruction* step in which the cell-averaged values are used to build a piecewise representation of the solution within each cell (typically a flat, linear, or higher-order polynomial is used). Additionally, a slope limiter is used (e.g., Van Leer, 1974) to suppress sharp numerical oscillations, avoiding issues such as additional extrema and overshooting near large gradients.

The fluxes at the boundaries during an interpolation step are estimated using solutions to the Riemann problem. At the core of finite-volume methods, it considers the advection of a jump discontinuity, using the analytical solution of the homogeneous problem. It allows for an analytical prediction of how much of the material (mass, momentum, energy, etc.) to be conserved enters into or leaves a cell. Approximate Riemann solvers, such as the HLL (Harten et al., 1983) or HLLC (Toro et al., 1994), can be used in PLUTO.

3.3.1 *Timestepping*

At the update stage, the conserved quantities are integrated in an "explicit" manner, where the solution in time $t + \Delta t$ is based on the known quantities from the current timestep t. This contrasts with "implicit" schemes, where the update equation contains the unknown future state, typically requiring a linear algebra operation for the time evolution, but can be useful for allowing arbitrary timesteps.

The stability of an explicit time-integration scheme is limited by a timestep constraint, a necessary criterion, that describes that waves cannot travel more than one grid cell per time-step. From von Neumann stability analysis, this so-called "Courant–Friedrichs–Levy (CFL)" condition can be written as:

$$\Delta t \leqslant \mathrm{CFL} \times N_{\mathrm{dim}} \frac{\Delta x}{\lambda_{\mathrm{max}}} \tag{19}$$

where Δx and λ_{max} are the grid spacing and the largest characteristic velocity, respectively, and N_{dim} is the number of spatial dimensions. CFL is then the Courant number used to control the simulation timestep. Additional time-step constraints might come from incorporating parabolic diffusive terms, such as radiative or dust diffusion in our case (see Chapters 4 and 5).

Part II

RESULTS

This part comprises the main results of this thesis. It is split into three chapters: Chapter 4 corresponds to the radiation hydrodynamic modeling of shadows in passive, irradiated protoplanetary disks. Chapter 5 discusses the fluid dust module of the `PLUTO` code, including both the one-fluid and multifluid implementations, along with tests. Finally, we discuss the outlook and summary of this work in Chapter 6. Each chapter will contain the corresponding information about the publications, whether published, under review, or submitted to the journal, and also mention the author contributions.

4

Starlight-driven flared disk geometry in radiation hydrodynamic models of protoplanetary disks

Based on the results from **Sudarshan, P.**, Flock, M., et al. 2026, A&A 706, A198 [ADS]
Authors: Prakruti Sudarshan, Mario Flock, Alexandros Ziampras, David Melon Fuksman and Tilman Birnstiel
Contributions: I set up the disk model with help from Mario Flock, developed the frequency-dependent ray-tracing routine, ran all the simulations, and wrote the manuscript. Alexandros Ziampras provided many suggestions to the code. All authors provided comments that helped improve the manuscript.

4.1 Overview

Protoplanetary disks observed in the millimeter continuum and scattered light show a variety of substructures. While embedded planets are a common explanation, various physical processes in the disk during the early planet formation phase might also trigger these features. One such possibility that has been theorized previously for passive disks is the thermal wave instability or the stellar irradiation instability, where the flared disk might become unstable as directly illuminated regions puff up and cast shadows behind them. This would manifest as bright and dark rings and as a staircase-like structure in the disk optical surface.

We provide a realistic radiation hydrodynamic model to test the limits of the thermal wave instability in starlight-heated protoplanetary disks. We make quantitative comparisons to existing results in the literature from simpler linear theory and 1D models to moment-transfer methods to elucidate the importance of correct numerical treatment for this problem.

We carried out global axisymmetric 2D hydrostatic and dynamic simulations that include radiation transport with frequency-dependent ray-traced irradiation and flux-limited diffusion. We varied the dust-to-gas ratios and surface densities. We also highlight the role of small grains and dust settling with the first radiation hydrostatic dust models to study starlight-driven shadowing.

We found that starlight-driven shadows are most prominent in optically thick, slowly cooling disks, which are shown by our models with high surface densities and dust-to-gas ratios of submicron grains $\epsilon_{\rm dg} = 0.01$. We found that thermal waves form and propagate inward in the hydrostatic limit. In contrast, our hydrodynamic models show bumps and shadows within 30 au that converge to a quasi-steady state on several radiative diffusion timescales, which indicates a long-lived staircase structure. Existing thermal pressure bumps might produce and enhance this effect and form secondary structures due to starlight-driven shadowing downstream. Hydrostatic models with self-consistent dust settling instead show a superheated dust irradiation absorption surface with a radially smooth temperature profile without staircases.

We conclude that we can recover thermally induced flared-staircase structures in radiation hydrodynamic simulations of irradiated protoplanetary disks using the flux-limited diffusion method. The shadowing effect is sensitive to the dust content in the disk. We highlight the importance of modeling dust dynamics consistently to explain starlight-driven shadows.

4.2 INTRODUCTION

Disk substructures are ubiquitous in modern observations of planet-forming disks at different wavelengths. These include millimeter thermal emission from ground-based instruments such as the Atacama Large Millimeter/submillimeter Array (ALMA, Andrews et al., 2018), scattered-light observations with the Spectro-Polarimetric High-contrast Exoplanet REsearch (SPHERE) instrument on the Very Large Telescope (VLT, Avenhaus et al., 2018; Benisty et al., 2023), and more recently, studies with the James Webb Space Telescope (JWST, Tazaki et al., 2025). Planets within the disk can carve out gaps (Zhang et al., 2018) that explain substructure, of which the planetary gaps in the PDS 70 and WISPIT 2 systems are direct observational constraints (Close et al., 2025; Haffert et al., 2019; Keppler et al., 2018; van Capelleveen et al., 2025). Many physical processes in the disk might also trigger substructures (Bae et al., 2023), such as zonal flows driven by the magneto-rotational instability (MRI, Flock et al., 2015; Uribe et al., 2011), spirals launched by gravitational instability (GI, Kratter and Lodato, 2016), gaps and vortices resulting from the vertical shear instability (VSI, Flock et al., 2020), and spirals and rings as a consequence of misaligned inner disks (Zhang and Zhu, 2024; Ziampras et al., 2025a). These substructures can invert the local pressure gradient and radially pile up dust in the disk, thereby acting as dust traps (Paardekooper and Mellema, 2004; Whipple, 1972). For a more detailed discussion on the origin and implications of substructures in disks, we refer the reader back to the Sect. 2.4 in the introduction.

The thermal wave instability (or TWI, also known as the irradiation instability or the self-shadowing instability) is one of the mechanisms proposed to explain rings and gaps in disks (Dullemond, 2000; Siebenmorgen and Heymann, 2012; Ueda et al., 2021; Watanabe and Lin, 2008; Wu and Lithwick, 2021) through induced pressure variations and subsequent dust concentrations. Historically, this instability was first studied as a consequence of unstable solutions (Dullemond, 2000) to the flared-disk model proposed to account for the observed infrared (IR) excess in spectral energy distributions (SEDs) of passive disks around T Tauri stars (Chiang and Goldreich, 1997; D'Alessio et al., 1999; Kenyon and Hartmann, 1987). This mechanism bring up an interesting possibility that substructures can form at the disk surface without invoking any additional ingredients, solely through starlight-driven shadowing through variations in the irradiation angle onto the surface of a flared disk. Indications of ripples in the flared-disk scattered-light observations of IM Lup with VLT-SPHERE (Avenhaus et al., 2018) are particularly exciting for this scenario.

The mechanism behind this starlight-driven shadowing is outlined as follows (see also Fig. 1 in Ueda et al., 2021, for an illustration): In vertical hydrostatic equilibrium, the balance between gravity and pressure (set by stellar irradiation heating) yields a passive (irradiated) disk with a flared geometry. The outer regions thus absorb substantially more stellar light than in an internally heated conical disk. For optically thick disks, scale-height perturbations might trigger thermal waves that travel inward, which expand regions in the disk and cast shadows behind them. This shows up as bright and dark bands or shadows in temperature or as staircases in the optical surface. They might be referred to in the literature as unstable

shadows to distinguish them from the stable shadow cast by the hot inner rim in transition disks (Dullemond et al., 2001) or due to an opacity transition at the snowline (Baillié et al., 2015).

Several methods have been used to study the problem of starlight-driven shadowing in the literature, such as, time-dependent linear theory (Dullemond, 2000; Wu and Lithwick, 2021), a simplified 1.5D treatment (i.e., mimicking vertical information with oblique radiation transfer, Okuzumi et al., 2022; Pavlyuchenkov et al., 2022a; Ueda et al., 2021; Watanabe and Lin, 2008), and 2.5D (i.e., 2D $\{R, z\}$ axisymmetric) Monte Carlo radiative transfer models (Siebenmorgen and Heymann, 2012; Wu and Lithwick, 2021). These models employed either a hydrostatic or a vertically isothermal approximation to the disk (i.e., temperature fluctuations at the atmosphere instantly propagate to the midplane). In the framework of linear analysis in the optically thick disk approximation (Chiang and Goldreich, 1997; D'Alessio et al., 1999; Dullemond, 2000), models predict that thermal waves form and propagate inward. Watanabe and Lin (2008) found quasi-periodic waves moving inward that decayed at 1 au. Their results also indicated that the disk might reach approximate steady states with bumps and shadows as the mass accretion rate $\dot{M}$ increases. The time-dependent analysis by Ueda et al. (2021) with a 1.5D energy equation, however, was focused on models with inwardly propagating waves and found a staircase structure radially outward up to ~100 au for high dust-to-gas mass ratios. Simulations by Wu and Lithwick (2021) with Monte Carlo models and a global two-layer model by Okuzumi et al. (2022) that accounted for radial reprocessing of starlight within the disk also showed inwardly propagating waves.

Based on the coupling of radial wave propagation and vertical radiative transfer, however, a full radiation hydrodynamic treatment is necessary to understand the existence and long-term implications of this instability in realistic disks. Radiation transport can be included in hydrodynamic simulations in several ways. These include moment methods such as flux-limited diffusion (FLD Levermore and Pomraning, 1981) in the gray (i.e., frequency-integrated) approach (Kley, 1989; Kolb et al., 2013), gray FLD coupled with frequency-dependent ray-traced irradiation (Flock et al., 2013; Kuiper et al., 2010), frequency-dependent multigroup FLD (Robinson et al., 2024), M1 moment-transfer method (Melon Fuksman et al., 2021), a three-temperature M1 scheme (Muley et al., 2023), and more complex discrete ordinate methods (Jiang, 2021, 2022).

There have been some efforts to model the thermal wave instability using 2.5D radiation–hydrodynamic simulations in the recent literature. Pavlyuchenkov et al. (2022b) used a diffusion solver for radiation transport, and Melon Fuksman and Klahr (2022) used M1 moment transfer coupled with frequency-dependent irradiation. While these works reported inwardly propagating thermal waves and shadows in the hydrostatic models, both studies concluded that the instability is damped in dynamical simulations. Melon Fuksman and Klahr (2022) attributed this to the inefficiency in transporting thermal perturbations from the atmosphere to the midplane in the disk, as radial advection is more dominant. Kutra et al. (2024) used a simpler radiative forcing method that enforced the vertically isothermal approximation and found staircase structures that did not propagate inward, but settled in the disk for several orbital timescales. Recently, Mori et al. (2025) also reported thermal wave instability-like behavior in their global nonideal magnetohydrodynamic models, particularly in the model that included the Hall effect and a magnetic field with an anti-aligned polarity.

We aim to provide limits to the thermal wave instability with a realistic radiation hydrodynamic model with frequency-dependent irradiation and cooling via FLD. We relaxed the

vertically isothermal assumption of previous models and modeled advection and thermodynamics in 2.5D self-consistently. While Kutra et al. (2024) also included mass advection in 2.5D with ATHENA++, our models treat radiation transfer more accurately. The ray-traced irradiation + FLD approach also modeled the transition from the optically thick to the thin regime better (Kuiper and Klessen, 2013) than Pavlyuchenkov et al. (2022b), which is more suited for optically thick disks. We also explored a higher resolution than they did, and we compared our models with the M1 moment-transfer method (Melon Fuksman and Klahr, 2022). Finally, we extended the gas-only disk simulations to include dust settling in the hydrostatic approach. This provided the first results on the effect of dust on starlight-driven shadows and the thermal wave instability.

For clarity, we adopt the following definitions for the words staircase, bump, and shadow: A staircase is a step-like modulation in the irradiation absorption surface as a function of radius, consisting of a heated puffed-up front (a bump) and a colder backside (a shadow). In this context, we note that a bump refers to the locally elevated irradiated region and not to the more traditional temperature or pressure gradient inversion commonly associated with the term. A shadow, in turn, corresponds to the dip in the irradiation flux downstream, which we trace through the midplane temperature decrease of at least 5% relative to the background power law. This is clearly illustrated by the sketch in Fig. 11. In Sect. 4.3 we outline the physics of our model, the model parameters, and the numerical methods we used. In Sects. 4.4–4.6 we show the results from our hydrostatic, dynamical, and dust models. We then discuss the implications of our results in the context of previous work in Sect. 5.7 and summarize our findings in Sect. 4.8.

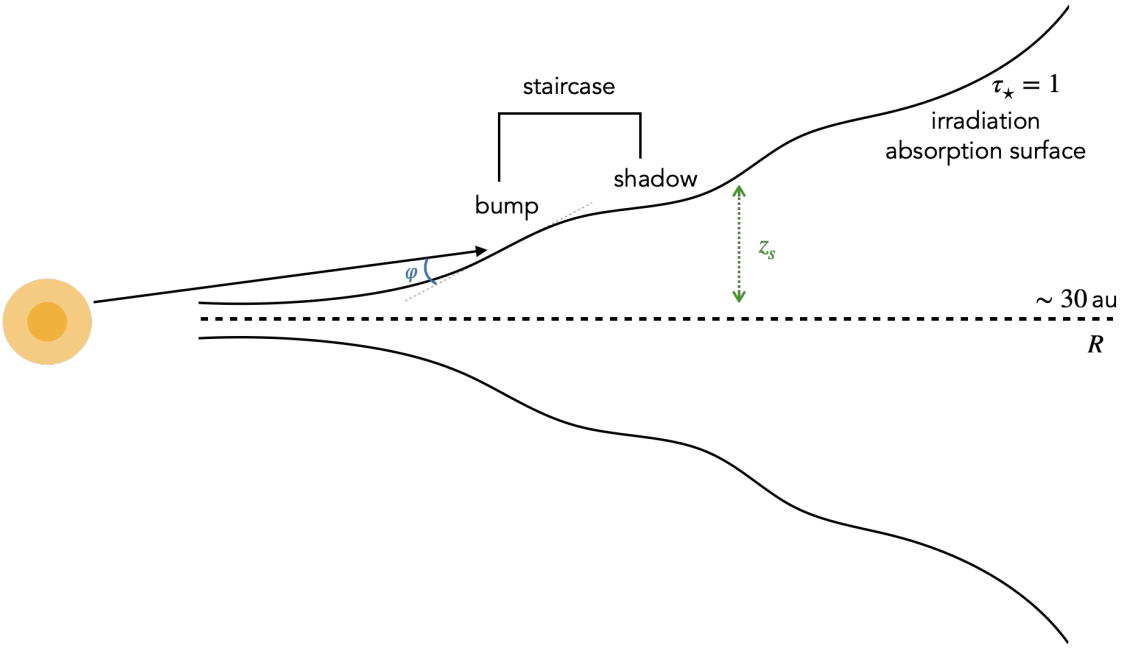


Figure 11: Schematic to illustrate the different terms bump, shadow, and staircase defined in the introduction. The star is denoted by the orange blob, and the stellar rays hit the irradiation absorption surface $\tau_\star = 1$ at a height $z_s = z\,(\tau_\star = 1)$ from the midplane and a flaring angle φ.

4.3 PHYSICS, MODEL SETUP, AND NUMERICS

In this section we outline the physics used in our model. We start with the gas dynamics and then describe the radiation method, whose self-consistent treatment is crucial for capturing how the disk responds to thermal perturbations.

4.3.1 *Radiation hydrodynamic equations*

We write down the conservative equations for gas dynamics and radiation transport in the gray flux-limited diffusion approximation following Flock et al. (2013). For gas density ρ, velocity $\boldsymbol{u}$, total energy density E_{tot}, radiation energy density E_{rad} and radiation flux $\boldsymbol{F}_{rad}$, they read

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{20a}$$

$$\frac{\partial (\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + P\mathbf{I}) = -\rho \nabla \Phi_\star, \tag{20b}$$

$$\begin{aligned} \frac{\partial E_{tot}}{\partial t} + \nabla \cdot [(E_{tot} + P)\,\boldsymbol{u}] &= -\rho \boldsymbol{u} \cdot \nabla \Phi_\star \\ &\quad - \kappa_P \rho c \left(a_R T^4 - E_{rad}\right) - \nabla \cdot F_\star \end{aligned} \tag{20c}$$

$$\begin{aligned} \frac{\partial E_{rad}}{\partial t} + \nabla \cdot \boldsymbol{F}_{rad} &= \kappa_P \rho c \left(a_R T^4 - E_{rad}\right), \\ \boldsymbol{F}_{rad} &= -\frac{\lambda c}{\kappa_R \rho} \nabla E_{rad}. \end{aligned} \tag{20d}$$

The gravitational potential of the star is given by $\Phi_\star = -GM_\star/r$ where G is the gravitational constant, $M_\star$ is the mass of the central star, and $r = \sqrt{R^2 + z^2}$ is the spherical radius distance to the star. The total energy is a function of kinetic energy and the specific internal energy e with $E_{tot} = \rho e + \frac{1}{2}\rho \boldsymbol{u}^2$. The gas pressure is denoted by P, and $\mathbf{I}$ is the unit matrix. The temperature-dependent Planck and Rosseland mean opacities are κ_P and κ_R respectively, a_R is the radiation constant, and c is the speed of light. The set of equations is complete with a closure relation, that is, the ideal gas equation $P = (\gamma - 1)\rho e$ with the adiabatic index chosen as $\gamma = 1.42$ (Decampli et al., 1978) and the mean molecular weight $\mu = 2.35$. The gas temperature T is given by $e = \rho c_V T$, with $c_V = R_{gas}/\mu(\gamma - 1)$ being the specific heat capacity at constant volume, and R_{gas} the gas constant. The isothermal sound speed is defined as $c_s = \sqrt{P/\rho}$ and the gas pressure scale height is given by $H = c_s/\Omega_K$ where $\Omega_K = \sqrt{GM_\star/R^3}$ is the Keplerian angular velocity.

The two coupled equations for radiative transfer are given by

$$\frac{\partial e}{\partial t} = -\kappa_P \rho c \left(a_R T^4 - E_{rad}\right) - \nabla \cdot F_\star, \tag{21a}$$

$$\frac{\partial E_{rad}}{\partial t} + \nabla \cdot \boldsymbol{F}_{rad} = \kappa_P \rho c \left(a_R T^4 - E_{rad}\right). \tag{21b}$$

We write the radiation flux above in terms of the radiation energy density with a flux limiter λ which accounts for the optically thick ($\lambda \to 1/3$) and thin regimes ($F_{rad} \to cE_{rad}$, see also Kolb et al., 2013). We use the flux limiter of Levermore and Pomraning (1981) which is given by $\lambda = \frac{2+X}{6+3X+X^2}$ with $X = \frac{|\nabla E_{rad}|}{\kappa_R \rho E_{rad}}$.

4.3.2 *Frequency–dependent irradiation*

The transition between the optically thick midplane of the disk to the optically thin surface is better captured by including ray-traced irradiation from the star at different frequencies (Kuiper et al., 2010). The frequency–dependent irradiation flux $F_\star$ from the central star is written in terms of the black body spectrum $B_\nu(T_\star)$,

$$F_\star(r) = \int_\nu \pi B_\nu(T_\star) \left(\frac{R_\star}{r}\right)^2 e^{-\tau_\nu(r)}\, d\nu\,, \tag{22}$$

with the optical depth τ_ν defined as an integral over the solid angle Ω and frequency ν (or wavelength $\lambda = c/\nu$),

$$\tau_\nu(r) = \int_{R_\star}^{r} \kappa_\nu \rho\, dr = \tau_{\nu,0}\,(r_{beg}) + \int_{r_{beg}}^{r} \kappa_\nu \rho\, dr. \tag{23}$$

The optical depth is a function of the frequency-dependent absorption opacity κ_ν and the density ρ. Following Melon Fuksman and Klahr (2022), we use tabulated values for opacities provided by the dust from Krieger and Wolf (2020, 2022). We plot κ_ν, κ_P, and κ_R, along with the blackbody function $B_\lambda(T_\star) = c/\lambda^2 B_\nu(T_\star)$ for $T_\star = 4000\,K$ in Fig. 12. We do not model the hot inner rim of the disk that otherwise casts a shadow, and instead compute an effective optical depth $\tau_{\nu,0}$ from the central star out to the edge r_{beg} of the disk region of particular interest for our problem, given by

$$\tau_{\nu,0}(r_{beg}) = \kappa_\nu \rho_{r_{beg}} (r_{beg} - R_\star), \tag{24}$$

following a similar prescription from Flock et al. (2013). A more detailed discussion on this initial optical depth profile and how it aligns with our current understanding of the inner disk structure can be found in Sect. 4.7.6. We then calculate the irradiation flux at each cell center by ray-tracing for different frequency bins.

4.3.3 *Model parameters*

For our reference run `MFID` we model a T Tauri central star, similar to previous models in the literature (e.g., Melon Fuksman and Klahr, 2022; Ueda et al., 2021; Watanabe and Lin, 2008), with a mass $M_\star = 1\,M_\odot$, radius $R_\star = 2.6\,R_\odot$, effective temperature $T_\star = 4000\,K$, and luminosity $L_\star = 1.56\,L_\odot$. We implement a surface density power law of $\Sigma(R) = \Sigma_0\,(R/1\,au)^{-1}$, with R being the cylindrical radius. We use $\Sigma_0 = 200\,g\,cm^{-2}$ for the `MFID` model, consistent with dust-mass based measurements at 1 au (van Boekel et al., 2017). To explore the effects of higher disk masses, we also consider a model with $\Sigma_0 = 2000\,g\,cm^{-2}$ (`MSIG2000`), which is more representative of classical minimum-mass solar nebula models (MMSN, Hayashi, 1981;

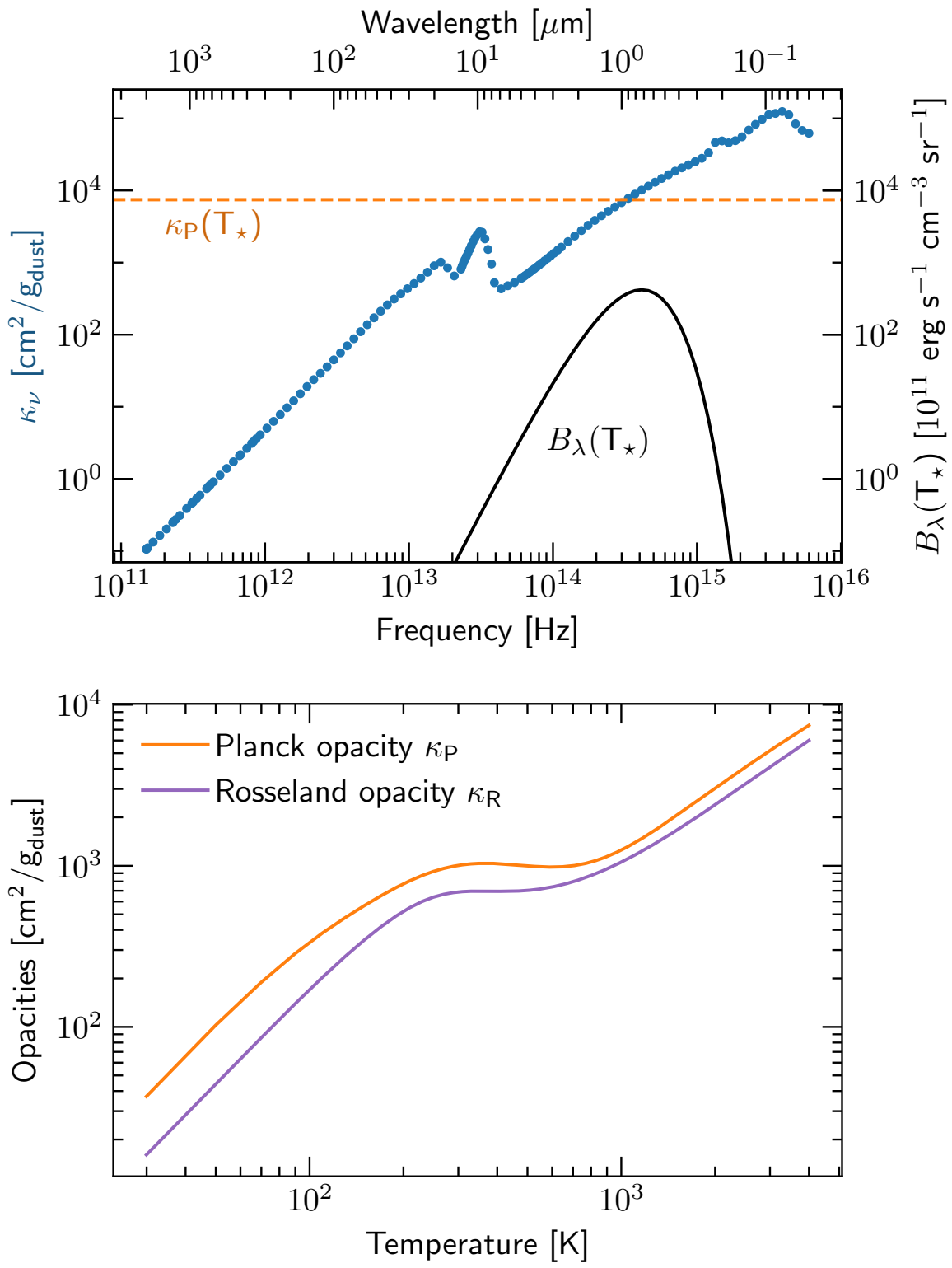


Figure 12: Top panel: Frequency-dependent dust absorption opacities (per gram of dust, g_{dust}) from tabulated values (Krieger and Wolf, 2020, 2022). The blue points denote 132 frequencies logarithmically sampled in the range $\nu \in [1.5 \times 10^{11}, 6 \times 10^{15}]$ Hz for the computation of irradiation flux in our models. The black line indicates the blackbody function, and the orange line shows the Planck opacity for $T_\star = 4000$ K (figure similar to Kuiper et al., 2010). Bottom panel: Temperature-dependent Planck and Rosseland mean opacities for the radiation transfer. The local maximum (≈ 250 K) and the subsequent dip in the mean opacities is due to the 10 µm silicate feature.

Table 1: Parameters for the different hydrostatic and dynamic runs. Each model is denoted by `M`, followed by a label indicating the parameter we altered relative to the fiducial run `MFID`. `ML1` is an additional model we set up that matches Melon Fuksman and Klahr (2022) with stellar parameters corresponding to $L_\star = 1\,L_\odot$. We only discuss this during the comparison in Sect. 6.2.

Model	ϵ_{dg}	$L_\star\,(L_\odot)$	$T_\star\,(K)$	$R_\star\,(R_\odot)$	$\Sigma_0\,(g/cm^2)$
`MFID`	0.01	1.56	4000	2.6	200
`MD0.001`	0.001	1.56	4000	2.6	200
`MSIG2000`	0.01	1.56	4000	2.6	2000
`ML1*`	0.01	1	4000	2.086	200

Weidenschilling, 1977). While dust mass estimates support the lower surface densities, we explore both models as gas mass estimates still remain poorly constrained. The higher surface density profile matches the one computed using a viscous disk model with a mass accretion rate $\dot{M} = 10^{-8}\,M_\odot\,yr^{-1}$ and viscosity parameter $\alpha = 10^{-3}$ following Shakura and Sunyaev (1973). This is a commonly used initial condition in disk theoretical studies (including the thermal wave instability, e.g., Flock et al., 2019; Steiner et al., 2021; Watanabe and Lin, 2008). However, our models ignore viscous heating (including it would yield a lower Σ in the inner regions, see Fig. 3 of Ziampras et al., 2020), which results in a disk that is marginally more massive in the innermost regions ($\lesssim 5$ au).

The initial guess for the temperature is determined by the optically thin solution for a passively irradiated disk (Chiang and Goldreich, 1997). We also sample dust-to-gas mass ratios $\epsilon_{dg} = \{0.01, 0.001\}$ (models `MFID` and `MD0.001`, respectively). A table with the different setups corresponding to the model parameters is given in Table 1. This excludes the hydrostatic dust models for which the parameters are described in detail in Sect. 4.6. Our models are inviscid and do not account for the effect of accretion heating.

4.3.4 *Numerics*

We solved the radiation hydrodynamic equations outlined in Eqs. (40) and (21) using the astrophysical finite–volume fluid code `PLUTO` (Mignone et al., 2007). We use the flux–limited diffusion module outlined in Flock et al. (2013) which uses a Jacobi preconditioned BiCGSTAB solver for the radiation matrix. We perform 2.5D axisymmetric simulations of gas assuming a constant dust-to-gas mass ratio ϵ_{dg} in spherical coordinates (r, θ) with a logarithmic grid in $r \in [0.41, 95]$ au and a uniform grid in $\theta \in [\pi/2 - 0.32, \pi/2 + 0.32]$. We use a grid of $N_r \times N_\theta = 1024 \times 128$ cells, which corresponds to a resolution of ≈ 6 cps (cells per scale height) in r and ≈ 7 cps in θ. We use a second-order Runge Kutta (RK2) time integrator, a linear reconstruction scheme, the HLLc Riemann solver (Toro et al., 1994) and a CFL (Courant–Friedrichs–Lewy) value of 0.4.

We implement strict outflow conditions for the radial velocity u_r at the radial inner boundary ($R < 0.5$ au), and use power-law extrapolation for the azimuthal velocity u_ϕ. The rest of the quantities are set to zero gradient. We also damp the density and velocities at the inner boundary using the de Val-Borro et al. (2006) prescription over a damping timescale of 0.01

inner orbits at R_0. For the radiation boundaries, we restrict the radiation energy density at the polar boundaries to a small value $E_{rad} = a_R T_0^4$, where $T_0 = 370\,\epsilon_{dg}\,(R/au)^{-0.5}$ K which is about 7% of the optically thin temperature profile for the model MFID (Chiang and Goldreich, 1997). Following the prescription in Flock et al. (2016, see their Eq. 19), we implement a modified gravitational potential in the inner radial boundary to improve numerical stability in the dynamical simulations. In very optical thin regions of the disk we modify the Rosseland mean opacity κ_R so that $\kappa_\nu \rho_{r_{beg}} \Delta r > 10^{-4}$ is fulfilled. This helps the convergence of the matrix solver without affecting the temperature significantly. Furthermore, we ensure that the radially innermost cell remains optically thin, preventing unphysical heating by the irradiation flux (Flock et al., 2013).

4.3.5 *Iterative solver for hydrostatic models*

Using the iterative method described in Flock et al. (2013) to compute solutions for irradiated disks in radiative hydrostatic equilibrium, we first run hydrostatic models with the parameters described above. Given an initial density profile $\rho(r,\theta)$, this solver iterates between each FLD step given by Eq. (21) over a typical radiative diffusion timescale of $\sim 10^6$ s (for typical solar parameters, see Eq. (67)). The extended runtimes with successive diffusion steps reaching up to 10^8 s in later iterations ensure thermal equilibrium, with temperatures reaching a converged state. Once the radiation solver converges, new densities (and azimuthal velocities) are computed by integrating the hydrostatic equations

$$\frac{\partial P}{\partial r} = -\rho \frac{\partial \Phi_\star}{\partial r} + \frac{\rho u_\Phi^2}{r}, \tag{25a}$$

$$\frac{1}{r}\frac{\partial P}{\partial \theta} = \cot\theta \, \frac{\rho u_\Phi^2}{r}, \tag{25b}$$

using a second-order Runge–Kutta integrator. The iterative process is continued until a converged density state is reached. The computed profiles are then used as initial conditions for the hydrodynamic simulations. While disk hydrostatics are more correctly treated in cylindrical coordinates (see, e.g., codes like cuDisc, Robinson et al., 2024), our approach works well and has been robustly tested and used for various problems (for validation tests, see Appendix B of Flock et al., 2013).

Previous models of the thermal wave instability suggest that unstable shadows form and propagate inward in such a hydrostatic iterative treatment (Okuzumi et al., 2022; Ueda et al., 2021). Our models further include vertical and radial radiative transfer, relaxing the commonly used isothermal approximation. We start by describing these hydrostatic models, followed by the hydrodynamic simulations in the following sections.

4.4 RESULTS I: FIDUCIAL MODEL mfid

We first outline results from our fiducial model MFID, for which the model parameters are listed in Table 1. This corresponds to a high dust-to-gas ratio case of $\epsilon_{dg} = 0.01$. Such a value represents the early times of disk evolution, where most of the dust material is still in the form of small ($\leqslant 10\,\mu$m) grains. For reference, a total dust-to-gas mass ratio $\epsilon_{tot} = 0.01$ for all

sizes and MRN dust distribution (Mathis et al., 1977), and a dust-to-gas ratio in small grains ϵ_{dg} of 0.01, 0.001 and 0.0001 would yield maximum grain sizes of a_{max} of 0.25 µm, 19 µm and 1.8 mm, respectively (see also Melon Fuksman et al., 2024). At later stages, dust grains grow, settle to the midplane, and small grains become depleted in the disk (Birnstiel et al., 2010; Testi et al., 2014). While $\epsilon_{dg} = 0.01$ may be on the higher end for a T Tauri disk, we adopt this value as our fiducial case in order to optimize the conditions under which staircase structures are most likely to develop.

4.4.1 *Hydrostatic model*

We see the emergence of alternating bright and dark regions in the disk with thermal waves forming and propagating inward, consistent with the expectations from hydrostatic models in the literature (Pavlyuchenkov et al., 2022a; Ueda et al., 2021; Watanabe and Lin, 2008). This is clearly depicted by the midplane temperature profiles plotted for different iteration times in Fig. 13 and also the 2D snapshot of temperature and irradiation heating rate $-\nabla \cdot F_\star$ at the end of the hydrostatic run in Fig. 14.

From Fig. 13, we see that as the disk tries to find a hydrostatic solution, a thermal wave forms and travels inward, and two form after $i = 15$ iterations that also propagate inward. The perturbations in temperature deviate from the background power law by up to 30%.

We plot the 2D snapshots at the last iteration ($i = 30$) in Fig. 14. To get a complete picture of where in the disk radiation is emitted and absorbed, we plot different optical depth surfaces, that is, $\tau_\nu = 1$ profiles corresponding to the starlight at different frequencies τ_ν and the emission of the disk itself τ_{disk}, defined by

$$\tau_\nu = \int_{R_\star}^{r} \kappa_\nu \rho dr, \quad \tau_{disk} = \int_{\theta=\pi/2}^{\theta} \kappa_\nu \rho r d\theta. \tag{26}$$

These surfaces exhibit a clear staircase pattern in contrast to the typical smooth flared-disk power-law solution. Nevertheless, it is important to investigate the existence of such a pattern in hydrodynamic simulations to assess their stability. This is discussed in the following section. The red line in Fig. 14 indicates the optical depth to the disk's own emission $\tau_{disk} = 1$, showing that the disk is optically thick up to $R \sim 50$ au.

4.4.2 *Hydrodynamics*

With initial conditions from the previously described hydrostatic model, we introduce dynamics, allowing the system to evolve over time. Plotted in Fig. 15 (top panel) are the profiles of the temperature and the irradiation source term at $t = 500$ inner orbits or 131.3 yrs. We do not find inwardly propagating waves. Instead, among the bumps in temperature that we started out with in the hydrostatic case, the innermost bump grows in amplitude, and as a result of this bump, the region behind it is shadowed, giving rise to the second one. After about 400 inner orbits, we see that this staircase structure reaches a quasi-steady state that remains stable for several hundreds of orbits. This is clearer in the top panel of Fig. 16, where we plot radial profiles of the midplane temperature normalized to the background power law for different times.

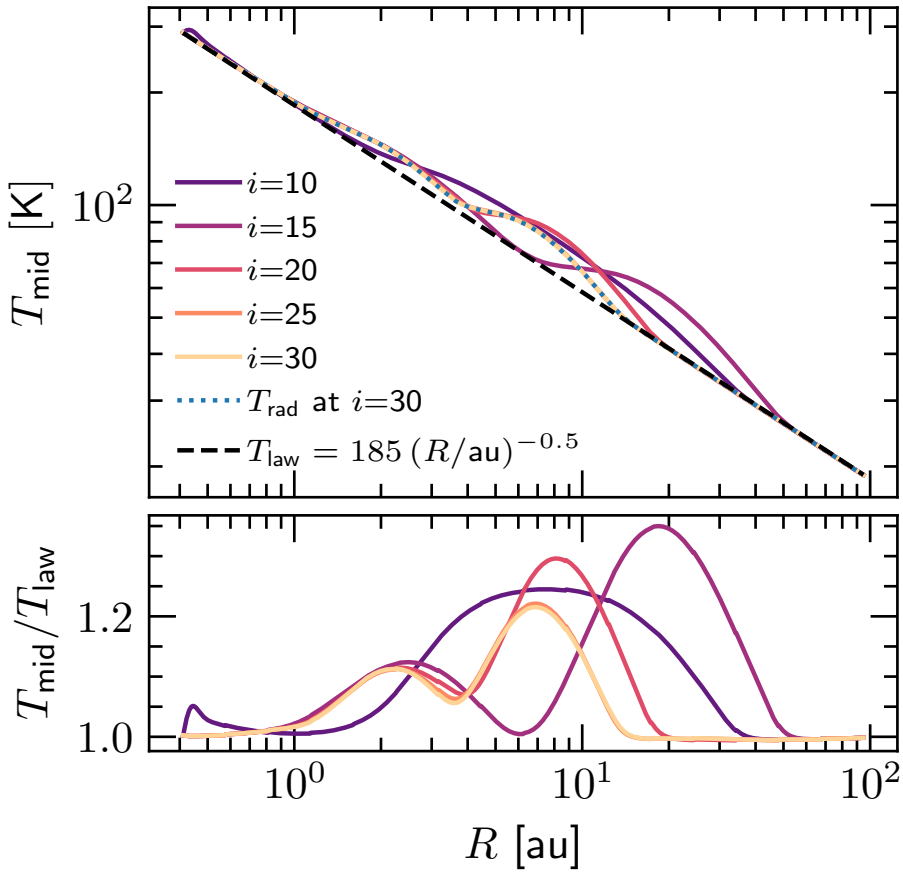


Figure 13: Top: Time evolution of the temperature profile at the disk midplane for iterations $\mathtt{i} = [10, 15, 20, 25, 30]$. The profiles indicate that thermal waves form and travel inward with time. There are two bump and shadow pairs at the end of the hydrostatic run for model `MFID`. The dotted blue line is the radiation temperature at $\mathtt{i} = 30$, and the dashed black line indicates the background power-law fit for the midplane $\mathtt{T}_{\mathrm{law}} = 185\,(\mathtt{R}/\mathrm{au})^{-0.5}$. Bottom: Deviation of the temperature profile from the background power law.

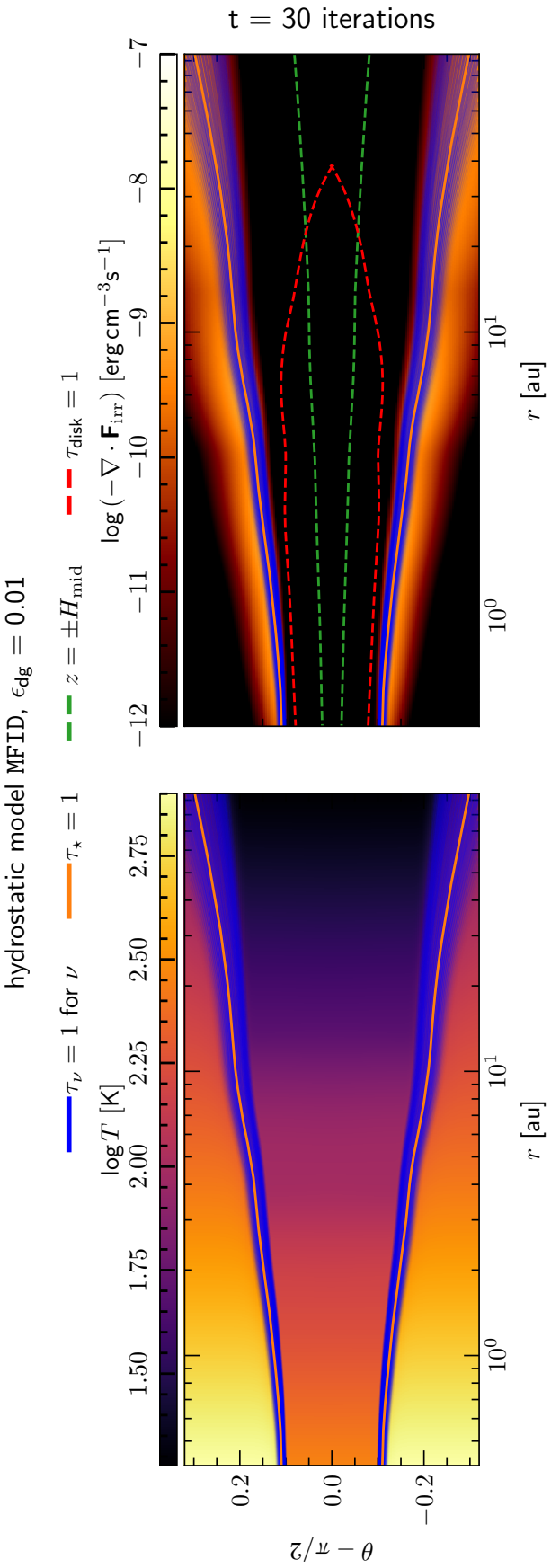


Figure 14: 2D profiles of temperature and the irradiation heating term for the hydrostatic model at i = 30 iterations. The orange line refers to the $\tau_\star = 1$ profile for the stellar radiation using the Planck mean opacity. Blue lines refer to $\tau_\nu = 1$ profiles for for different frequencies ν (same as in Fig. 12), with the line opacity weighted by B_ν at that frequency. (see also Fig. 7 in Melon Fuksman and Klahr, 2022). The red line indicates the $\tau_{disk} = 1$ profile for the disk's radiation using $\kappa_P(T)$. The green lines indicate the midplane gas scale height of the disk.

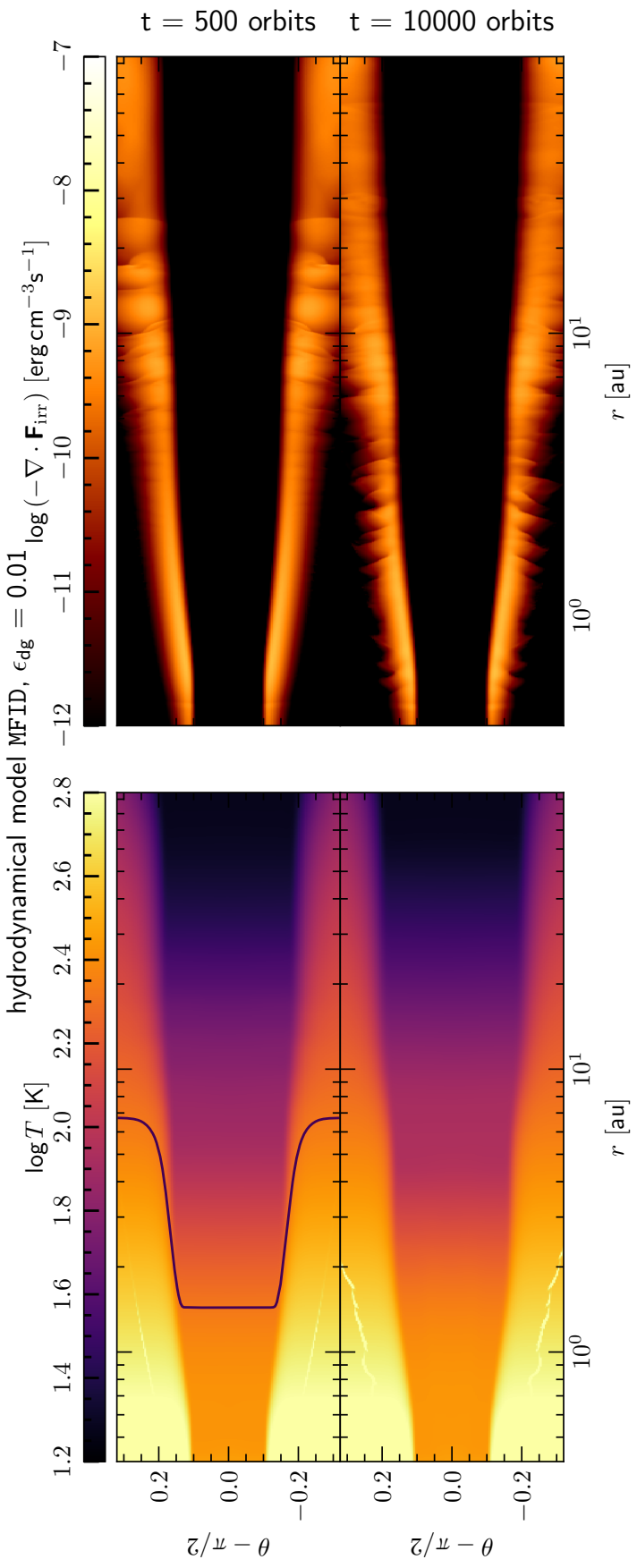


Figure 15: 2D profiles of temperature and the irradiation heating term for the hydrodynamic model at t = 500 orbits (top) and t = 10000 orbits (bottom). The solid black line denotes the temperature corresponding to the opacity transition in Fig. 12.

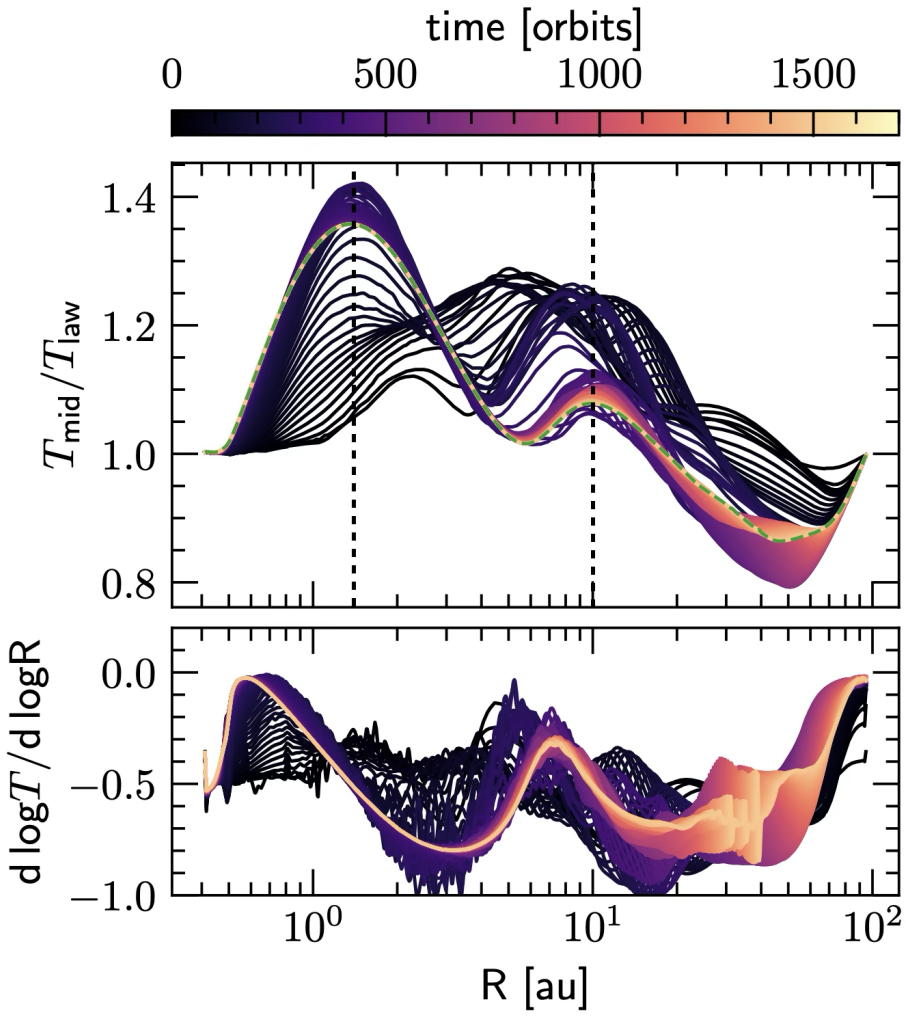


Figure 16: Top: Time evolution of the temperature profile at the disk midplane against the background power law. The dashed lines (the first corresponds to the opacity transition) indicate the final locations of the temperature bumps. Bottom: Time evolution of the radial temperature gradient indicating a less steep profile at the shadow locations and no sign flip.

4.4.2.1 *The inner bump*

The innermost bump has slightly grown, and shifted inward compared to its radial location in the static models. There are two factors to consider while interpreting this first bump as a consequence of the thermal wave instability. First, its location between 1–2 au, which corresponds to a temperature of ≈ 250 K, coincides with the 10 µm silicate transition[1] in the temperature-dependent opacities (see the bottom panel of Fig. 12). This is further indicated by a solid black line in Fig. 15, and the first dashed line in the time evolution plot in Fig. 16. Regions in the disk with opacity transitions can locally change the heating rate, and a strong enough transition could also flatten the pressure gradient causing bumps (Garaud and Lin,

1 This is more of a local change in the power-law dependence of the opacity on temperature due to variations in the optical constants near the silicate resonance rather than a jump due to a change in dust composition (e.g., near a snowline).

2007). While this may not be enough for the formation of this inner bump, it may have an influence. Second, the inner disk is modeled with an effective optical depth profile $\tau_{\nu,0}$, which can affect the first bump through a nonuniform heating rate. In our simulations, we do find a slight enhancement in the irradiation power inward of 1 au (see Fig. 45) with the prescription of $\tau_{\nu,0}$. A more detailed discussion on this is included in Appendix A.2.

4.4.2.2 *Secondary structure around a bump*

Regardless of the origin of the first bump, we would like to highlight that the key consequence of this inner bump is that it casts a shadow behind, driving a second one. This second bump is a natural consequence of the first, driven by starlight-driven shadowing. This behavior is unlike that expected by the "thermal wave instability" discussed in Wu and Lithwick (2021) and Ueda et al. (2021), where the staircase pattern propagates inward. Instead, the disk seems to relax to a quasi-steady state characterized by bumps and shadows formed sequentially from the inside out. Forming secondary structure around existing bumps creates interesting implications for disk studies focused on pressure traps (see Sect. 4.7.7 for a discussion on snowlines).

We plot the radial temperature gradient in the bottom panel of Fig. 16. The bumps and shadows have a clear imprint with modulations of the radial temperature profile. However, neither the first bump and shadow pair nor the subsequent ones in the `MFID` model produce local extrema ($^{dT_{mid}}/_{dR} = 0$). The profile remains radially monotonic, albeit with a varying slope.

4.4.3 *Long-term evolution of the staircase structure*

We run the fiducial model `MFID` for a long time, the second panel of Fig. 15 shows the 2D profiles after $t = 10000$ inner orbits or 2625.3 yrs. We see that the two bump and shadow pairs remain in the disk after several thousands of years. This brings up the interesting possibility that they could correspond to long-lived substructure in the disk. Both panels showing temperature in Fig. 15 show hot regions in the upper layers of the inner disk, owing to numerical artifacts that arise due to a combination of strong shocks at very low-density regions and some compressional heating due to the modified gravitational potential (seen also in other works like Cecil and Flock, 2024; Melon Fuksman and Klahr, 2022).

Our results clearly show that starlight-driven staircases are strongly affected by the hydrodynamics in the disk and also show that the predictions from linear theory and hydrostatic models do not fully come true. Our results for the $\epsilon_{dg} = 0.01$ `MFID` model are different from such predictions where inwardly propagating thermal waves were seen (Pavlyuchenkov et al., 2022a; Ueda et al., 2021; Wu and Lithwick, 2021). This is also different from the dynamical models of Melon Fuksman and Klahr (2022), who found that the thermal waves completely damp away, although our setups have a different treatment of the inner disk profile. We show that in the dynamical case, while thermal waves do not propagate inward, they grow and reach a quasi-steady state. While Kutra et al. (2024) hint toward a similar theory of staircases settling in the disk, there are clear differences in the physics we model. We make comparisons to previous works in detail in Sect. 5.7.

4.5 RESULTS II: CHANGING THE DUST CONTENT

We explore the two models `MD0.001` and `MSIG2000` with lower dust-to-gas ratios of $\epsilon_{dg} = 0.001$ and a higher surface density $\Sigma_0 = 2000$ g/cm^2 respectively. Both of these vary the dust content in the disk, which controls the efficiency of radiation transport and hence affects the thermal structure of the disk. The growth rate of the instability, predicted by Ueda et al. (2021), Wu and Lithwick (2021) depends on the height at which the irradiation is absorbed on the surface of the disk. Disks with a higher dust opacity, be it due to a higher ϵ_{dg} or gas mass, achieve this. Previous work also verified that for lower dust content, the TWI was suppressed (Melon Fuksman and Klahr, 2022; Ueda et al., 2021).

4.5.1 *Low dust-to-gas ratio model `MD0.001`*

In this section we outline our results from our hydrostatic and dynamical models for the low dust-to-gas ratio case of $\epsilon_{dg} = 0.001$. As outlined above, these values are more typical to the dust content observed in later stages (i.e., Class II protoplanetary disks Birnstiel, 2024; Rilinger and Espaillat, 2021). Figure 17 shows the temperature and the irradiation source term after 30 iterations for the hydrostatic model (top) and the hydrodynamic model (bottom) after $t = 500$ orbits. We do not see any temperature bumps or shadows forming in the hydrostatic model, but recover an inner bump in the dynamic model. While studies in the literature point toward the TWI being suppressed for very low dust-to-gas mass ratios $\epsilon_{dg} \lessapprox 10^{-4}$ (Ueda et al., 2021), we do not recover any secondary shadowing already for the $\epsilon_{dg} = 0.001$ model. The combined effect of the reduced dust content and a lower radiation boundary (see T_0 in Sect. 4.3.4) could enhance the cooling efficiency of the disk in the outer parts, affecting the formation of staircases. An outline of the cooling timescales in the disk can be found in Appendix A.1. We speculate that the smaller amplitude of the shadowing with the FLD method might also affect this limit. We discuss this further in Sect. 4.7.5.

4.5.2 *High surface density model `MSIG2000`*

We now present the results for the model with a surface density of $\Sigma_0 = 2000\,\mathrm{g/cm^2}$ and a dust-to-gas ratio of $\epsilon_{dg} = 0.01$, motivated in Sect. 4.3.3. A higher surface density not only implies a higher dust content in the disk, with 10 times the amount of dust at a certain radius compared to the fiducial model but also a longer cooling timescale (in the hydrodynamic case the diffusion-regulated cooling timescale scales with $\Sigma^2\epsilon_{dg}$; see also Eq. (67)). Such a disk would be overall more optically thick than the `MFID` model, which would have an effect on the temperature structure.

Figure 18 shows the temperature and the irradiation heating term for the hydrostatic and dynamical `MSIG2000` models. We see three temperature bumps at the end of the 30 hydrostatic iterations, all within 10 au of the disk. The time evolution of the staircase structure is clearer in Fig. 19, showing how the system readjusts from this static initial condition with three TWI bumps. In the hydrodynamic model, the inner bump continues to grow, altering the structure downstream: the staircases shift inward, and the first bump eventually grows sufficiently to merge with the second. The thermal response time for the disk is 100 times slower than the `MFID` model, and after about 7000 inner orbits, we see a final state with two bumps (see also the bottom panel of Fig. 18, where the final state is plotted at 12000 inner orbits). On

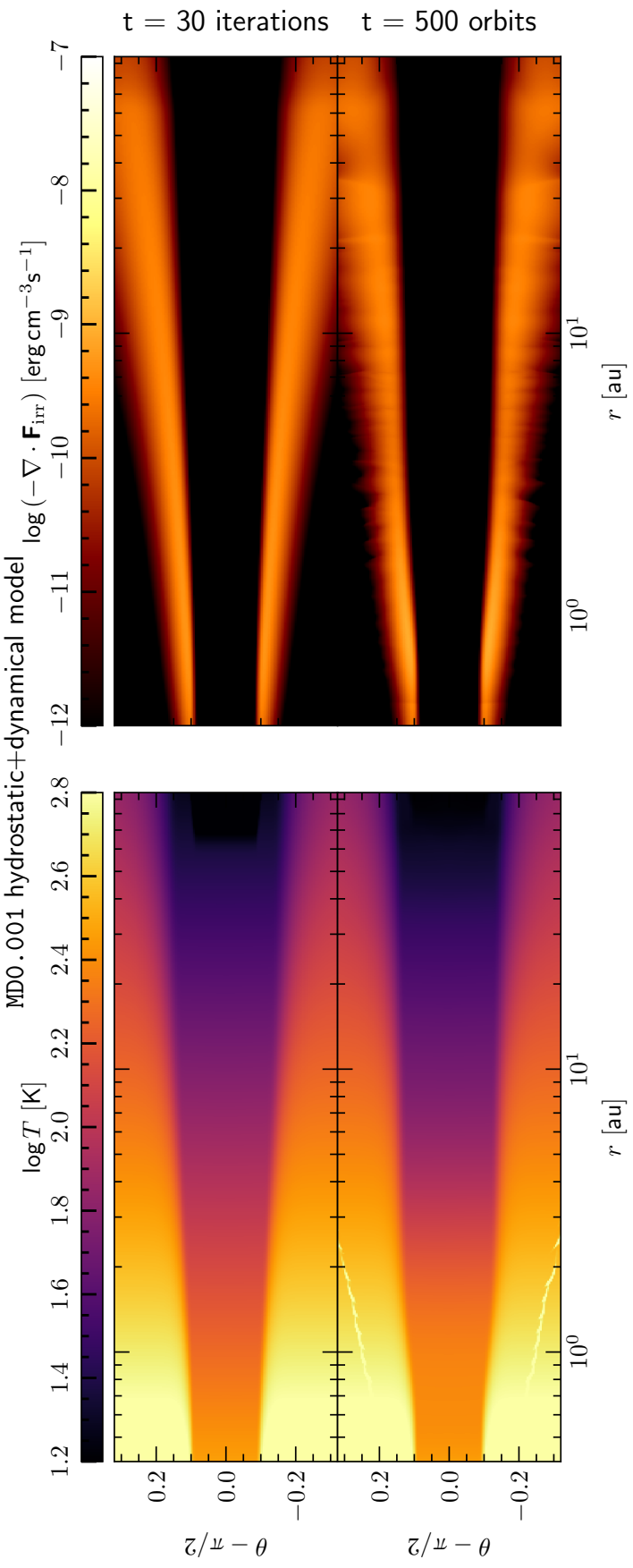


Figure 17: 2D profiles of temperature and the irradiation source term for the `MD0.001` hydrostatic+dynamic model with $\epsilon_{dg} = 0.001$. Top panel: Hydrostatic model after t = 30 iterations does not show any staircase structure. Bottom panel: Hydrodynamic model after t = 500 orbits shows the innermost bump.

more careful observation, we can also spot a smaller bump at $\sim 20\,\mathrm{au}$ in the disk, with a relative temperature difference of 10%, less noticeable in the 2D profiles (Fig. 18) compared to the more prominent ones (see Sec. 4.7.5 for a discussion on the amplitude of the bumps in our models). The profiles of the radial temperature gradient in the bottom panel of Fig. 19 show that the innermost bump in the hydrodynamic model causes a temperature gradient inversion i.e., a local maximum in the profile, something that was absent in the fiducial `MFID` model. The bumps downstream to it are more comparable, with the shadowing causing deviations in the slope of the temperature profile.

Overall, it is clear that the dust content in the disk has a significant effect on the staircase structure: a higher dust content correlates not only to more bumps that are more pronounced, but also show signs of inward motion. This hints at more favorable conditions for the staircase structure in opaque disks, further enhanced by a driving inner bump.

Interestingly, our `MSIG2000` model shows inwardly moving waves similar to the predictions of the TWI, until it reaches a steady state after long-term evolution. This behavior could imply that optically thick disks provide more favorable conditions for the TWI itself, but identifying these conditions is beyond the scope of this work.

4.6 RESULTS III: ADDING DUST SETTLING

With the dust content controlling radiation transport in the bulk of the disk via its opacity, modeling dust consistently is crucial to correctly capturing the shape of the thermal surface of the disk, as the latter determines the irradiation absorption surface. Dust settles at different heights and interacts with the gas through drag forces. In almost all previous models studying the thermal wave instability (including the results in our previous sections), there is an inherent assumption that the dust is perfectly coupled to the gas, and that the dust content scales with the gas mass through a constant dust-to-gas ratio. We present the first physically motivated hydrostatic model with dust by introducing a dust scale height within the hydrostatic radiation solver in `PLUTO` (see Sect. 4.3.5). In the Epstein regime, the dust–gas drag force depends on the Stokes number $\mathrm{St} = t_{\mathrm{stop}}\Omega_K$, which describes how well dust is coupled to the gas. Given a dust grain size of a and a dust density ρ_d, the stopping time t_{stop} is defined as

$$t_{\mathrm{stop}} = \frac{\rho_d a}{\rho v_{\mathrm{th}}}, \tag{27}$$

where $v_{\mathrm{th}} = \sqrt{8/\pi}c_s$ is the mean thermal velocity (Birnstiel, 2024). The vertical profile for the dust can be determined by the settling–diffusion equilibrium. A simple model usually imposes a Gaussian profile for the dust similar to the gas, although now with the dispersion being the dust scale height H_d (e.g., observations of IM Lup in Pinte et al., 2008). A more general approach involves modeling dust turbulence as a diffusive process (Dubrulle et al., 1995; Dullemond and Dominik, 2004b; Mulders and Dominik, 2012). A steady state solution for vertical settling and turbulent diffusion can be written in terms of the diffusion coefficient D which quantifies diffusivity. For the simplest case, we utilize the solution assuming $D = \alpha_d c_s H$, where α_d is a dimensionless diffusion parameter. The dust density structure is then given by Fromang and Nelson (2009)

$$\rho_d = \rho_{d,\,\mathrm{mid}}\ \exp\left[-\frac{\mathrm{St}_{\mathrm{mid}}}{\alpha_d}\left(\exp\left(\frac{z^2}{2H^2}\right) - 1\right) - \frac{z^2}{2H^2}\right]. \tag{28}$$

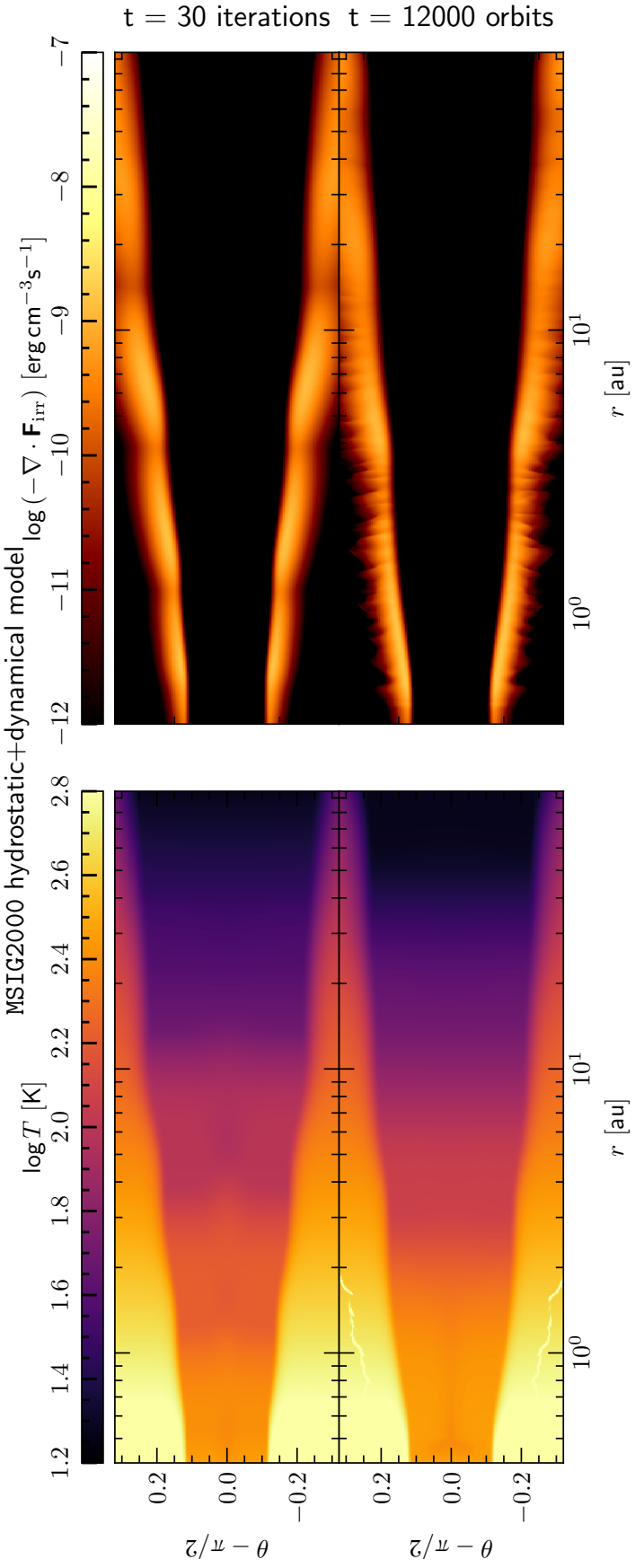


Figure 18: 2D profiles of temperature and the irradiation source term for the high surface density `MSIG2000` hydrostatic+dynamic model. The hydrostatic run shows three prominent bumps (top), hydrodynamic run shows two bumps (bottom). Midplane temperature reached equilibrium after several radiative diffusion timescales, and hence the longer run time of t ≈ 12000 orbits.

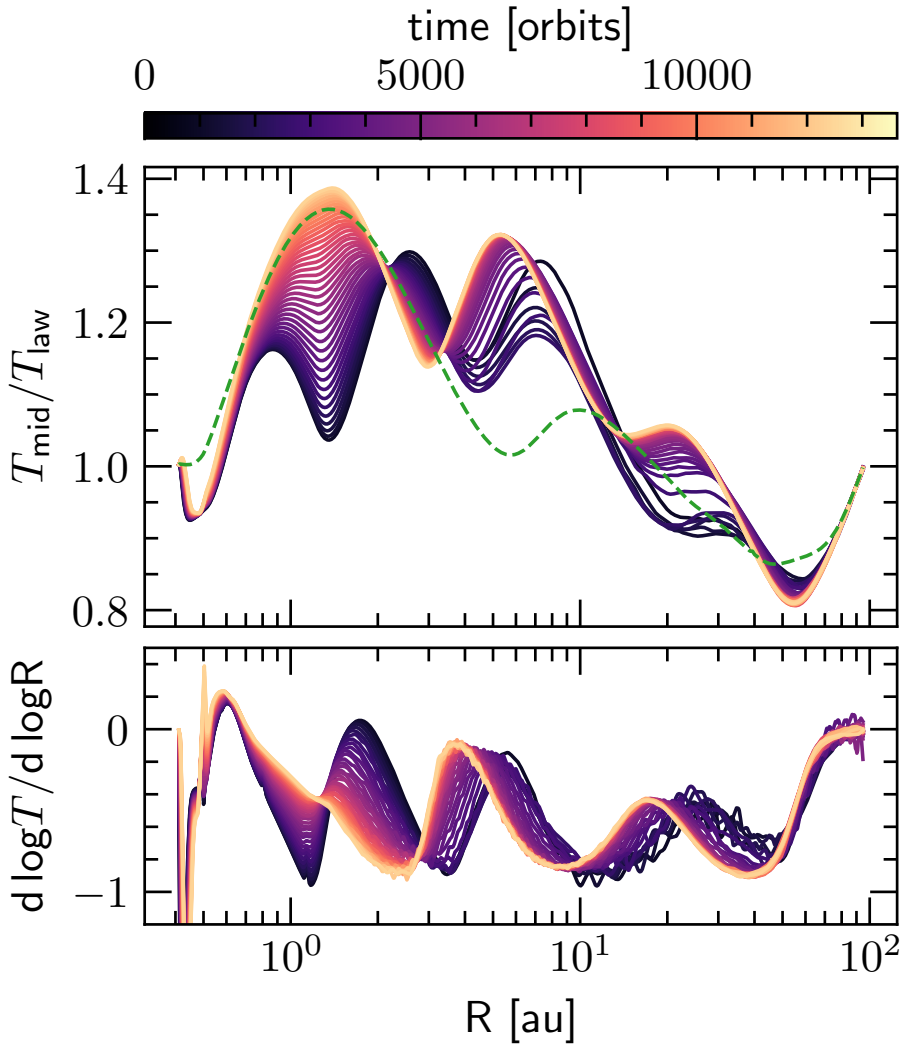


Figure 19: Top: Long-term time evolution of the midplane temperature deviation from the background power law for the dynamical model `MSIG2000`. After about $t = 700$ orbits, we see that the first and the second bump merge together. We can also spot an additional bump at ≈ 20 au. The dashed green line corresponds to the final state of the `MFID` model in Fig. 16, for comparison. Bottom: The evolution of the radial temperature gradient in time similar to Fig. 16. Here, we see that the innermost bump is a local maximum, producing a temperature gradient inversion, while the other two shadows produce slope changes in the profile.

We compute the dust density at the midplane from this formula at the end of every hydrostatic iteration once the new temperatures and gas densities are determined. In our models, we consider a fixed grain size of $a = 10\,\mu m$ and dust density $\rho_d = 3\,g\,cm^{-3}$.

Figure 20 shows 2D profiles of temperature and the irradiation source term for the dust models, for different diffusion parameters $\alpha_D \in [0.0001, 0.001, 0.01]$. We see a superheated dust layer and a very sharp irradiation surface for all cases, with dust settling more for a lower α_D (Zsom et al., 2011). For a given grain size, the dust–gas coupling is stronger where the gas density is higher. While the dust and gas are well coupled in the midplane, there is a sharp transition layer at the height where the dust is no longer well-coupled to the gas, and the dust density drops rapidly. Since the dust layer dictates the irradiation surface, this results in sharp layers where the majority of the irradiation from the star is absorbed (Chiang and Goldreich, 1997). The height at which the radiation is absorbed is thus substantially closer to the midplane, an effect usually neglected in previous work on this topic. This affects the flaring angle of the disk, and hence the temperature structure. We see that staircases are not formed in any of these hydrostatic models with the dust scale height included.

While these models are already a first step to evolve dust in the disk, this iterative static settling–diffusion approach breaks down at a few gas scale heights. Our radiation transfer method accounts for small grain opacities through the precomputed opacity tables based on a grain size distribution, but the assumption of a constant 10 μm mass-carrying component, which would experience moderate settling is a simplification (see Sect. 4.6 for a more detailed discussion on caveats of the existing dust models). To fully consistently model the transition of coupled-to-decoupled dust grains, we would have to model dust–gas interaction with drag and turbulent diffusion, along with the opacity feedback on radiation transport. We note that including this could affect the preliminary results with dust discussed in this section. Our future work will focus on this aspect in more detail, modeling dust as a pressureless fluid with the PLUTO code (outlined in Ziampras et al., 2025b).

4.7 DISCUSSION

In this section, we discuss some additional aspects of our models, such as the physical implications of the shadows and their observability, the effect of the initial optical depth profile, the dust scale height, and the cooling timescales, which might be important for the long-term evolution of shadows in the disk. We will also draw qualitative comparisons to previous work on the thermal wave instability, the linear theory and hydrostatic models of Wu and Lithwick (2021), Ueda et al. (2021), and the hydrodynamic models of Pavlyuchenkov et al. (2022b), Melon Fuksman and Klahr (2022), and Kutra et al. (2024). Finally, we highlight some caveats of our work and future directions.

4.7.1 *Physical implications of the bumps and shadows*

To understand if the staircase structure we see in our radiation hydrodynamic simulations can produce observable features, in addition to the relative midplane temperature profiles and the temperature gradients plotted in Figs. 16 and 19, we provide further analysis here. The first two panels of Fig. 21 show the radial profiles of the luminosity factor $\varphi = \sin\alpha$, where α is the flaring (grazing) angle and the height of the irradiation surface normalized to the local radius z_s/R where $z_s = z\ (\tau_\star = 1)$. The flaring angle is the angle at which

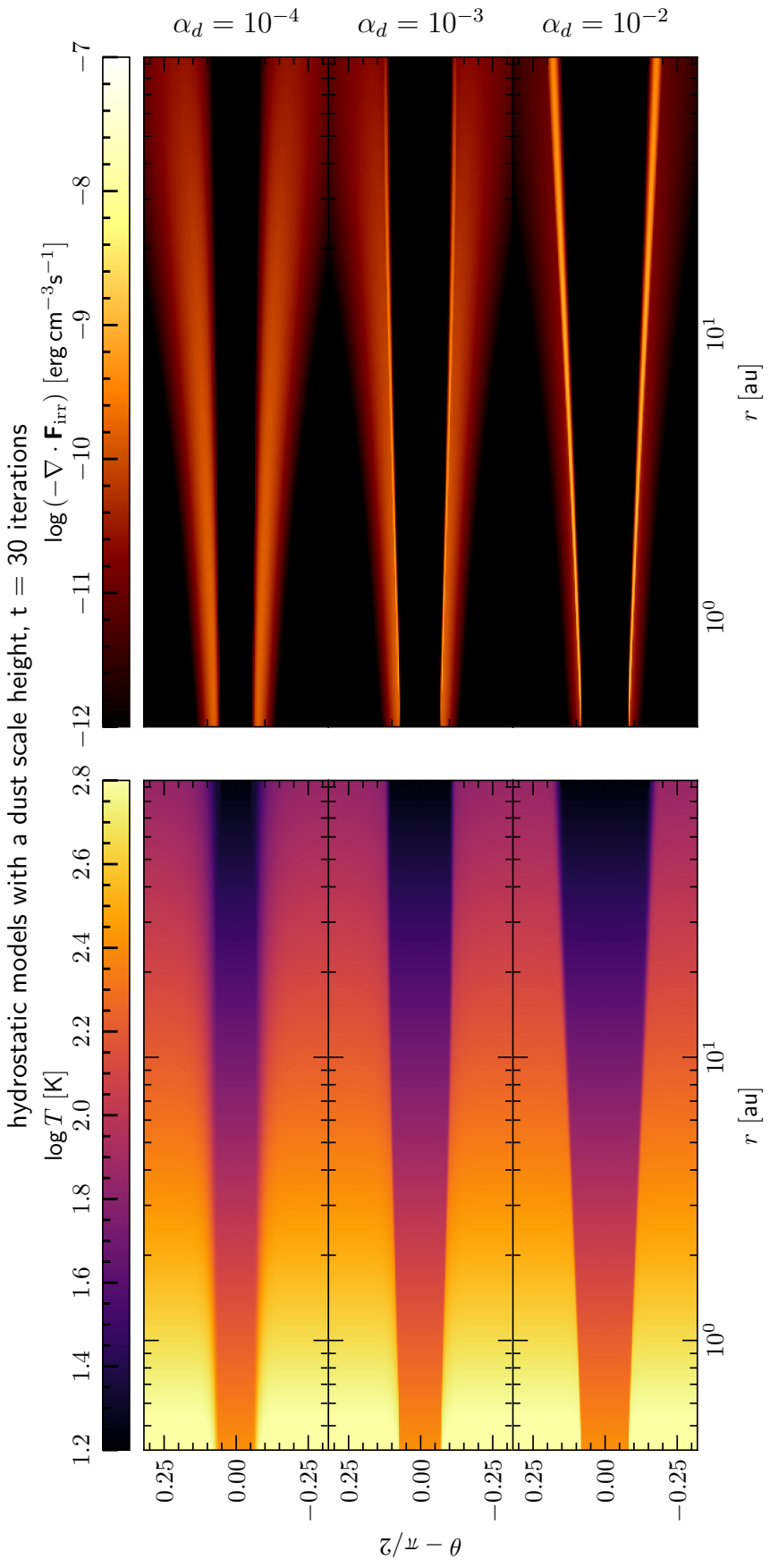


Figure 20: 2D profiles of temperature and the irradiation heating term for the hydrostatic model with a dust tracer at $i = 30$ iterations for diffusion coefficients $\alpha_D \in [0.0001, 0.001, 0.01]$. We see a very sharp irradiation surface but there are no staircases formed in any of these models.

stellar rays hit the surface of the disk defined similar to the classical approach of Chiang and Goldreich (1997) and Dullemond (2000), where $\alpha = \arctan(dz_s/dR) - \arctan(z_s/R)$ and the irradiation factor φ is the fraction of luminosity intercepted by the disk surface, essentially approximating Eq. (22) as

$$F_\star \approx \phi \frac{L_\star}{4\pi r^2}. \tag{29}$$

For small angles (i.e., razor-thin disks), $\varphi \simeq \alpha$. We use the above mentioned quantities to trace whether the bump and shadow pairs produced in our models can affect the optical or near-infrared (NIR) intensities producing observable signatures. We see that φ decreases downstream of the bumps, blocking stellar rays to efficiently reach the shadows. This is also reflected by the nonuniform profiles of z_s/R, deviating from a smooth power-law profile. This provides evidence that the optical/NIR intensities are indeed reduced in the shadowed regions, but to also illustrate this further we provide synthetic scattered-light images in the H Band (VLT-SPHERE) with RADMC-3D (Dullemond et al., 2012) in Sect. 6.1 showing that our models produce bright rings and subsequent dark shadows. We also plot the midplane pressure gradient for our models in the bottom panel of Fig. 21 to understand the implications of the shadows/bumps in our models in the mm continuum. We do not see the inversion of the pressure gradient (i.e., dust traps, Paardekooper and Mellema, 2004; Whipple, 1972) for any of our models, but we see deviations in the pressure gradient at the shadow locations. This implies that while the staircase structure due to the starlight-shadowing mechanism is not strong enough to trap dust, they could act as dust traffic jams that could slow down dust grain drift, and also act as seeds for secondary dust accumulation mechanisms (Jiang and Ormel, 2021). We highlight that the self-consistent modeling of dust–gas dynamics coupled with radiation transfer is required for more accurate constraints on observable signatures of shadows, which is why providing state-of-the-art synthetic observations both in NIR and the mm continuum is not the focus of this work.

4.7.2 *Vertical velocity profiles of the long-term evolved models*

As mentioned in Sect. 4.4.3, we ran the fiducial model `MFID` for several radiative diffusion timescales and saw long-lived staircases. Figure 22 shows the vertical velocity profile of the disk at $t = 10\,000$ orbits. In the outer disk ($\gtrsim 10$ au), we see characteristic vertical motions extending from the top and bottom surfaces to the midplane (the so called finger modes) of the vertical shear instability (VSI, Nelson et al., 2013; Pfeil and Klahr, 2019). We also see some fully developed (corrugation) modes extending throughout the whole disk. The VSI is a hydrodynamic instability that could occur in outer protoplanetary disks where the cooling timescale is short, and can generate turbulence levels that match the ones observed in disks (Dullemond et al., 2022; Villenave et al., 2025). Although typical studies of the VSI require a higher resolution (~ 16 cps) to capture saturated states, and even higher (~ 50 cps) for accurate growth rates (Flores-Rivera et al., 2020), we can still resolve the characteristic motions for grid sizes used in this study (Ziampras et al., 2023). A more detailed calculation of the cooling timescales in our model can be found in Appendix A.1. The VSI is expected to mix small grains in the disk efficiently (Dullemond et al., 2022). A combination of a VSI-active outer disk and a vertically settled dust layer in the inner regions could further result in a sharp radial transition of the height of the dust layer, affecting or even promoting structure due

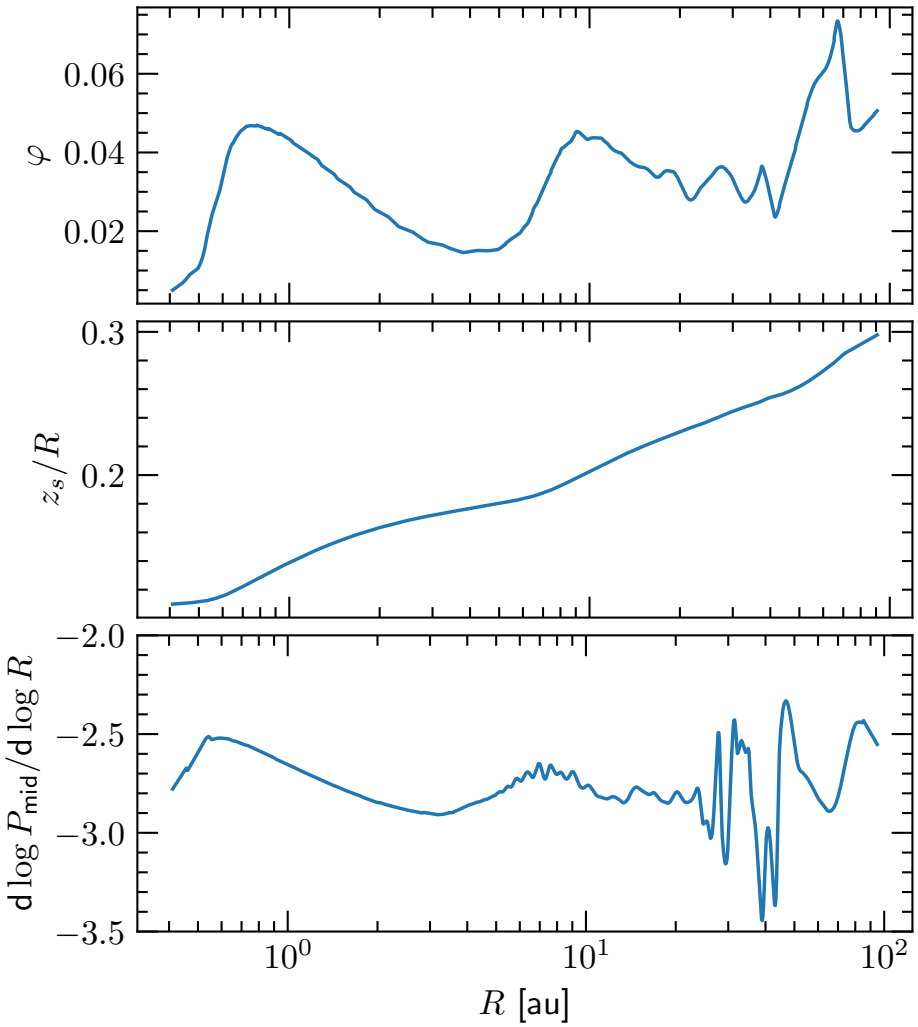


Figure 21: Top: Radial profile of the luminosity factor φ at the irradiation absorption surface showing a decreased incidence of stellar rays in the shadowed regions. Middle: Radial profiles of the surface aspect ratio h at the $\tau_\star = 1$ surface. We see that the profiles deviate from a smooth power-law. Bottom: Midplane radial pressure gradient shows deviations in the pressure profile. We note the flaring angle φ and the midplane pressure gradient d log P_{mid}/d log R are smoothed with a rolling average over 20 cells to eliminate noise.

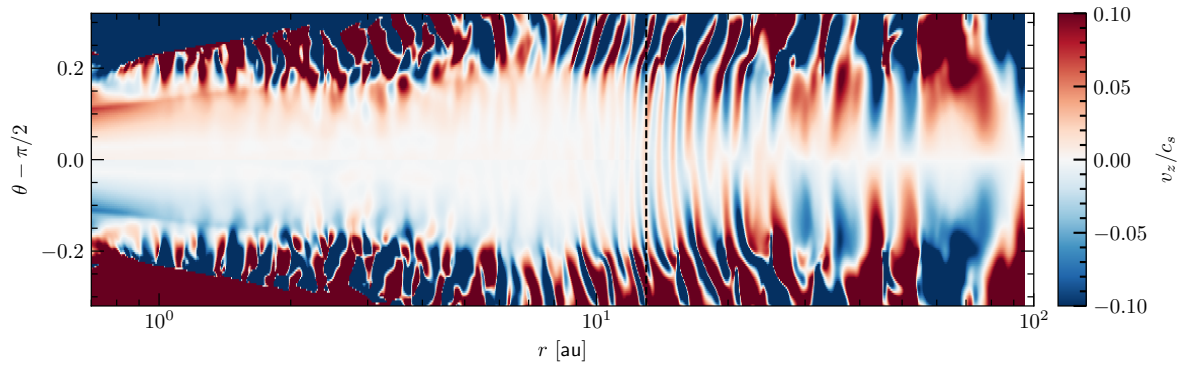


Figure 22: Vertical velocity profiles of the disk at t = 10000 orbits normalized to the sound speed. We see the development of the VSI modes in the outer disk. The dashed line indicates the location where VSI is expected according to the Lin and Youdin (2015) criterion.

to starlight-driven shadowing. This can only captured by models with self-consistent dust dynamics.

4.7.3 *Hydrodynamic modeling changes the picture: Comparisons to linear theory and hydrostatic studies*

Within the framework of linear analysis in the optically thick disk approximation, D'Alessio et al. (1999) showed that the disk is stable to temperature perturbations for very long cooling times (e.g., in the inner disk) and Dullemond (2000) found unstable thermal waves for short cooling times (e.g., in the outer disk). We find that temperature bumps and shadows are most likely between 1–30 au, corresponding to intermediate-to-long cooling times (similar to the findings of Wu and Lithwick, 2021). Time-dependent analysis by Watanabe and Lin (2008) and Ueda et al. (2021) found shadows radially outward up to ~100 au for dust-to-gas mass ratios up to $\epsilon_{dg} = 0.001$. While both linear theory (Dullemond, 2000), and hydrostatic models (Okuzumi et al., 2022; Ueda et al., 2021; Wu and Lithwick, 2021) found inwardly propagating waves, we find temperature bumps that grow and reach a quasi-steady configuration in our models, similar to models of Watanabe and Lin (2008) with stabilized shadows for high mass accretion rates $\dot{M}$. While we see some inward motion of bumps in the `MSIG2000` model, they also reach a quasi-steady state after a few thousand orbits.

We show that linear theory and vertically isothermal hydrostatic models do not accurately reflect the conditions in the disk. Consistent radial and vertical transport of radiation and advection indicate a more quasi-setady picture than a disk susceptible to a thermal wave or a self-shadowing instability. While this nonlinear quasi-steady state remains to be fully constrained, the natural formation of long-lived shadows in the disk is an interesting explanation for the origin of some of the ubiquitously observed substructures in protoplanetary disks.

4.7.4 *Comparisons to existing hydrodynamic models*

Kutra et al. (2024) performed hydrodynamic simulations with a simplified cooling model, where they relaxed the local temperature of the disk toward a forcing temperature calculated

based on a vertical integration of the irradiation source term, over the thermal relaxation timescale. This restricts their model effectively to a vertically isothermal approximation, implying changes in the temperature at the surface are propagated downward almost instantaneously likely favoring the formation of staircases (Melon Fuksman and Klahr, 2022). They perform hydrodynamic simulations with gray irradiation, constant opacities for the dust, and find staircases forming in the disk that stall.

While our hydrodynamic models with frequency-dependent irradiation and the flux-limited diffusion method also predict a similar quasi-steady state, it is important to note key differences in physics. Solving 2D axisymmetric hydrodynamics with a simplified cooling model (one that is effectively vertically isothermal) constrains thermal evolution to being a purely radial process. Heating and cooling due to vertical compression, expansion, or radiative diffusion is affected by the imposed vertical temperature profile to which the system is forced to relax. We believe that this can cause unphysical propagation or even amplification of temperature perturbations, exaggerating the staircase structure (to similar amplitudes like in 1.5D simulations, e.g., Ueda et al., 2021). Moreover, the thermal relaxation timescale they use is vertically constant throughout the disk (see Fig. 1 of Kutra et al., 2024), not capturing the transitions in optical depth from the disk surface to the midplane nor vertical heat diffusion. For context, their model matches closely to our `MSIG2000` model, but while their midplane reaches equilibrium after a few hundred orbits, the same takes several thousand orbits in our model.

Pavlyuchenkov et al. (2022b) also performed hydrodynamic simulations with a diffusion method, one that handles radiation transport in the optically thick regime accurately, but is artifically too diffusive in the optically thin and transition regions. This results in excessively rapid cooling near the optically thin disk atmosphere, where dust densities and therefore cooling rates, that are proportional to $\rho\epsilon_{dg}\kappa_P$ are extremely low. Our model uses a flux limiter to capture both regimes, and also includes frequency-dependent ray tracing which improves the treatment of the radiation field (Kuiper and Klessen, 2013). Their initial conditions differ from ours and more closely resemble a disk around a Herbig star with a luminosity of $L = 5\,L_\odot$. They include the inner rim, which when heated by such a bright star, puffs up and casts a wide shadow radially that extends outward into the optically thin region, thereby suppressing other bumps in the domain. Kutra et al. (2024) showed that this shadow extended to about 20 au, and recovered the temperature bumps as the shadow was smaller when they used stellar properties similar to our work. They use a sparse grid resolution (especially in the vertical direction), which might have prevented them from capturing the irradiation surface accurately, inhibiting the formation of shadows in the disk.

Melon Fuksman and Klahr (2022) used the M1 closure for radiation transport along with frequency-dependent irradiation. While in hydrostatic models we see that the M1 method produces temperature bumps that are more pronounced in amplitude than the ones we see with FLD, they always found that the thermal waves damp away when including hydrodynamics. This is in contrast to our findings, with the two shadows stabilizing for several hundreds of orbits in our simulations. We nevertheless compare our radial midplane temperature profiles and find that they match quite well (see Sect. 6.2). Our prescriptions of the inner disk and the initial optical depth differ, where we adopt a simpler prescription following Eq. (24) and Flock et al. (2013). We discuss the implications of using different initial optical depth profiles in Sect. 4.7.6 and the differences between our models in Appendix A.2.

The differences in the radiation methods among all of these existing hydrodynamic studies make it difficult to establish direct comparisons. This acts as motivation for future studies to compare the different methods in a systematic way to constrain the reliability of the irradiated disk studies. More complicated methods like the half-moment (HM) method (Melon Fuksman et al., 2025) and the discrete ordinate methods Jiang (2021) could help resolve the discrepancy.

4.7.5 *The amplitude of the bumps with FLD*

We see that the largest amplitude of the bumps (with the inner bump being the strongest) is about 40% relative to the background power law. The shadows resolved with our FLD model are less steep compared to those resolved with other radiation methods. In comparison, bumps in the M1 hydrostatic models of Melon Fuksman and Klahr (2022) showed amplitudes of order 50–60% (see Fig. 41). The models of Kutra et al. (2024) show bumps that stand out by a factor of 2, more consistent with a 1D thermodynamic approach. With FLD, we also see that the bumps are not strong enough for a temperature gradient inversion (except for the inner bump in the `MSIG2000` model) as opposed to the above mentioned methods, where they are locations of local extrema. These differences are expected with different radiation methods, details of are explored in code comparison works like Menon et al. (2022) and Thomas and Pfrommer (2022).

4.7.6 *The inner disk structure and the initial optical depth*

The structure and evolution of the inner rim of a protoplanetary disk is still an active field of research. The disk at the magnetospheric radius, studied by previous works like Dullemond et al. (2001) and in radiation hydrodynamical models like Chrenko et al. (2024) and Flock et al. (2025, 2019), is expected to feature a hot, puffed-up inner rim which casts a shadow upon the outer disk. Hence, there is still uncertainty about choosing the exact inner disk profile and whether the prescription for the optical depth of the inner disk $\tau_{\nu,0}$ (see Eq. (24)) is realistic. Our prescription of the initial optical depth is similar to the one used in Flock et al. (2013) and could introduce a small hotspot in the inner disk (see Fig. 45) which was not the case in the approach of Melon Fuksman and Klahr (2022). Nevertheless, this parameter cannot be constrained without more informed modeling of the inner disk temperature profile, and whether such a profile is smooth or structured. Studies further show that owing to material pileup at the inner edge of the dead zone, which can periodically trigger flares in MRI activity and accretion bursts, the inner disk can be hydrodynamically active and likely structured (Cecil and Flock, 2024; Vorobyov and Basu, 2006). This can cause dynamical shadowing in the outer disk (e.g., outwardly moving shadows along with static ones found in Schneider et al., 2018).

4.7.7 *Snowlines and opacity transitions*

The possibility that opacity transitions could trigger bumps, causing subsequent structure around it due to starlight-driven shadowing, could have physical and chemical implications on the structure in the outer disk close to snowlines, which are regions of high opacity shifts (Baillié et al., 2015; Garaud and Lin, 2007; Oka et al., 2011). Some viscous models in the

literature that include this show evidence of downstream shadowing (Savvidou et al., 2020). Moreover, we know that snowlines might not be thermally stable, and could be susceptible to the snowline instability (Owen, 2020). Dynamic changes of the snowline location, combined with secondary shadows downstream, might create ringed structures as in this work that could show variability in the disk. Secondary structures due to starlight-driven shadowing could also be important for planet formation models that invoke shadows around the snowline in the solar system (for e.g., modeling the formation of Jupiter in the shadow cast by dust pileup near the water snowline in Ohno and Ueda, 2021).

4.7.8 *The effect of dust*

Our hydrostatic dust models in Sec. 4.6 show very sharp irradiation absorption layers, and no temperature bumps regardless of the level of turbulent diffusion. This is in contrast with the models that take into account dust-settling using RADMC-3D in Wu and Lithwick (2021). The authors of this work show shadows due to the TWI extending out to 100 au, but their approach relied on an *ad hoc* prescription of removing dust from the upper layers (above two scale heights). While this emulates a dust distribution with vertical settling, it does not explicitly account for turbulent diffusion, size dependence on settling and is not entirely physically self-consistent with the radiation transfer iterations. The qualitative impact of settling in determining the accurate irradiation absorption surface is robust from the dust models included in this chapter. However, we note that relaxing the constant grain size of 10 μm for the mass-carrying dust component could have an effect on the amplitude of the shadows, something worth exploring in the future.

Self-consistent hydrodynamic models of dust–gas interaction are an important next step toward understanding the shadows and the overall thermal structure of the disk. Our hydrostatic dust scale-height models, even though they take into account a settling–diffusion equilibrium, can be improved further to account for the dust transition layer, that is, the height beyond where dust decouples dynamically from the gas. Moreover, the effect of dust backreaction and opacity feedback (e.g., Krapp et al., 2024b; Ziampras et al., 2025b) will be crucial to understanding the irradiation absorption surface, and subsequently any possible starlight-driven shadowing. For regions where dust–gas coupling is weak, decoupled dust temperatures with schemes like the three-temperature approach of Muley et al. (2023) might also become relevant.

4.8 SUMMARY

We ran radiation hydrostatic and hydrodynamic simulations using frequency-dependent ray-traced irradiation and the flux-limited diffusion approach to study regions of the disk that are susceptible to starlight-driven shadowing and to explore the thermal wave instability (TWI). We also presented the first models that self-consistently capture the vertical structure of dust in the hydrostatic approximation. We outline our key findings below.

1. The best-case scenario for recovering starlight-driven shadows corresponds to optically thick, slowly cooling disks that are characterized by a high dust-to-gas ratio in small grains $\epsilon_{dg} = 0.01$ or high surface densities (models `MFID`, `MSIG2000`). In the hydrostatic limit, we recovered the TWI with thermal waves that grew and traveled inward.

2. The growth and final state of the shadows in the disk are strongly affected by dynamics. We report that temperature bumps within 30 au of the disk grow, reach a quasi-steady state, and remain stable for several thousand orbits. The largest relative amplitude of 40% was found for the bumps with respect to the background power law in the midplane. The inner bump was about $1.7 - 2$ au wide, and the second bump was 10 au wide in the `MFID` model and 5 au wide in the `MSIG2000` model, where even a third bump formed.

3. The staircase structure did not propagate inward at the timescales predicted by linear theory and 1D models, and hence, we did not recover the instability referred to by these models, but a long-lived stable structure.

4. While the innermost bump in the disk might be affected by the opacity transition at 250 K or the optical depth profile of the inner disk, the consequent bumps were natural consequences of an existing first bump, driven by starlight-driven shadowing.

5. The amplitude of the shadows is less pronounced than the amplitudes recovered in the vertically isothermal limit (Kutra et al., 2024; Ueda et al., 2021). This might be a result of more accurate thermodynamic modeling and indicates the importance of accounting for vertical diffusion of temperature in any study that relies on temperature transitions.

6. For lower dust-to-gas ratios $\epsilon_{dg} \leqslant 0.001$, the shadowing is suppressed. We only recovered the innermost bump in the dynamic runs.

7. We presented the first physically motivated hydrostatic models that include dust settling. Our models with 10 µm dust grains showed a superheated dust layer without any staircases, indicating that dust affects the irradiation absorption layer, and hence, any subsequent shadowing. Further, the $\tau_v = 1$ layer substantially moves closer to the midplane. We highlight that coupling the dust motion and dynamics to the opacity remains important.

8. Better constraints on the structure and optical depth of the inner disk, an important initial condition in our model in light of recent self-consistent radiation hydrodynamic modeling of the inner rim casting a shadow on the outer and less turbulent disk (Cecil and Flock, 2024; Flock et al., 2025, 2017b) will help us to address the robustness of these starlight shadows in a more realistic scenario.

We note that a more detailed parametric study of different stellar properties and opacities, including snowlines, will help us to further constrain the limits of starlight-driven shadowing in protoplanetary disks. Furthermore, the differences in the results from various radiation methods need to be explored in detail and are decisive for our current understanding of the temperature structure in protoplanetary disks. More refined methods with multigroup radiative transfer or angle-dependent discrete ordinate methods will constrain our models better.

It is also interesting to explore the observability of these shadows with synthetic observations. This mechanism might be a simple explanation of rather common surface substructure, such as rings and gaps found in scattered-light observations with VLT-SPHERE (and possibly in the future, with the facilities on the Extremely Large Telescope). The subsequent effect on the dust dynamics will also give us insight into how these shadows might manifest in the

millimeter continuum, which will add to our understanding of the observed substructures in protoplanetary disks.

5

The fluid dust module of the PLUTO code

Based on the results of **Sudarshan, P.**, Ziampras, A., et al., submitted to ApJS, "Multifluid dust dynamics in PLUTO: Benchmarking dust–gas drag and diffusion against existing implementations"
Authors: Prakruti Sudarshan, Alexandros Ziampras, Mario Flock, Thomas Pfeil, Dhruv Muley, Giancarlo Mattia and Andrea Mignone
Contributions: I wrote the dust diffusion module, the test problems, ran all the simulations, and wrote the manuscript. Alexandros Ziampras added the multispecies loop, wrote the implicit solver, provided the setup for the 2D axisymmetric disk setup and wrote the sections on the implementation of the implicit drag solver and the Riemann solver. Mario Flock wrote the explicit method. All authors provided comments that helped improve the manuscript.

and

the Appendix of
Ziampras, A., **Sudarshan, P.**, et al., 2025, MNRAS, 536, 4, pp.3322-3337 [ADS]
Authors: Alexandros Ziampras, Prakruti Sudarshan, Cornelis P. Dullemond, Mario Flock, Vittoria Berta, Richard P. Nelson and Andrea Mignone
Contributions: Alexandros Ziampras is the main author of this article, who carried out the simulations and wrote the main text, but the Appendix section, which is included as part of this chapter, is written by me. I wrote the implementation of the one-fluid dust module and ran all the simulations and tests for it. Alexandros Ziampras wrote the implicit solver, and Mario Flock wrote the explicit method. All authors provided comments that helped improve the manuscript.

5.1 Overview

Modeling dust is foundational to our understanding of different astrophysical processes. Modern hydrodynamical codes employ different methods to treat dust dynamics, ranging from passive scalars and particle-in-cell (PIC) methods to Lagrangian superparticles and fluid approaches. In this paper, we present the multispecies dust–gas drag coupling and dust diffusion modules for the pressureless dust fluid method of the PLUTO code. We test both the explicit and implicit one-fluid and multifluid drag solvers, demonstrate their convergence properties, and also provide numerical benchmarks for some standard non-linear problems in the literature. We model the micro-scale turbulence in the dust fluid using a concentration diffusion formalism with an explicit, unsplit scheme, with super-timestepping as an alternative for very diffusive problems. We provide comparisons to formalisms used in codes FARGO3D and Athena++ with test problems. With our set of benchmark problems motivated by protoplanetary disks and planet formation, we show that our generalized multifluid

drag and diffusion framework can be readily coupled to additional physical processes, such as dust coagulation and radiative transfer, making it a useful tool for problems at the cutting edge of protoplanetary disk and planet formation studies.

5.2 INTRODUCTION

The role of dust has been studied extensively at different scales and in different astrophysical environments. From the galactic scales to molecular clouds, dust controls extinction, heating/cooling processes, chemistry, and star formation (Cazaux and Tielens, 2004; Draine, 2003; Galliano et al., 2018; McKee and Ostriker, 2007). At smaller scales, dust becomes dynamically significant in circumstellar environments around AGB stars (Höfner and Olofsson, 2018) and in protoplanetary disks (Birnstiel, 2024). In these disks, dust grains grow from submicron-sized monomers to centimeter-sized pebbles, which are needed to form planetary cores with radii of several tens of thousands of kilometers (Drażkowska et al., 2023; Johansen et al., 2007). The transport and evolution of dust is thus fundamental to understanding planet formation. Both observational and theoretical studies have shown the importance of dust growth and dynamics in shaping substructures seen in high-resolution direct imaging (Andrews et al., 2018; Pinilla et al., 2012) and the overall chemical content and distribution in disks (e.g., Krijt et al., 2016). In the outer regions, dust dominates the opacity and sets the temperature structure (Dullemond and Dominik, 2004a) as well as the cooling timescale, which is important for various hydrodynamic instabilities as well as many other dynamical processes operating in the disk (for reviews, see Bae et al., 2023; Lesur et al., 2023b).

With an explicit treatment of individual grains being computationally prohibitive, modern hydrodynamic codes instead employ representative dust elements coupled to the gas through a drag force. In the Epstein regime (Epstein, 1924) where the mean free path of a gas molecule is larger than the typical size of the dust grain a, the drag force is determined by the momentum exchange between dust grains and the gas molecules. A commonly used assumption for disks is that the dust–gas relative velocity remains subsonic such that the collisional velocity is set by the thermal velocity of the gas. The characteristic timescale at which dust is dynamically coupled to the gas, or the stopping time t_s, can then be written down as:

$$t_s = \frac{a\rho_a}{\bar{v}\rho_g}, \tag{30}$$

where ρ_a is the density of the dust grain and ρ_g is the density of the gas. The mean thermal velocity $\bar{v}$ is given by $\bar{v} = \sqrt{8/\pi}c_s$ where $c_s = \sqrt{P/\rho_g}$ is the isothermal sound speed, and P is the gas pressure. This provides the definition for the dimensionless Stokes number $St = t_s\Omega_K$, relating the stopping time to the orbital timescale Ω_K^{-1} and providing a more useful quantity to measure the aerodynamic drag coupling between dust particles and gas.

In existing grid-based codes, where gas is evolved as a fluid on a Eulerian grid, dust is typically modeled as an ensemble of Lagrangian superparticles (Bai and Stone, 2010b; Mignone et al., 2019) or as one or more additional, pressureless fluids, with examples of such implementations found in `LA-COMPASS` (Fu et al., 2014), `FARGO3D` (Benítez-Llambay et al., 2019), and `Athena++` (Huang and Bai, 2022). Both approaches have their respective advantages: the former resolves appropriately the dynamics of grains with $St \gg 1$ (Bai and Stone, 2010b) and can correctly capture the crossing of particle trajectories (Johansen et al., 2007); the latter is instead computationally cheaper, parallelizable, and allows for an easier implementation of

dust feedback while typically remaining accurate for $\mathrm{St} \lesssim 1$ (Binkert et al., 2021; Krapp et al., 2024b). For $\mathrm{St} \ll 1$ dust–gas interaction can happen on very short timescales (i.e., smaller than the typical simulation timestep), resulting in a stiff operator that necessitates (semi-)implicit solvers that can circumvent this issue (Benítez-Llambay et al., 2019). In the context of dust as a pressureless fluid, this approach has been widely applied for protoplanetary disk problems, including recently to scenarios of pebble accretion (Konijn and Paardekooper, 2026) that are sensitive to the underlying dust dynamics. While the fluid approximation is not without its limitations (Binkert, 2023; Lynch and Laibe, 2024), its versatility, wide range of applications in the literature, and ease of implementation motivate its use.

Recent numerical advances to the pressureless dust fluid method include specialized Riemann solvers to handle delta shocks (Huang and Bai, 2022; Verrier et al., 2025), efficient matrix inversion algorithms (Krapp and Benítez-Llambay, 2020) to solve momentum coupling between gas and dust, and novel time-integration schemes (Krapp et al., 2024a; Tedeschi-Prades et al., 2025) to improve the convergence order. One other aspect is the inclusion of dust diffusion to model the effect of turbulence (due to an arbitrary source). This is typically done by modeling a concentration diffusion source term (Morfill and Voelk, 1984), but there are differences in how these equations are implemented across codes that are worth quantifying.

Here, we present one-fluid and multi-fluid pressureless dust modules for the astrophysical gas dynamics code `PLUTO` (Mignone et al., 2007), with second-order explicit and implicit drag solvers. To verify the accuracy of our implementation, we provide a suite of numerical tests, including benchmarks of the non-linear streaming instability problem presented in Johansen and Youdin (2007). In addition, we implement a method for concentration and momentum diffusion similar to the one introduced in `cuDisc` (Robinson et al., 2024), with comparisons to the methods used in the codes `FARGO3D` and `Athena++`. The fluid dust module presented here can be readily extended to include additional physics, such as radiation hydrodynamics with opacities dynamically computed from the evolving dust distribution (Krapp et al., 2024b; Ziampras et al., 2025a), as well as dust growth, coagulation, and fragmentation (Drążkowska et al., 2019; Pfeil et al., 2024; Ziampras et al., 2026a).

This chapter is organized as follows: We first describe the one-fluid implementation of the dust module in Sect 5.3. This includes the numerical implementation of the explicit and implicit drag solvers, and tests. Further, from Sect. 5.5, we present the multifluid equations including the drag term, and provide tests for the drag solvers. In Sect. 5.6, we discuss the existing formulations of dust diffusion and provide comprehensive tests, comparing them to existing approaches in the literature. In Sect. 5.7, we present discussions and conclusions. Additional clarifications and details are provided in Appendices B.1, B.2, B.3 and B.4.

5.3 NUMERICAL IMPLEMENTATION OF THE ONE-FLUID MODULE

We describe the algorithm and tests for a two-fluid module for dust and gas implemented in the `PLUTO v4.4` hydrodynamics code. Dust interacts with gas via aerodynamic drag following dynamics outlined by the continuity and momentum equations. Following the prescription in

Benítez-Llambay et al. (2019) for an arbitrary number of species, we write down the equations for two species (i.e., gas or dust) where index $i \in [g, d]$,

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\rho_i \boldsymbol{u}_i) = 0 \tag{31a}$$

$$\frac{\partial (\rho_i \boldsymbol{u}_i)}{\partial t} + \nabla \cdot (\rho_i \boldsymbol{u}_i \otimes \boldsymbol{u}_i + \delta_{ig} P_i \mathbb{I}) = f_i, \tag{31b}$$

with the identity matrix $\mathbb{I}$, and the drag force defined in terms of the collision rate α_{ij}

$$f_i = -\rho_i \alpha_{ij} (\boldsymbol{u}_i - \boldsymbol{u}_j), \; i \neq j \tag{32}$$

where $\alpha_{ij} = a_i \delta_{jg} + \epsilon_j a_j \delta_{gi}$, with δ being the Kronecker delta function. The collision rate is parameterized using the stopping time t_s, giving $a_i = t_s^{-1}$. We only include the contribution of dust in momentum transfer and hence solely focus on conservation of momentum, and not total energy. Dust diffusion is currently not part of the module.

The way transport and source terms are solved for in the PLUTO code (see Fig. 1 in Mignone et al. (2007) for the flowchart) can be described using the reconstruct-solve-average (RSA) method:

1. volume-averaged conserved quantities $\left(\rho, \rho \boldsymbol{u}, e + \rho u^2/2\right)$ are converted to their primitive counterparts $(\rho, \boldsymbol{u}, e)$,
2. left- and right-interface values are reconstructed from volume averages by means of piecewise polynomial interpolands in each zone,
3. a Riemann problem is solved to calculate numerical fluxes for the gas
4. fluxes are used to update the conserved quantities keeping the whole scheme unsplit and explicit, and the solution is advanced with a timestep determined by the Courant–Friedrichs–Levy condition (CFL).

PLUTO employs Strang splitting to handle source terms where physical processes, such as viscosity and cooling, operate on different timescales compared to transport. We integrate our dust–drag terms into this framework as described by the flowchart in Fig. 23. For the dust, numerical fluxes are determined by the Riemann solver in LeVeque (2004).

The public version of the dust module uses an explicit integrator, including Eq. (32) as an additional source term and thereby achieving second-order temporal accuracy.

We also implemented an unconditionally stable implicit scheme. The latter updates the velocities of both fluid components with an operator-split step that is exact to first order in time, following equation (31a):

$$\left.\begin{aligned} \rho_g \frac{\boldsymbol{u}_g' - \boldsymbol{u}_g}{\Delta t} &= -\rho_d \frac{\boldsymbol{u}_g' - \boldsymbol{u}_d'}{St} \Omega_K \\ \rho_d \frac{\boldsymbol{u}_d' - \boldsymbol{u}_d}{\Delta t} &= -\rho_d \frac{\boldsymbol{u}_d' - \boldsymbol{u}_g'}{St} \Omega_K \end{aligned}\right\} \Rightarrow \begin{aligned} \boldsymbol{u}_g' &= \frac{(1+b)\boldsymbol{u}_g + b\epsilon \boldsymbol{u}_d}{1 + b + b\epsilon} \\ \boldsymbol{u}_d' &= \frac{b\boldsymbol{u}_g + (1+b\epsilon)\boldsymbol{u}_d}{1 + b + b\epsilon} \end{aligned} \tag{33}$$

where $\boldsymbol{u}'$ indicates the velocity after a timestep Δt, $b \equiv \Delta t / t_s$ is the ratio of timestep to stopping time, and $\epsilon \equiv \rho_d / \rho_g$ is the dust-to-gas ratio.

To ensure that the correct terminal velocity limit is recovered, the dust–gas drag operator in Eq. (33) is always applied after the transport and source terms regardless of whether

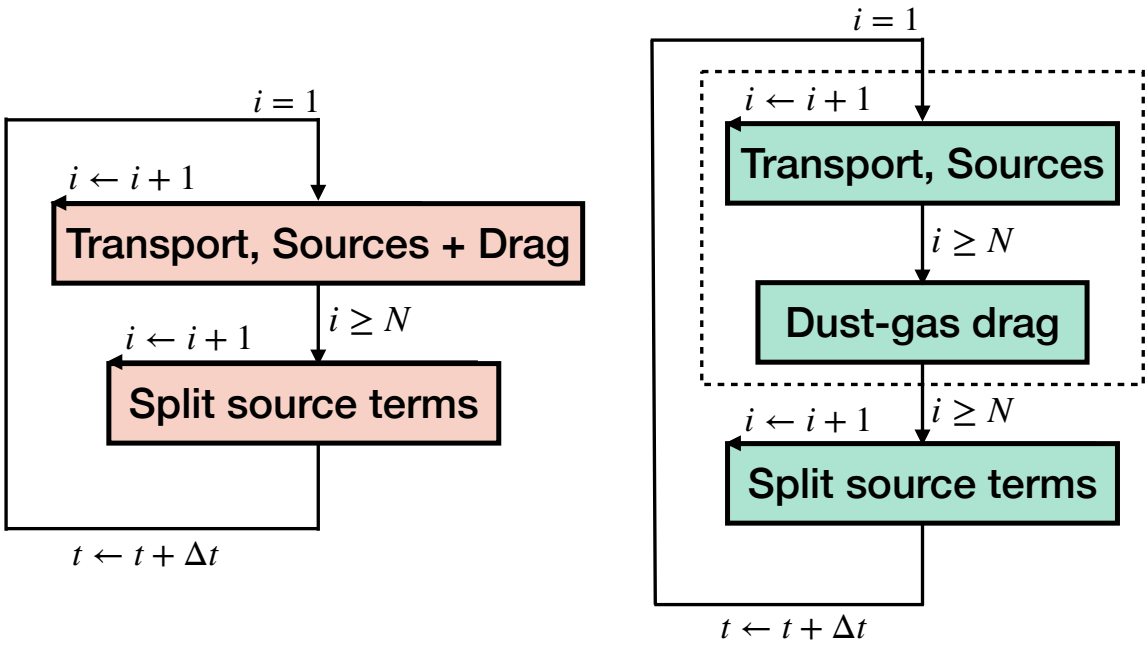


Figure 23: A simplified flowchart of the steps involved in the explicit (left) and implicit (right) schemes including dust drag. The drag term is added to the transport and source term stages in the case of the explicit solver, while for the implicit solver, operator splitting is used. The dotted box is used to indicate that transport, source terms, and dust–gas coupling are treated as a single, "unsplit" step for the purposes of Strang splitting in order to recover the terminal velocity limit for the dust (Booth et al., 2015).

additional, split-source terms are considered (Booth et al., 2015, see also Fig. 23). Although it reduces the accuracy of the scheme to first-order, benefits of employing an unconditionally stable, fully implicit integrator are compared to those of a fully explicit one in Sect. 5.4.3.

5.4 ONE-FLUID DUST: TESTING

In this section, we validate our solver using some well-known tests with checks for stability, accuracy and convergence. Unless otherwise stated, tests have been performed using the HLLC Riemann solver (Toro et al., 1994), linear spatial reconstruction, the limiter by Van Leer (1974), and a second-order Runge–Kutta time integrator.

5.4.1 *Radial drift test*

An unperturbed disk rotates at sub-Keplerian velocities due to the background pressure gradient. Dust does not feel pressure and therefore typically feels a head-/tailwind, drifting in-/outwards. For a simple accreting disk in the absence of dust feedback the dust velocity can be written analytically (Takeuchi and Lin, 2002) as

$$u_{R,d} = \frac{\eta R \Omega_K + St^{-1} u_{R,g}}{St + St^{-1}}, \qquad \eta = \frac{d\log P}{d\log R} h^2, \quad P = \Sigma c_s^2, \tag{34}$$

where $St = t_s \Omega_K$ is the Stokes number.

We set up a 1D vertically integrated, inviscid, adiabatic disk in polar coordinates, with 1024 logarithmically-spaced radial cells and $R \in [0.4, 2.5]\ R_0$. As initial conditions, we use power laws for the surface density $\Sigma = \Sigma_0 (R/R_0)^{-1/2}$, temperature $T = T_0 (R/R_0)^{-1}$, and a constant aspect ratio $h = 0.05$. We set the radial boundaries to the initial profile, and use a linear damping function at the edges of the domain. We switch off dust feedback and run the model for different $St \in [10^{-3}, 100]$ in intervals of powers of 10. We then integrate for $100\,\Omega_K^{-1}$ to ensure that the disk has reached a steady state, with a CFL of 0.4. Under these conditions, Eq. (46) simplifies to

$$\frac{u_{R,d}}{u_K} = -\frac{3}{2}\frac{h^2}{St + St^{-1}} = \text{const.}, \quad u_K = R\Omega_K. \tag{35}$$

We therefore compute the radial average of $u_{R,d}/u_K$ between 1–2 R_0 and compare it to the theoretical expectation from Eq. (35).

We plot the results of the comparison after $100\,\Omega_K^{-1}$ in Fig 27. Both the implicit and explicit solvers match with the analytical profile very well, with relative errors within 2%, except for $St > 20$ which is reasonable as the fluid approximation breaks down for weak dust–gas coupling. The errors are smaller for the second-order explicit solver, but we note that for $St < 0.01$, the explicit solver needs significantly smaller time steps to maintain stability, in contrast to the unconditionally stable implicit scheme. We conclude that our solvers perform well without dust feedback.

5.4.2 *Linear* `DUSTYWAVE` *test*

We test dust–gas coupling in our solvers using the `DUSTYWAVE` test (Laibe and Price, 2011, 2012) that involves adding a perturbation to the two-fluid system and comparing its oscillation and damping rate with expectations from linear theory. Similar to the prescription in equations (45) & (46) of Benítez-Llambay et al. (2019), for two species, a perturbation of the form $\delta p = \delta\hat{p} e^{ikx - \omega t}$ where k is a wavenumber gives the dispersion relation

$$\omega^3 t_s - \omega^2(1+\epsilon) + \omega\omega_s^2 t_s - \omega_s^2 = 0 \tag{36}$$

where $\omega_s = k c_s$. This is solved for different values of the stopping time t_s (which, in dimensionless units, represents the Stokes number St) and the complex roots which describe the oscillation and damping rate of the sound wave are calculated. For $\omega \neq t_s^{-1}$, the components of the eigenvectors for all quantities are given by

$$\frac{\delta\hat{u}_g}{c_s} = -i\frac{\omega}{\omega_s}\frac{\delta\hat{\rho}_g}{\rho_g^0} \tag{37a}$$

$$\frac{\delta\hat{u}_d}{c_s} = -i\frac{\omega}{\omega_s}\frac{1}{(1-\omega t_s)}\frac{\delta\hat{\rho}_g}{\rho_g^0} \text{ and} \tag{37b}$$

$$\frac{\delta\hat{\rho}_d}{\rho_d^0} = \frac{1}{(1-\omega t_s)}\frac{\delta\hat{\rho}_g}{\rho_g^0}. \tag{37c}$$

For different Stokes numbers, we list the different eigenvectors for the dust density, gas and dust velocities in Table 3. We calculate numerical solutions for this problem with a setup with

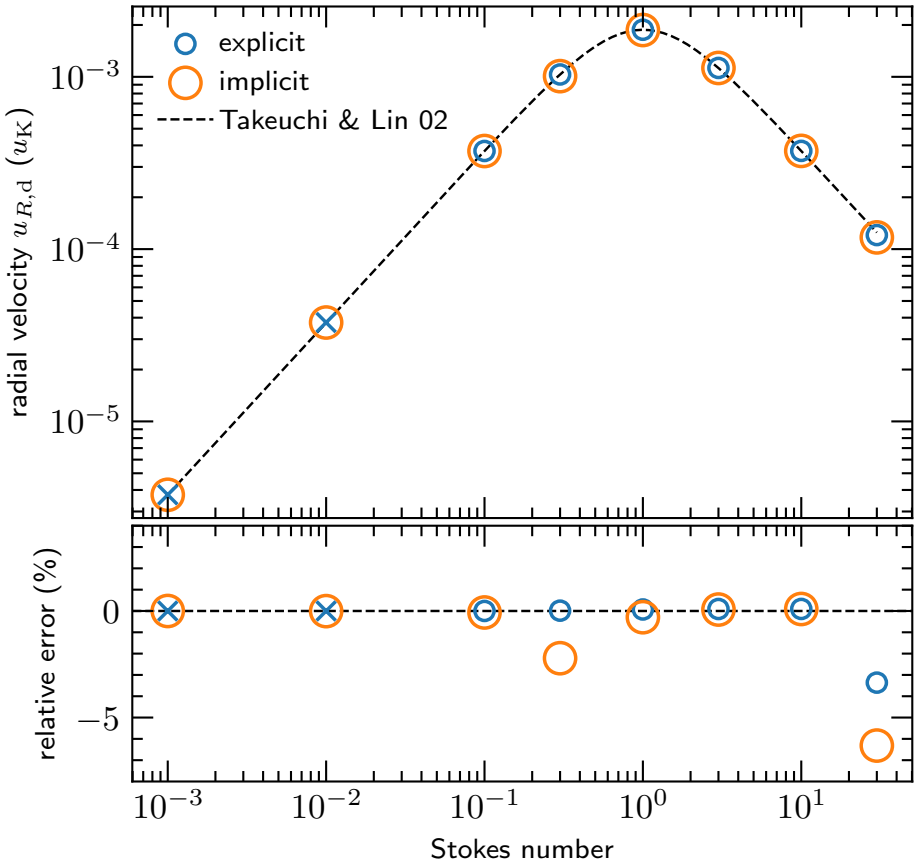


Figure 24: Comparison of radial velocity values of the dust component between the explicit PLUTO dust fluid module (release version `4.4-patch2`, blue) and the implicit module (used in this paper, orange) against analytical profiles (dashed lines) from Takeuchi and Lin (2002) for different Stokes numbers, measured by averaging $u_{R,d}/u_K$ between 1–2 R_0. For $St \lesssim 0.001$ (indicated by crosses), the explicit method is only stable for smaller time steps (we use a CFL = 0.2 for the $St = 10^{-2}$ model, and a fixed time step of $\Delta t = 10^{-5}$ for the $St = 10^{-3}$ model). The bottom panel shows the relative errors between each result and its theoretical value. The explicit solver performs better overall, while both models show higher errors for $St > 10$.

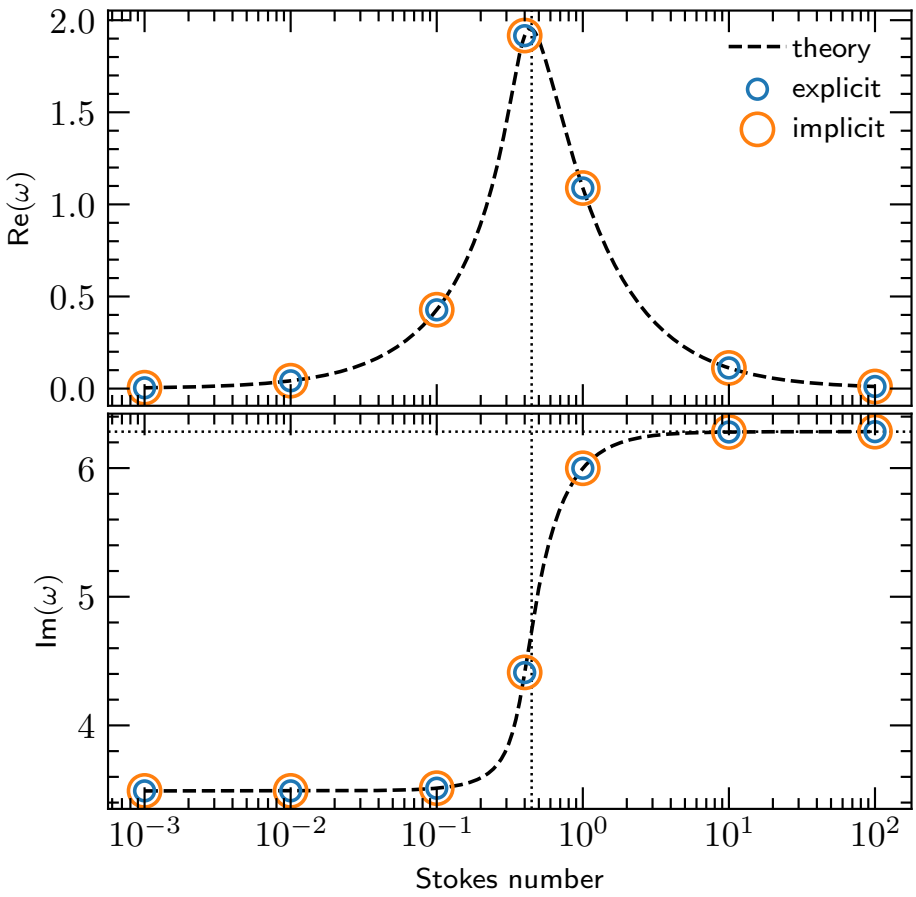


Figure 25: Real and imaginary components of the linear dusty wave for the explicit and implicit dust fluid modules. The dashed curves correspond to the analytical profiles, and a vertical dotted line corresponds to the value of St for which the wave damps the fastest. The eigenvectors for the different Stokes numbers can be found in Table 3.

a constant background density $\rho_d^0 = 2.24$, $\rho_g^0 = 1$, and zero velocities, adding a perturbation of the form,

$$\delta p = A\ [\mathrm{Re}(\delta\hat{p})\cos(kx) - \mathrm{Im}(\delta\hat{p})\sin(kx)]. \tag{38}$$

where a small amplitude $A = 10^{-4}$ is used. We use periodic boundary conditions, a domain of size $x \in [0, L]$ with $L = 1$ and 1024 cells, and set the sound speed $c_s = 1$ and the wavenumber $k = 2\pi/L$. We then compare the damping rate $\mathrm{Re}(\omega)$ and oscillation frequency $\mathrm{Im}(\omega)$ to the analytical solution and find very good agreement with our implicit and explicit solvers over the range of Stokes numbers as illustrated in Fig. 29. The second-order explicit scheme shows lower errors overall, with relative errors below 0.1%. The implicit scheme shows larger errors in the damping rate $\mathrm{Re}(\omega)$ for the low Stokes regime, with a highest relative error of 10% for $\mathrm{St} = 10^{-3}$.

Stokes No. (St)	$\delta\hat{\rho}_d/\rho_d^0$	$\delta\hat{u}_g/c_s$	$\delta\hat{u}_d/c_s$
0.001	$0.7590 + 1.062i$	$0.5556 - 6.704 \times 10^{-4}i$	$0.5556 + 0.001269i$
0.01	$0.7551 + 1.073i$	$0.5556 - 0.006705i$	$0.5554 + 0.01269i$
0.1	$0.6980 + 1.183i$	$0.5589 - 0.06802i$	$0.5376 + 0.1262i$
0.4	$0.1653 + 1.248i$	$0.7020 - 0.3049i$	$0.2216 + 0.3685i$
1	$0.2081 + 0.8846i$	$0.9547 - 0.1733i$	$0.02653 + 0.1595i$
10	$0.2962 + 0.7790i$	$0.9996 - 0.01782i$	$2.534 \times 10^{-4} + 0.01592i$
100	$0.3052 + 0.7704i$	$1.000 - 0.001783i$	$2.500 \times 10^{-7} + 0.001592i$

Table 2: Initial conditions for the DUSTYWAVE test, corresponding to eigenvectors for perturbations in the dust density, gas and dust velocities for $St \in [10^{-3}, 100]$. Perturbations in gas density are always assumed to be $\delta\hat{\rho}_g/\rho_g^0 = 1$. These values are used in Eq. (50).

5.4.3 *Convergence of the one-fluid module*

To evaluate the spatial convergence of our solvers, we repeat the DUSTYWAVE tests in Sect. 5.4.2 for different grid resolutions. To achieve this, we vary the number of grid cells in the range $N_x \in [128, 2048]$ in intervals of powers of 2. Similarly to Benítez-Llambay et al. (2019), we allow the time-step to vary by choosing a CFL of 0.4. We then calculate the root mean square error (RMSE) for any quantity, here ρ_g, defined by

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{\mathrm{cells}}\left(\rho_g - \rho_g^{\mathrm{theory}}\right)^2} \tag{39}$$

We see very good convergence for both the implicit and explicit solvers, with the overall errors being lower for the second-order explicit solver, as seen in Fig. 26 The scaling of the RMSE with spatial resolution agrees very well with that of Fig. 6 in Benítez-Llambay et al. (2019), which shows a mean square error scaling with $N_x^{-2.2}$ (i.e., $\mathrm{RMSE} \propto N_x^{1.1}$). We therefore conclude that both the explicit and implicit dust fluid modules implemented in the PLUTO code are well-tested in terms of stability, accuracy, and convergence rate.

5.5 MULTIFLUID DUST–GAS DRAG

In this section, we write down the multi-fluid equations including the drag terms, discuss the numerical method, and provide tests for the drag solvers. We also include the non-linear streaming instability problem for an unstratified disk (Johansen and Youdin, 2007; Youdin and Johansen, 2007), and compare it to solutions found by other codes.

5.5.1 *Multi-fluid equations*

The continuity and momentum equations for a gas fluid with density ρ_g and velocity $\boldsymbol{v}_g$, and a set of N pressureless dust fluids with densities $\rho_{d,n}$ and velocities $\boldsymbol{v}_{d,n}$ in conservative form can be written as follows:

$$\frac{\partial \rho_g}{\partial t} + \nabla \cdot (\rho_g \boldsymbol{v}_g) = 0, \tag{40a}$$

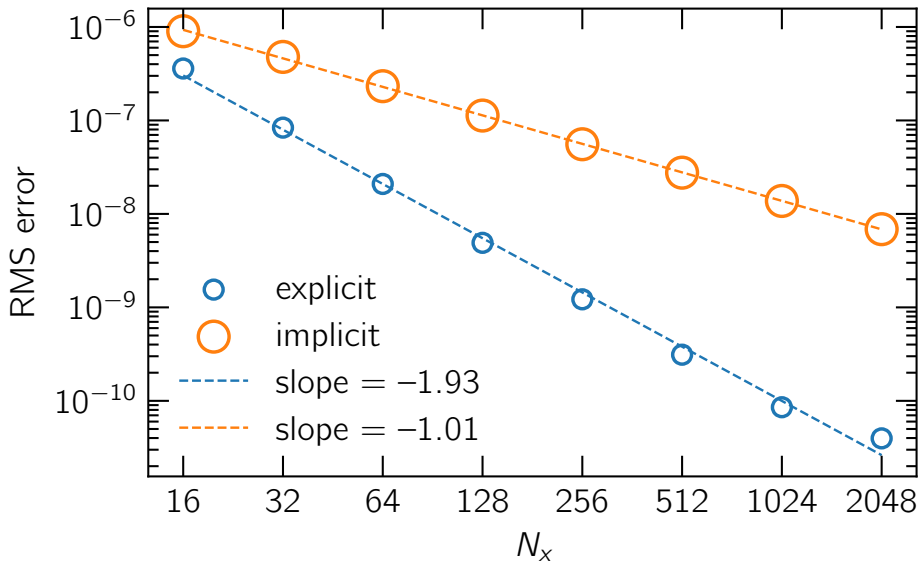


Figure 26: Root mean square errors for the dusty wave test models with different spatial resolutions. The implicit module shows an error that scales linearly with N_x, while the fully second-order explicit module scales quadratically.

$$\frac{\partial(\rho_g \boldsymbol{v}_g)}{\partial t} + \nabla \cdot (\rho_g \boldsymbol{v}_g \otimes \boldsymbol{v}_g + P\mathbf{I}) = \sum_{n=1}^{N} \frac{\rho_{d,n}}{t_{s,n}} (\boldsymbol{v}_{d,n} - \boldsymbol{v}_g) + S_g, \tag{40b}$$

$$\frac{\partial \rho_{d,n}}{\partial t} + \nabla \cdot \left(\rho_{d,n} \boldsymbol{v}_{d,n}\right) = 0, \tag{40c}$$

$$\frac{\partial(\rho_{d,n} \boldsymbol{v}_{d,n})}{\partial t} + \nabla \cdot \left(\rho_{d,n} \boldsymbol{v}_{d,n} \otimes \boldsymbol{v}_{d,n}\right) = \frac{\rho_{d,n}}{t_{s,n}} (\boldsymbol{v}_g - \boldsymbol{v}_{d,n}) + S_d. \tag{40d}$$

Here, $t_{s,n}$ is the stopping time for the n-th dust species, P is the gas pressure determined by the appropriate equation of state, and $\mathbf{I}$ is the identity matrix. For most simulations included in this paper, we adopt a locally isothermal equation of state, and hence $P = \rho_g c_s^2$. For the adiabatic model, we use the ideal gas equation of state defining $P = (\gamma - 1)\rho_g e$ where $\gamma = 7/5$ is the adiabatic index, and e is the specific internal energy. Finally, S_g and S_d are source terms for the gas and dust, respectively (e.g., gravity, viscosity, etc.). The drag terms are integrated within `PLUTO` following the implementation detailed in Sect. 5.4.1.

To compute dust fluxes, we have implemented two Riemann solvers: the exact solver of LeVeque (2004) and a simpler Lax–Friedrichs solver. Following Mignone et al. (2007), we define the vector of "primitive" and conserved variables $\boldsymbol{V}$ and $\boldsymbol{U}$ for a given dust species (here dropping the subscript 'd') as

$$\boldsymbol{V} = (\rho,\ v_1,\ v_2,\ v_3), \quad \boldsymbol{U} = (\rho,\ \rho v_1,\ \rho v_2,\ \rho v_3), \tag{41}$$

with subscripts denoting each of the three spatial directions. We can then define the flux tensor $\bar{\boldsymbol{T}}(\boldsymbol{U})$ with elements $T_{ij} = v_i U_j$ and its projection onto direction d as $\boldsymbol{f}$, with $f_j = \delta^{di} T_{ij}$. The vectors $\boldsymbol{V}$, $\boldsymbol{U}$, and $\boldsymbol{f}$ are then reconstructed onto cell interfaces from their volume-averaged counterparts using the reconstruction–limiting scheme chosen by the user, yielding

a "left" and "right" value for each cell (e.g., U_L, U_R). The Lax–Friedrichs flux $\boldsymbol{F}_{LF}$ through a cell interface is then

$$\boldsymbol{F}_{LF} = \frac{1}{2}\Big[\boldsymbol{f}_L + \boldsymbol{f}_R - |\lambda_{max}|\,(\boldsymbol{U}_R - \boldsymbol{U}_L)\Big], \tag{42}$$

with $\lambda_{max,d} = \max(v_{d,L}, v_{d,R})$ being the largest signal velocity in direction d. While diffusive, the Lax–Friedrichs solver is quite robust and easy to compute.

The solver by LeVeque (2004) follows a Roe-like approach:

$$\boldsymbol{F}_{LV} = \begin{cases} \boldsymbol{f}_L, & \lambda_d > 0, \\ \boldsymbol{f}_R, & \lambda_d < 0, \\ \frac{1}{2}(\boldsymbol{f}_L + \boldsymbol{f}_R), & \lambda_d = 0 \end{cases} \qquad \lambda_d = \frac{\sqrt{\rho_L}v_{d,L} + \sqrt{\rho_R}v_{d,R}}{\sqrt{\rho_L} + \sqrt{\rho_R}}.$$

In the case that $v_{d,L} < 0$ and $v_{d,R} > 0$, the flux is set to $\boldsymbol{F}_{LV,d} = 0$.

Implementing gas–dust drag as an implicit step requires solving a linear algebra problem for the system of equations

$$\begin{aligned} \frac{\boldsymbol{v}_g' - \boldsymbol{v}_g}{\Delta t} &= -\sum_n \epsilon_n \frac{\boldsymbol{v}_g' - \boldsymbol{v}_{d,n}'}{t_{s,n}}, \qquad \epsilon_n \equiv \frac{\rho_{d,n}}{\rho_g}, \\ \frac{\boldsymbol{v}_{d,n}' - \boldsymbol{v}_{d,n}}{\Delta t} &= \sum_n \frac{\boldsymbol{v}_g' - \boldsymbol{v}_{d,n}'}{t_{s,n}}, \end{aligned} \tag{43}$$

where a prime superscript denotes velocities after a timestep Δt. Expressed in matrix form and defining $b_n \equiv \Delta t / t_{s,n}$, we get:

$$\begin{bmatrix} 1 + \sum_n \epsilon_n b_n & -\epsilon_1 b_1 & -\epsilon_2 b_2 & \cdots & -\epsilon_N b_N \\ -b_1 & 1 + b_1 & 0 & \cdots & 0 \\ -b_2 & 0 & 1 + b_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -b_N & 0 & 0 & \cdots & 1 + b_N \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_g' \\ \boldsymbol{v}_{d,1}' \\ \boldsymbol{v}_{d,2}' \\ \vdots \\ \boldsymbol{v}_{d,N}' \end{bmatrix} = \begin{bmatrix} \boldsymbol{v}_g \\ \boldsymbol{v}_{d,1} \\ \boldsymbol{v}_{d,2} \\ \vdots \\ \boldsymbol{v}_{d,N} \end{bmatrix}. \tag{44}$$

The matrix can be inverted analytically following the method documented in Krapp and Benítez-Llambay (2020) (implemented into PLUTO for one dust fluid in Sect. 5.3). However, for our multispecies implementation, the velocity update is modified slightly as follows:

$$\boldsymbol{v}_g' = \boldsymbol{v}_g + \frac{\sum_n a_n(\boldsymbol{v}_{d,n} - \boldsymbol{v}_g)}{1 + \sum_n a_n}, \qquad a_n \equiv \frac{\epsilon_n b_n}{1 + b_n}, \tag{45a}$$

$$\boldsymbol{v}_{d,n}' = \boldsymbol{v}_{d,n} - \frac{b_n}{1 + b_n}\left(\boldsymbol{v}_{d,n} - \boldsymbol{v}_g'\right). \tag{45b}$$

With this formulation, the velocity updates are expressed entirely in terms of relative velocities. As a result, the scheme is independent of any background flow, making it directly compatible with a rotating frame, FARGO orbital advection, or shearing boxes. If dust back-reaction is disabled, Eq. (45a) is omitted, as $\boldsymbol{v}_g' = \boldsymbol{v}_g$ in the limit that all $\epsilon_n \to 0$.

We note that, while the dust–gas drag term is treated implicitly as described above, the advective fluxes for all dust species are integrated explicitly with the regular time-stepping scheme chosen by the user. In that sense, the complete implementation of dust advection and drag formally follows an "implicit–explicit" (IMEX) scheme. For the sake of clarity, we will henceforth refer to this scheme as "implicit", to contrast this with the explicit dust–gas drag implementation.

We then apply the implicit drag operator in an operator-split fashion at the end of every Runge–Kutta stage rather than only after the last stage. A flowchart of the timestepping scheme is provided in Appendix B.1. We found that this approach yields 2nd-order accuracy or higher for the problems shown here, as outlined in the convergence tests in Section 5.5.2. Unless otherwise mentioned, the standard test setups below use an isothermal equation of state, the LeVeque (2004) dust solver, a piecewise-linear (PLM) reconstruction scheme, and second-order Runge–Kutta (RK2) timestepping. For the timestep, we set a Courant–Friedrichs–Levy (CFL) value of 0.4.

5.5.2 *Radial drift test*

We generalize the radial drift test performed in Sect. 5.4.1 to include multiple dust species. For a vertically integrated, axisymmetric accreting disk at cylindrical radius R from the central star, the radial dust velocity in the absence of dust feedback or diffusion can be written in analytical form given by Takeuchi and Lin (2002),

$$v_{R,d,n} = \frac{St_n \eta R \Omega_K + v_{R,g}}{1 + St_n^2}, \qquad \eta = \frac{d\log P}{d\log R} h^2, \quad P = \Sigma c_s^2, \tag{46}$$

where $v_{R,d,n}$ and St_n are the radial velocity and the Stokes number of the dust species n, respectively. Using the disk scale height $H = c_s/\Omega_K$, the disk aspect ratio is defined as $h = H/R$.

We use the same initial conditions as described in Sect. 5.4.1 with an inviscid, adiabatic disk in cylindrical polar coordinates, with 1024 cells logarithmically spaced between radii $R \in [0.4, 2.5]\, R_0$. We use power laws for the surface density and temperature and a constant disk aspect ratio of $h_0 = 0.05$. We pick the HLLC Riemann solver for the gas Toro et al. (1994). We prescribe 15 dust species sampled between Stokes numbers $St_n \in [10^{-3}, 3]$, excluding the four species with $St < 0.1$ when using the explicit solver because the dust–gas drag operator becomes stiff and significantly reduces the timestep. Nevertheless, as seen in Sect. 5.4.1, one could lower the CFL criterion to lower the timestep and get reasonable solutions for such values of St.

Figure 27 shows the radial dust velocity normalized to the Keplerian velocity v_K. We see that results from both the explicit (indicated by the marker '$\times$' in blue) and the implicit (indicated by the marker '$+$' in orange) drag solvers show very good agreement with the analytical solution by Takeuchi and Lin (2002) denoted by the dashed black line. The root mean square errors (RMSE) are plotted in the bottom panel, defined by:

$$\text{RMSE} = \sqrt{\frac{1}{N_x} \sum_{\text{cells}} \left(v_{R,d,n} - v_{R,d,n}^{\text{ref}} \right)^2} \tag{47}$$

where N_x is the number of cells and $v_{R,d,n}^{\text{ref}}$ is the reference analytical solution following Eq. (46). We find that RMSE values are of order $\lesssim 10^{-4}$ for $St \lesssim 0.3$, remaining below 0.1%

even for St $\gtrsim 1$. With this, we conclude that the multi-fluid implementation of the drag forces acting on the dust works as intended. In the next sections, we test the full dust–gas coupling and dust feedback.

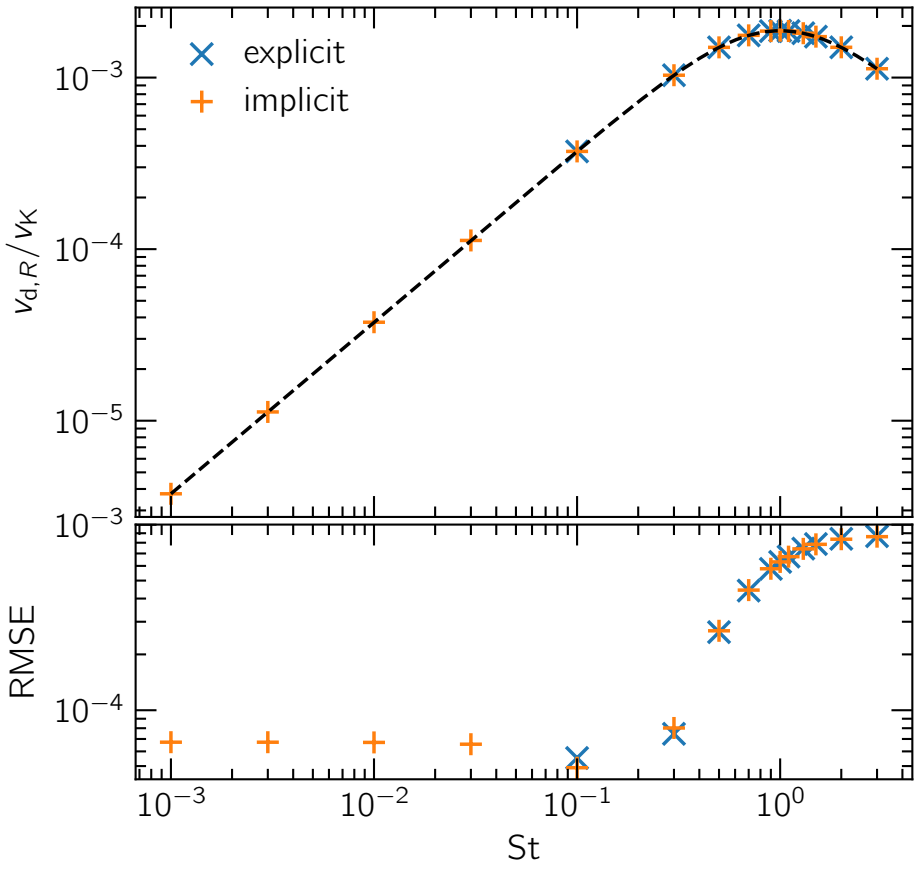


Figure 27: Comparison of radial velocity values of the dust component between the explicit (marker '×' in blue) and the implicit (marker '+' in orange) drag solvers against analytical profile (black dashed line) from Takeuchi and Lin (2002) for different Stokes numbers, measured by averaging $u_{R,d}/u_K$ between 1–2 R_0. For St $\lesssim 0.001$, the explicit method is only stable for smaller time steps, but here, we omit those. The bottom panel shows the relative mean square errors between each result and its theoretical value. The explicit solver performs better overall, while both models start to show higher errors for St $\gtrsim 1$ but give reasonable answers up to St ~ 30.

We also measure the convergence rate of both explicit and implicit drag solvers for the radial drift test by executing the setup for different radial resolutions $N_x \in [16, 256]$. In Fig. 28 we show the spatial convergence of both solvers, for St $= [0.1, 1]$ (only the case of St $= 0.1$ is shown) by plotting the RMSE against N_x, demonstrating clearly that both the implicit and the explicit solvers show second-order convergence in space.

5.5.3 *Linear* `DUSTYWAVE` *test for multiple species*

To test the dust–gas coupling with the multi-fluid solver, we extend the single-species linear `DUSTYWAVE` test in Sect. 5.4.2 to include two additional dust fluids, setting St $= \{0.001, 1, 10\}$ to

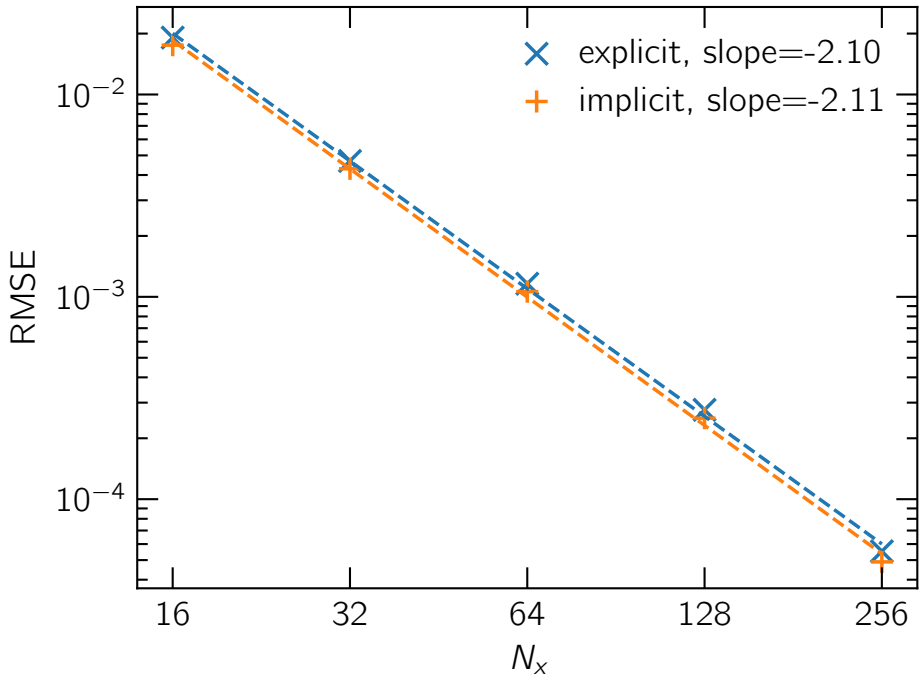


Figure 28: RMS error plotted against the grid resolution. The blue crosses correspond to the explicit scheme, and the orange plus markers, the implicit scheme. The colored dashed lines are the respective fits, and the slopes are -2 for both schemes. As in Benítez-Llambay et al. (2019), we do not fix the timestep; hence, both the spatial and temporal errors add to one another, but the implicit solver still achieves second-order convergence.

sample the strong, moderate, and weak coupling regimes. For N dust species, and a perturbation of the form $\delta p = \delta\hat{p}e^{ikx-\omega t}$, one can write the dispersion relation following Benítez-Llambay et al. (2019) as

$$\omega^2\left(1+\sum_{n=1}^{N}\frac{\epsilon_n}{1-\omega t_{s,n}}\right)+\omega_s^2=0, \tag{48}$$

where k is a real wavenumber, $t_{s,n}$ is the stopping time for a given species n (in dimensionless units, such that it also represents the Stokes number St_n for the species), ω is the angular frequency, and $\omega_s = kc_s$. This dispersion relation yields a polynomial of degree $N+1$ for N species, and can be solved to get the required complex roots for the oscillation and damping rate of the waves. We have provided an `ipython` notebook[1] that can be used to compute this polynomial and obtain the initial conditions for the `DUSTYWAVE` test along with this manuscript. We rewrite the eigenvectors from Sect. 5.4.2,

$$\frac{\delta\hat{v}_g}{c_s}=-i\frac{\omega}{\omega_s}\frac{\delta\hat{\rho}_g}{\rho_g^0}, \tag{49a}$$

$$\frac{\delta\hat{v}_{d,n}}{c_s}=-i\frac{\omega}{\omega_s}\frac{1}{(1-\omega t_{s,n})}\frac{\delta\hat{\rho}_g}{\rho_g^0}, \tag{49b}$$

1 https://github.com/prakrutisudarshan/dustywave-solution

St	$\delta\hat{u}_g/c_s$	$\delta\hat{\rho}_d/\rho_d^0$	$\delta\hat{u}_d/c_s$
0.001	$-0.7486 - 0.03667i$	$0.7468 - 0.003514i$	$-0.7489 - 0.03315i$
1	...	$0.02530 - 0.1546i$	$-0.03295 + 0.1538i$
10	...	$-4.397 \times 10^{-4} - 0.01586i$	$-3.381 \times 10^{-4} + 0.01592i$

Table 3: Initial conditions for the DUSTYWAVE test, corresponding to eigenvectors for perturbations in the dust density, gas and dust velocities for $St \in [0.001, 1, 10]$. Perturbations in gas density are always assumed to be $\delta\hat{\rho}_g/\rho_g^0 = 1$. These values are used in Eq. (50).

$$\frac{\delta\hat{\rho}_{d,n}}{\rho_{d,n}^0} = \frac{1}{(1 - \omega t_{s,n})} \frac{\delta\hat{\rho}_g}{\rho_g^0}. \tag{49c}$$

Of the obtained roots, we choose one of the oscillatory modes to initialize the numerical setup. Like in Benítez-Llambay et al. (2019) and Sect. 5.4.2, we use a perturbation of the form,

$$\delta p = A\ [\Re(\delta\hat{p})\cos(kx) - \Im(\delta\hat{p})\sin(kx)]. \tag{50}$$

We initialize the background dust density to $\rho_{d,n}^0 = 0.747$ for three species, which adds up to 2.24, the value used for the single-dust fluid test case in Sect. 5.4.2. We set $A = 10^{-4}$, the background gas density to $\rho_g^0 = 1$, and initialize the dust velocities to zeroes. We use a Cartesian domain with $x \in [0, 1]$ using 1024 cells, $c_s = 1$, and $k = 2\pi$. The eigenvector solutions for different St are given in Table 3.

Figure 29 shows the propagation of the perturbations in dust density, the density residuals $(\delta\hat{\rho}_{d,n} - \delta\hat{\rho}_{d,n}^{ref})$ with $\delta\hat{\rho}_{d,n}^{ref}$ following the analytical solution in Eq. (49), and the velocities in time at an arbitrary location $x = 0.005$. Solid lines denote the analytical solutions for the different Stokes numbers St_i. The data from the simulation are indicated by the markers '×' for the model with the explicit solver and '+' for the implicit one, with different colors referring to different St. We show that all quantities follow the analytical solution with great accuracy, both by visual inspection and after comparing the density residuals, indicating that the dust–gas coupling with the multi-fluid implementation behaves correctly. As we have already shown in Fig. 28 for the radial drift test and in Sect. 5.4.2 for the single-fluid DUSTYWAVE test that the drag solvers converge with second-order accuracy, we omit a convergence test for this problem.

5.5.4 *Non-linear benchmarks of the streaming instability*

The streaming instability (SI, Johansen et al., 2007; Lim et al., 2024; Youdin and Goodman, 2005b) is a hydrodynamic dust–gas drag instability that is a widely accepted pathway to planetesimal formation in the midplane dust layers of protoplanetary disks. Linear and non-linear regimes of the streaming instability in the unstratified, compressible gas case (also known as the AB and BA problems, first constructed by Johansen and Youdin, 2007; Youdin and Johansen, 2007) are widely used by numerical codes to compare their drag solvers (Bai and Stone, 2010b; Benítez-Llambay et al., 2019; Huang and Bai, 2022; Mignone et al., 2019). Recently, Baronett et al. (2026) provided a large-scale code comparison of the BA configuration for seven different codes, including particle and fluid formulations for dust, showing that all

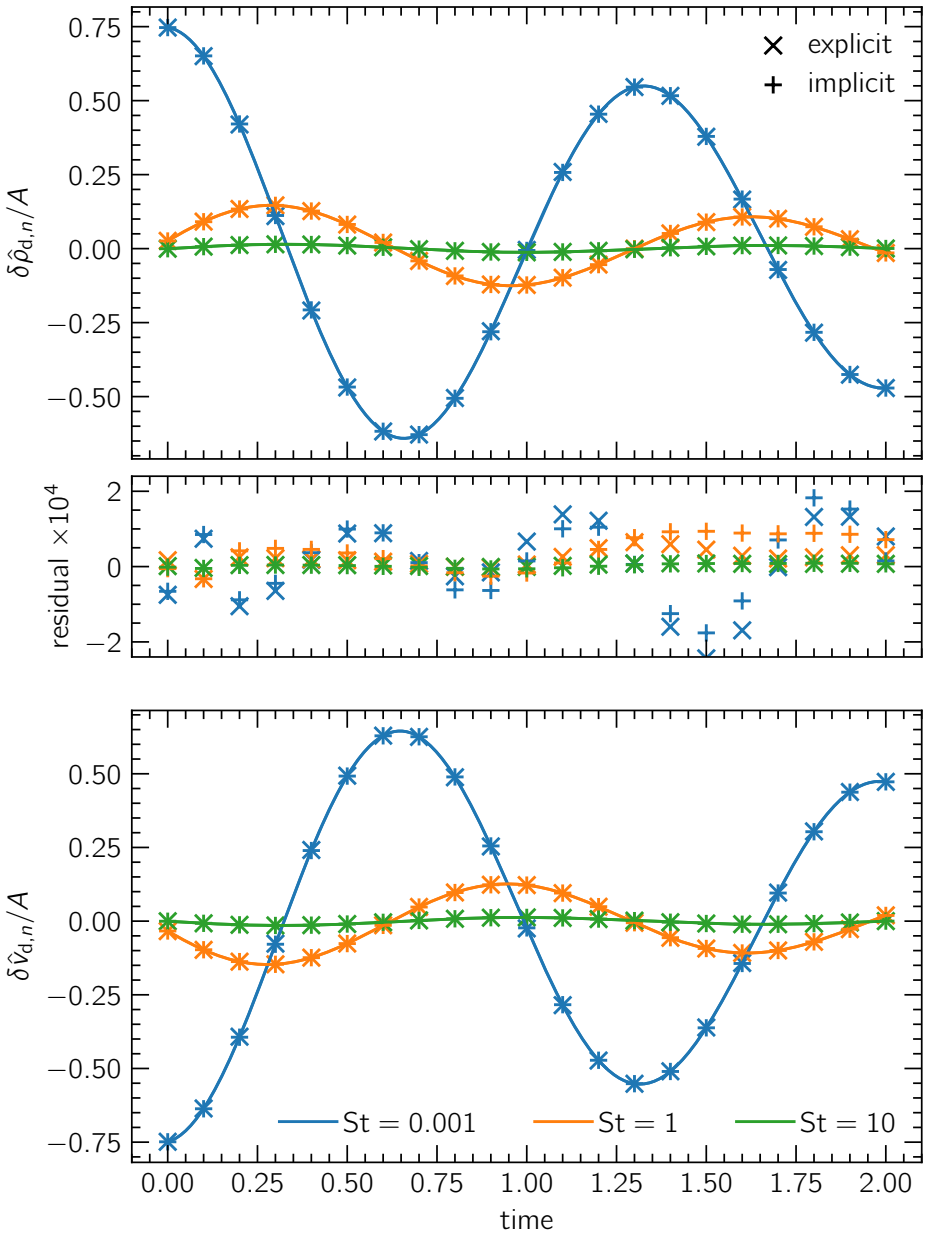


Figure 29: Perturbations in dust density, the density residuals, and velocity (i.e., the eigenvectors described in Eq. (49)) normalized to the amplitude A for the different Stokes numbers outlined in Table 3 using both explicit and implicit dust fluid solvers. The analytical solution is described by the solid lines in blue, orange, and green for $\mathrm{St} = 0.001, 1, 10$ respectively. The data from the simulation is marked by the '$\times$' for the explicit and '$+$' for the implicit solver. The values match the analytical solution with great accuracy.

codes achieve good convergence at high enough resolution. Following this, we benchmark the non-linear streaming instability AB and BA problems using previously existing codes with full resolution analysis.

In the shearing box approximation, one considers a small patch of the disk at a certain radius R_0 corotating at the local orbital frequency Ω_0. The continuity and momentum equations for the gas and dust species can be written following Hawley et al. (1995) and Mignone et al. (2019), which for our case are outlined in Appendix B.2. In the absence of vertical gravity (i.e., in the unstratified case), with constant background densities for the dust and gas species, there exists an equilibrium solution following the NSH (Nakagawa–Sekiya–Hayashi) solutions (Nakagawa et al., 1986), which was later generalized to include multiple dust species in Benítez-Llambay et al. (2019). By defining the quantities

$$A_N = k^2 \sum_{n=1}^{N} \frac{\epsilon_n t_{s,n}}{1 + k^2 t_{s,n}^2} \quad \text{and} \quad B_N = 1 + \sum_{n=1}^{N} \frac{\epsilon_n}{1 + k^2 t_{s,n}^2}, \tag{51}$$

the steady-states for the gas velocities are given by

$$v^0_{g,x} = A_N \chi_0 (A_N^2 + k^2 B_N^2)^{-1}, \tag{52a}$$

$$v^0_{g,y} = -q\Omega_0 x - \frac{k^2}{2} B_N \chi_0 (A_N^2 + k^2 B_N^2)^{-1}, \tag{52b}$$

where $k^2 = \kappa^2 \Omega_0^{-2}$ and κ is the epicyclic frequency. The dust velocities can now be expressed in terms of the gas velocities, following, for species n,

$$v^0_{d,n,x} = \frac{v^0_{g,x} + 2 t_{s,n} (v^0_{g,y} + q\Omega_0 x)}{1 + k^2 t_{s,n}^2} \tag{53a}$$

$$v^0_{d,n,y} = -q\Omega_0 x + \frac{(v^0_{g,y} + q\Omega_0 x) - (2 - q) t_{s,n} v^0_{g,x}}{1 + k^2 t_{s,n}^2}. \tag{53b}$$

The vertical velocities are $v^0_{g,z} = v^0_{d,n,z} = 0$. We use these as our initial condition for the non-linear SI setup. On top of these solutions, the gas and dust velocities are initialized with a uniform random noise in the interval $[-0.01 c_{s,o}, 0.01 c_{s,o}]$, where $c_{s,o}$ is the sound speed in the box.

The numerical setups for the non-linear AB and BA tests are initialized with a shearing box in the x–z directions using the conditions outlined in Table 4, with any additional details in Appendix B.2 where the equations are written down. We use the Roe Riemann solver for the gas, an implicit treatment for dust–gas drag including backreaction, the MC_LIM flux limiter, and a CFL criterion of 0.3.

Figure 30 shows snapshots of the dust-to-gas ratios for both the BA (top panel) and AB (bottom panel) problems at different grid resolutions: $128^2, 256^2, 512^2, 1024^2$, and 2048^2. The snapshots correspond to an orbital time of 100 Ω_0^{-1} for the BA test and 40 Ω_0^{-1} for the AB test. On qualitative comparison to previous studies, both the BA and AB problems exhibit the expected characteristic features. The BA test shows the extended, zig-zag dust filaments, with more intricate features being resolved as we go to higher grid resolutions, but the overall characteristic imprint remaining the same (Baronett et al., 2026). The AB test, in contrast,

Problem	t_s	ϵ	$L_{x,z}/H_g$
BA	1.0	0.2	2.0
AB	0.1	1.0	0.1

Table 4: Initial conditions for the BA and the AB problems. Here, t_s is the stopping time, ϵ the dust-to-gas ratio, and $L_{x,z}/H_g$ the box size in $x-z$ directions normalized to the gas scale height.

shows the characteristic web-like dust filaments and cavities that trickle down to smaller scales as the resolution increases.

Figure 31 shows the time series of the maximum dust density $\max(\rho_d)$ normalized to the initial gas density $\rho_{g,o}$ for both the BA and AB runs, respectively. We see that for the BA runs (left panel), the models reach saturation by about 20 Ω_0^{-1}. While we see some convergence in resolution for the BA problem, the maximum dust density increases as we go to higher resolutions, exceeding 100 at 2048^2. The obtained values are consistent with the results in Baronett et al. (2026) for PLUTO and other codes, except for differences due to the initial noise distribution. We refer the reader to Appendix B.3 for a comparison of two BA setups with the same concentrations but divided between 1 and 10 dust species, respectively, for the resolution 1024^2, where we find that in the absence of crossing trajectories in the fluid approximation, the evolution looks the same.

For the run AB, we see that the SI reaches saturation much faster (within 10 Ω_0^{-1}) except for the lowest-resolution run 128^2, which seems to take longer (up to 27 Ω_0^{-1}). This is consistent with what is reported in previous works like Benítez-Llambay et al. (2019) and Huang and Bai (2022). We see a steady increase in the maximal dust densities with grid resolution without signs of convergence, consistent with trends reported in previous fluid works. While Baronett et al. (2026) explored the comparison of the unstratified BA test, a comparison of the AB test between particle and fluid codes could prove very useful to see whether the particle codes reaching higher maximum dust densities is an inherent property of the method or is specific to that problem.

We also plot the cumulative dust density distribution functions (CDF) for the dust densities at the saturated state averaged over time $\overline{\rho_d}$ ($t > 20\Omega_0^{-1}$) to understand the convergence properties of the two problems in Fig. 32. We use the same definition and follow the notations as in Baronett et al. (2026), where densities are measured at cell centers, as in Benítez-Llambay et al. (2019) and Huang and Bai (2022) (in contrast to measuring an effective particle count as in Bai and Stone, 2010b; Johansen and Youdin, 2007). For the BA test, we see convergence for lower dust densities, but this breaks at higher dust densities, see $P(< 10^{-3})$, as also reported in Fig. 31. The lack of convergence for the AB problem, as reported above, is also evident in the CDF plot throughout, consistent with the findings of other fluid codes (Benítez-Llambay et al., 2019; Huang and Bai, 2022), warranting a comparison with particle methods.

5.6 DUST DIFFUSION

A prescription for the diffusion of dust is very helpful, especially for problems geared towards protoplanetary/accretion disks. Protoplanetary disks are believed to exhibit turbulence, and depending upon the region of the disk, various magneto- and hydro-dynamic instabilities are proposed as sources of turbulence. However, since the exact source of the tur-

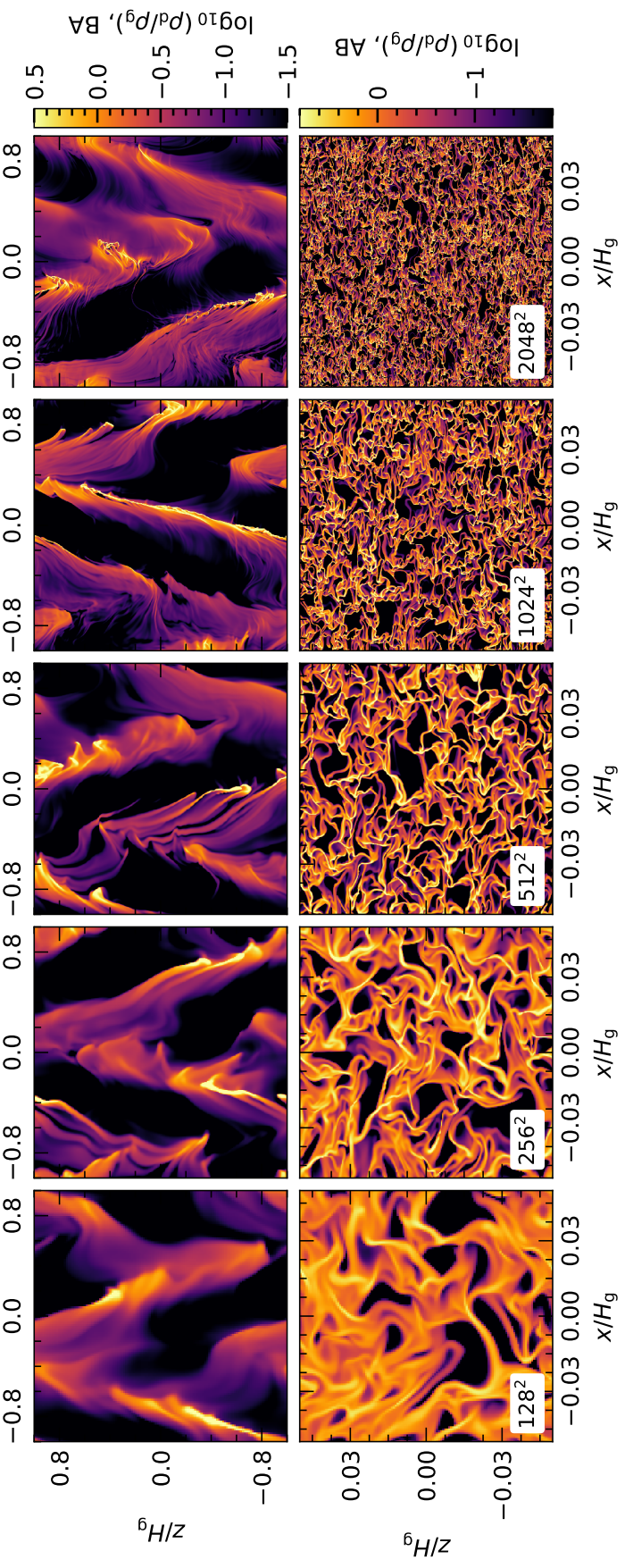


Figure 30: Top row: 2D plots of the dust-to-gas mass ratio $\epsilon = \rho_g/\rho_d$ in the logarithmic scale for the BA problem for different grid resolutions starting from 128^2 to 2048^2 plotted at the saturation states of SI (at 100T). Bottom row: Similar plot for the AB problem but at the saturation state after 40T. The x- and z- axes are normalized to the gas scale height H_g. For details as to the dust-to-gas ratio, the box size, and the Stokes number for the two problems, we refer the reader to Table 4.

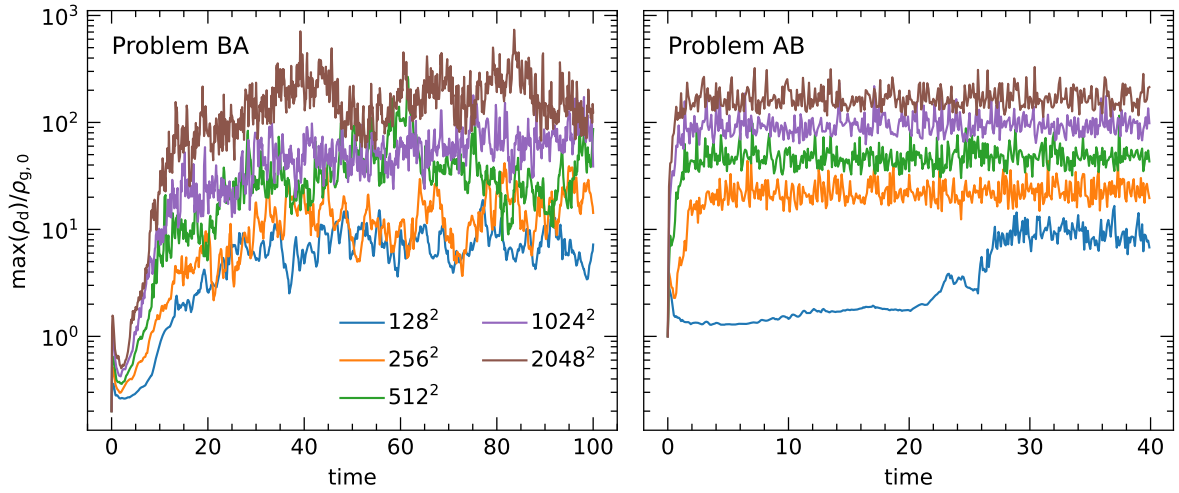


Figure 31: Left: Time series of the maximum dust density normalized to the initial gas density $\max(\rho_d)/\rho_{g,0}$ for different resolutions for the BA problem. We see some convergence for the resolutions 512^2 and 1024^2, but for the highest resolution 2048^2 the clumping is higher. Right: Same plot as the left but for the AB problem; we see the lack of convergence with increasing resolution for the dust fluid method.

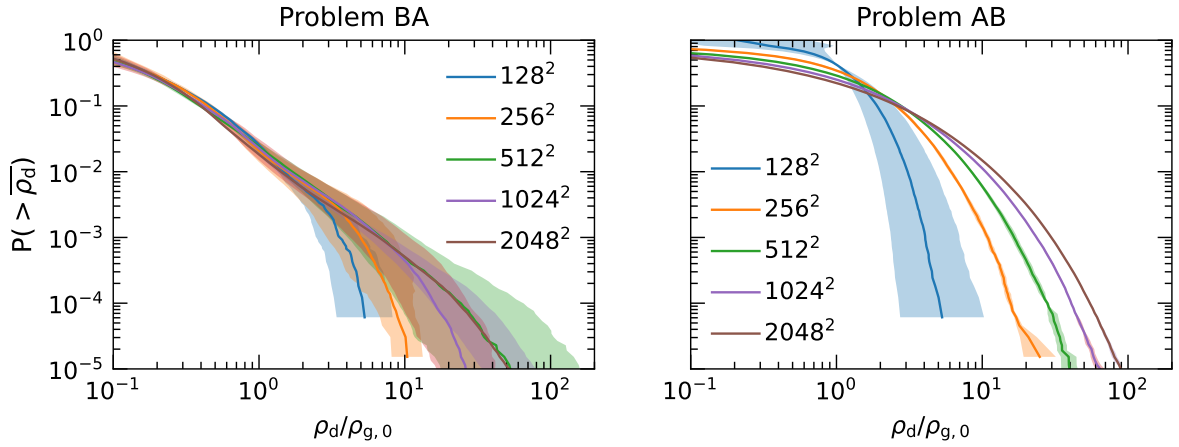


Figure 32: Cumulative distribution functions (CDF) for the dust densities at the saturated state averaged over time $\overline{\rho_d}$ ($t > 20\Omega_0^{-1}$) for different grid resolutions for the BA (left panel) and AB problems (right panel). The shaded area corresponds to 1σ time variability throughout the saturated state. The BA problem shows convergence for smaller dust densities, but breaks for higher densities, as also seen in the plot above in Fig. 31. Similarly, the AB problem does not converge as well.

bulence is not well established, a generalized model for dust diffusion is used, similar to the α disk model for gas turbulence by Shakura and Sunyaev (1973). In particle codes, this can easily be added by adding random kicks to the particles (e.g., Ormel and Liu, 2018). In the Eulerian approach, a gradient diffusion method is generally used. The underlying assumption is that the flux due to turbulent diffusion is proportional to the gradient of the dust density or concentration (depending on the definition). The continuity equation then becomes an advection–diffusion equation. Modern codes like `FARGO3D`, `cuDisc` (Robinson et al., 2024), and `Athena++` all provide a gradient diffusion-based model. However, based on the exact underlying assumptions employed, there are some differences in the equations these codes implement. The effect of these changes is unknown, and along with the equations and the numerical implementation of dust diffusion in `PLUTO`, we address some of the discrepancies in the following section.

5.6.1 *The equations*

Assuming that the diffused quantity is the dust-to-gas ratio, also called the concentration $\epsilon_n = \rho_{d,n}/\rho_g$, a diffusion flux $\boldsymbol{F}_{\mathrm{dif},n}$ for species n can be written based on Fick's law as,

$$\boldsymbol{F}_{\mathrm{dif},n} = -\rho_g D_n \nabla \epsilon_n. \tag{54}$$

Here, $D_n = \nu/\mathrm{Sc}$ is the diffusivity for dust species n, related through the Schmidt number $\mathrm{Sc} = 1 + \mathrm{St}^2$ (Youdin and Lithwick, 2007) to the gas kinematic viscosity ν. For this reason, we include dust diffusion only when gas viscosity is enabled, which in turn appears in Eq. (40b) through the viscous stress tensor $\bar{\sigma}_{ij}$ as

$$S_g = \nabla \cdot \bar{\sigma}_{ij}, \quad \bar{\sigma}_{ij} = \rho_g \nu \left(\frac{\partial v_{g,i}}{\partial x_j} + \frac{\partial v_{g,j}}{\partial x_i} + 2\delta_{ij} \nabla \cdot \boldsymbol{v}_g \right). \tag{55}$$

The dust continuity and momentum equations in Eq. (40c) and (40d) including dust diffusion then read

$$\frac{\partial \rho_{d,n}}{\partial t} + \nabla \cdot \left(\rho_{d,n} \boldsymbol{v}_{d,n} + \boldsymbol{F}_{\mathrm{dif},n} \right) = 0, \tag{56a}$$

$$\begin{aligned} \frac{\partial}{\partial t} \left(\rho_{d,n} \boldsymbol{v}_{d,n} \right) + \nabla \cdot \left(\rho_{d,n} \boldsymbol{v}_{d,n} \otimes \boldsymbol{v}_{d,n} + \boldsymbol{v}_{d,n} \otimes \boldsymbol{F}_{\mathrm{dif},n} \right) \\ = \frac{\rho_{d,n}}{t_{s,n}} \left(\boldsymbol{v}_g - \boldsymbol{v}_{d,n} \right) + S_d. \end{aligned} \tag{56b}$$

We note that, formally, if Eq. (56b) were to be derived from its advective form (i.e., evolving $\boldsymbol{v}_{d,n}$ rather than $\rho_{d,n}\boldsymbol{v}_{d,n}$) alongside Eq. (56a) it would instead read

$$\begin{aligned} \frac{\partial}{\partial t} \left(\rho_{d,n} \boldsymbol{v}_{d,n} \right) + \nabla \cdot \left(\rho_{d,n} \boldsymbol{v}_{d,n} \otimes \boldsymbol{v}_{d,n} \right) \\ = \frac{\rho_{d,n}}{t_{s,n}} \left(\boldsymbol{v}_g - \boldsymbol{v}_{d,n} \right) + S_d - \boldsymbol{v}_{d,n} \nabla \cdot \boldsymbol{F}_{\mathrm{dif},n}. \end{aligned} \tag{57}$$

However, following Binkert (2023), we add a term of the form $-\left(\boldsymbol{F}_{\mathrm{dif},n} \cdot \nabla\right) \boldsymbol{v}_{d,n}$ to the right-hand side of Eq. (57), finally arriving at Eq. (56b), which has the additional property of conserving the global momentum of each dust species n. For an extended discussion regarding this topic, we refer the reader to Binkert (2023).

In FARGO3D, Weber et al. (2019) state that they only model the additional term in Eq. (56a) and rely on the operator-split nature of the code to account for the corresponding momentum diffusion flux, essentially recovering Eq. (57) rather than Eq. (56b). Since PLUTO is a dimensionally unsplit code, we account for this term explicitly by implementing Eq. (56b), similar to cuDisc. In doing so, we also include geometric source terms that emerge in curvilinear geometries on the right-hand side of Eq. (56b). The implementation is further compatible with the super-timestepping (STS) scheme (Alexiades et al., 1996) available in PLUTO, where the above terms are integrated in an operator-split fashion, by choosing the optimal number of substeps to maintain stability. This offers a considerable speed-up compared to the standard explicit treatment for very diffusive problems where the timestep would otherwise be dominated by the parabolic contribution ($\propto \Delta x^2/D$) rather than its hyperbolic counterpart ($\propto \Delta x/v_x$).

Huang and Bai (2022) perform Reynolds mean-field averaging and include the advection of the diffusive flux and the time evolution of the diffusive velocity, arguing that the latter is required for Galilean invariance. They write the momentum equation (excluding the drag term) as follows:

$$\frac{\partial}{\partial t}\left[\rho_{d,n}\left(\boldsymbol{v}_{d,n}+\boldsymbol{v}_{d,dif,n}\right)\right]+\nabla\cdot\left(\rho_{d,n}\boldsymbol{v}_{d,n}\otimes\boldsymbol{v}_{d,n}+\boldsymbol{\Pi}_{dif,n}\right)=S_{d,n}, \tag{58}$$

with $\boldsymbol{\Pi}_{dif,n}=\boldsymbol{v}_{d,n}\otimes\boldsymbol{F}_{dif,n}+\boldsymbol{F}_{dif,n}\otimes\boldsymbol{v}_{d,n}$ and $\boldsymbol{v}_{d,dif,n}=\boldsymbol{F}_{dif,n}/\rho_{d,n}$.

In the following section, we test the implementation of Eqs. (56) that we have included in PLUTO and finally also test the Athena++ version (i.e., Eq. (58)) with the Gaussian test in the discussion section 5.7.1.

5.6.2 *Steady state test: settling–vertical equilibrium*

In this section, we describe a steady-state settling–mixing test in 1D Cartesian geometry. Solving for the z- component of the momentum equation (Eq. (56b)), along with a gravity source term from the star included $\nabla\phi_\star=-GMz/R^2\,\hat{z}$ yields an expression for the settling velocity of the dust $v_{zd,\,set}=-\Omega_K St z$. Substituting this back into Eq. (56a) one can write

$$\rho_{d,z}v_{d,set}+F_{dif,z}=0. \tag{59}$$

With $F_{dif,z}=-\rho_g D\;\partial\epsilon/\partial z$, $D=\nu/(1+St^2)$, and $\nu=\alpha c_s H_g$ (Shakura and Sunyaev, 1973), solving for the dust-to-gas mass ratio gives us the expression

$$\epsilon=\epsilon_0\exp\left(-\frac{z^2}{2H_d^2}\right) \tag{60}$$

where H_d is the dust scale height and here, $H_d=\sqrt{\alpha H_g^2/(1+St^2)St}$. We set up a 1D Cartesian simulation with $z\in[0,4]$ H_g uniformly spaced using 400 cells, a constant viscosity parameter $\alpha=10^{-2}$, and three dust species. We initialize the dust density with a Gaussian profile, and use reflective boundaries. We use the HLLC Riemann solver for the gas, an explicit treatment for dust–gas drag, and an isothermal equation of state. Dust feedback is turned off.

Figure 33 shows the 1D profiles of the dust-to-gas ratios normalized to the initial dust-to-gas ratios for the three dust fluids in blue, green, and orange in increasing order of coupling $St=\{0.1,0.01,0.001\}$. The dotted black lines correspond to the analytical solution for the

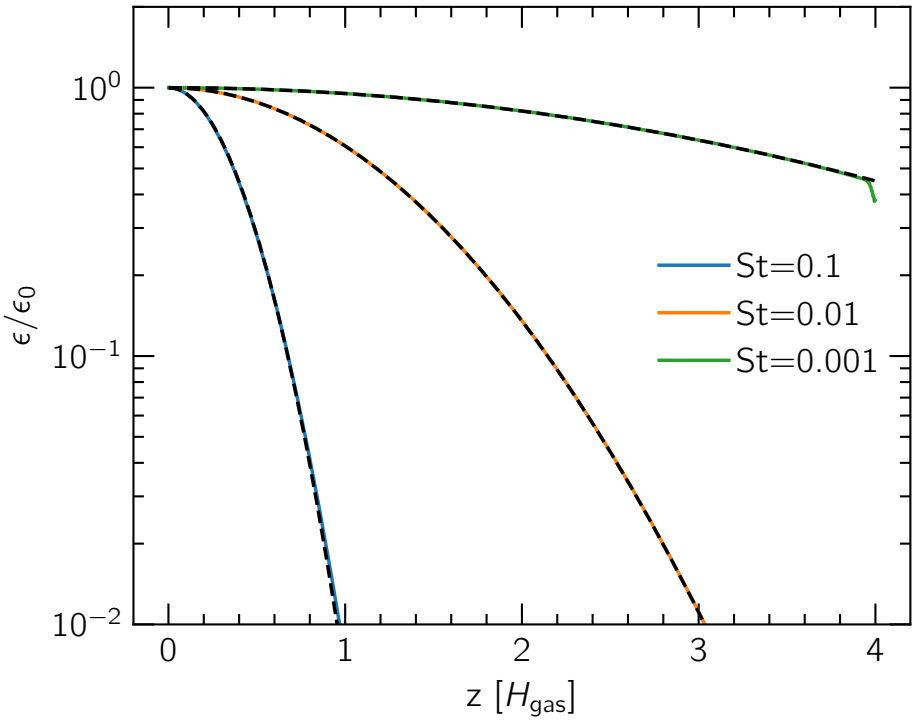


Figure 33: Dust-to-gas mass ratio ϵ normalized to the initial midplane dust-to-gas mass ratio ϵ_0 plotted against the z- coordinate in units of the gas density H_{gas} for the vertical settling test problem. The different colors correspond to the simulation data for the different Stokes numbers St $= 0.1, 0.01$ and 0.001. The dashed black lines correspond to the respective analytical solution for the Stokes number.

corresponding dust fluid given by Eq. (60). The z- coordinate is normalized to the gas scale height, and we model up to 4 gas scale heights. We show very good agreement with the analytical solution for the dust fluids, confirming that our implementation of dust diffusion in Cartesian geometry works as intended. For the case of St $= 0.1$, we recover the analytical solution with very good accuracy up to a height of 6 dust scale heights, beyond which we reach the density floor.

5.6.3 *1D Gaussian test in Cartesian coordinates*

As the previous test only verifies that the flow converges to the correct steady state, we further set up a time-dependent problem with a moving, diffusing 1D Gaussian bump.

We initialize two dust fluids with St $= \{0.1, 0.3\}$ alongside a constant gas background, with two Gaussian profiles for the dust-to-gas mass ratios ϵ_n, given by

$$\epsilon_n(x, 0) = \epsilon_{n,0} \exp\left(-\frac{(x_n - x_{n,0})^2}{2\sigma_n^2}\right) \tag{61}$$

centered at $x_{n,0}$ with an initial width of σ_n and amplitude of $\epsilon_{n,0}$. The respective Gaussian profiles have widths $\sigma_n = \{0.02, 0.05\}$ and amplitudes $\epsilon_{n,0} = \{0.1, 0.6\}$ and both are centered at $x_{n,0} = 0.5$. We set a constant gas diffusivity $\nu = 10^{-3}$ and impose a background velocity

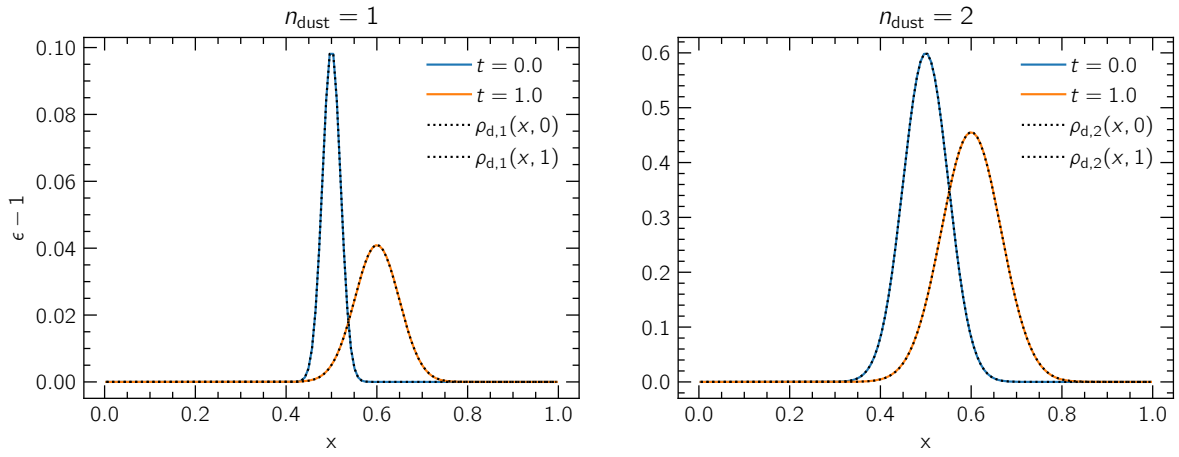


Figure 34: 1D profiles for the moving Gaussian test for the two dust species (left and right). Plotted are relative dust densities against the gas densities and the x-coordinate. The blue line corresponds to the initial state of the Gaussian, and the orange line after time t=1. The Gaussian is kicked with a velocity $v_x =$ in the x-direction, and we see that the Gaussian moves and diffuses in time. The dashed black lines correspond to the analytical solution at both times, and the data matches theory well.

$v_0 = 0.1$ on the gas and both dust species. The corresponding analytical solution in time for the moving and diffusing Gaussian bumps is given by,

$$\epsilon_n(x, t) = \frac{\epsilon_{n,0}\sigma_n}{\sqrt{\sigma_n^2 + 2D_n t}} \exp\left[-\frac{(x - x_{n,0} - v_0 t)^2}{2(\sigma_n^2 + 2D_n t)}\right]. \tag{62}$$

In our numerical setup, we use a grid of $x \in [0, 1]$ with 128 uniformly spaced cells and periodic boundaries, the implicit drag solver, and the explicit diffusion solver.

Figure 34 shows the 1D profiles of the Gaussian bump in ϵ_n for the two fluids in the left and right panels, respectively. The initial profiles are indicated in blue, and at time $t = 1$ in orange. The dashed black lines are the respective analytical solutions for the two different Gaussian dust bumps. The agreement between the numerical and analytical solutions indicates that the diffusion implementation captures time-dependent evolution correctly. These models include both concentration and momentum diffusion, and we found that the correction in the momentum equation is essential for accurately reproducing time-dependent solutions. Neglecting this correction does not recover the correct solution, an effect that would not be apparent in a diffusion-only Gaussian test. We include a comparison test without momentum diffusion in Appendix B.4.

We then test the convergence rate of the combined drag and diffusion solvers by running this setup for different resolutions N_x with $v_0 = 0$ and computing the RMSE, defined for this problem as

$$\text{RMSE} = \sqrt{\frac{1}{N_x} \sum_{\text{cells}} \left(\epsilon_n(x, t) - \epsilon_n^{\text{ref}}(x, t)\right)^2} \tag{63}$$

where $\epsilon_n^{\text{ref}}(x, t)$ is the reference analytical solution following Eq. (62). Figure 35 shows the RMSE values as a function of N_x for the two dust species in blue and orange, with both explicit and implicit drag solvers (indicated by markers '×' and '+' respectively, with the

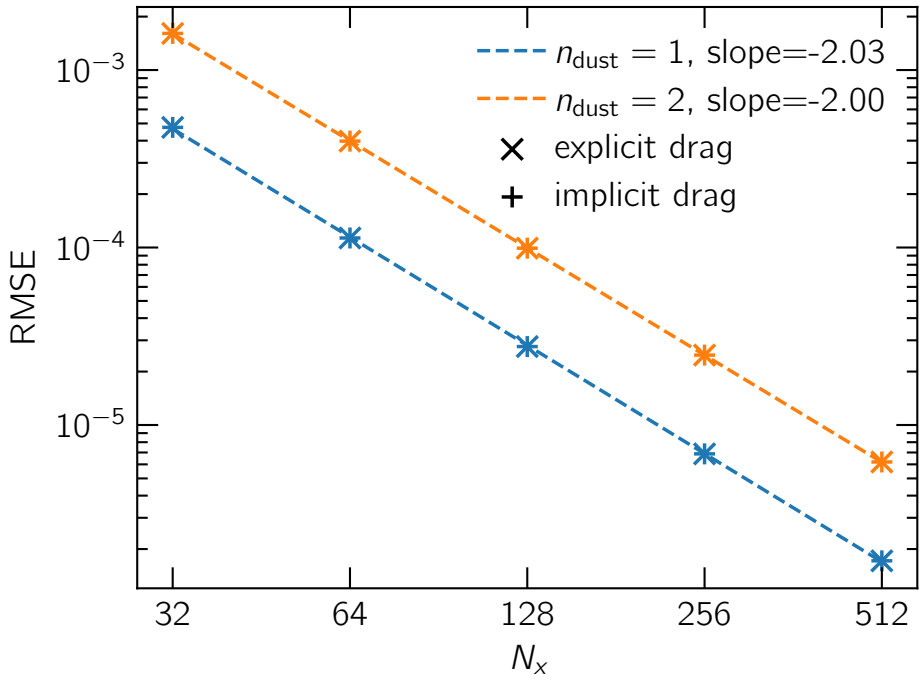


Figure 35: RMSE plotted with respect to the grid resolution. The marker '×' corresponds to the explicit drag scheme and the marker '+' to the implicit drag scheme, the implicit scheme for the two dust species (in orange and blue). The colored dashed lines are the respective fits, and the slopes are second order for both schemes.

explicit diffusion solver. We reach second-order convergence for both dust species and drag solvers, with the errors being identical between the two drag solvers. This indicates that the evolution in this test is diffusion-dominated. We conclude that the combination of our drag and diffusion solvers, with both implicit and explicit drag schemes, achieves second-order convergence in line with our expectations.

5.6.4 *2D Gaussian test in Cartesian coordinates*

We now extend the above test to 2D in Cartesian coordinates $\{x, y\}$. The time-dependent analytical solution can now be written similarly to the 1D case as,

$$\begin{aligned}\epsilon_n(x, y, t) = &\frac{\epsilon_{n,0}\sigma_{n,x}\sigma_{n,y}}{\sqrt{\sigma_{n,x}^2 + 2D_n t}\sqrt{\sigma_{n,y}^2 + 2D_n t}} \\ &\times \exp\left[-\frac{(x_n - x_{n,0} - v_{x,0}t)^2}{2(\sigma_{n,x}^2 + 2D_n t)} - \frac{(y - y_{n,0} - v_{y,0}t)^2}{2(\sigma_{n,y}^2 + 2D_n t)}\right].\end{aligned} \tag{64}$$

For this test, the Gaussian profiles have widths $\sigma_{n,x} = \{0.03, 0.05\}$ and $\sigma_{n,y} = \{0.04, 0.06\}$, amplitudes $\epsilon_{i,0} = \{0.01, 0.36\}$, and are centered at $\{x_{1,0}, y_{1,0}\} = \{0.2, 0.3\}$ and $\{x_{2,0}, y_{2,0}\} = \{0.4, 0.5\}$. We also impose a constant background velocity field of $v_{x,0} = 0.1$ and $v_{y,0} = 0.2$ on all fluids. The grid in y is the same as what was prescribed for x in the 1D test, along with the rest of the initial conditions.

Figure 36 shows heatmaps of the concentration of dust species with $\mathrm{St} = 0.1$, with this particular test for both codes PLUTO (top row) and FARGO3D (bottom row). From left to right, we show the simulation output at $t = 1$, the analytical solution following Eq. 64, and the relative error between the two. We find that both codes, regardless of implementation, yield reasonably good results, with errors of very similar magnitude. It is nevertheless interesting to note that the morphology of the error is different between the two codes, with PLUTO showing a more antisymmetric error about the center of the Gaussian compared to FARGO3D. This might be related to the latter using an operator-split scheme, or due to the slightly different implementation of momentum diffusion between the codes (see Eqs. (56b) and (57)). We do not show the evolution of the second dust species, as its behavior is identical to the one shown.

5.6.5 *2D Gaussian test in polar coordinates*

This test can now be very similarly implemented for cylindrical polar coordinates to verify whether the implementation of the geometric source terms is correct. We set up a grid with $R \in [0.1, 5]$ and $\phi \in [0, 2\pi]$ with 1024 cells each, and transform the 2D Cartesian bumps to cylindrical coordinates such that the initial conditions and analytical solution in Eq. (64) remain comparable. For this test, the Gaussian profiles have widths $\sigma_{n,x} = \{0.09, 0.15\}$, $\sigma_{n,y} = \{0.12, 0.3\}$, amplitudes $\epsilon_{n,0} = \{0.01, 0.36\}$ and are centered at $\{x_{1,0}, y_{1,0}\} = \{0, 2\}$ and $\{x_{2,0}, y_{2,0}\} = \{0.1, 2.1\}$. We use outflow (i.e., zero-gradient) boundary conditions at the radial and periodic boundaries in the azimuthal direction, and the other initial conditions follow the tests above in this section. We note that due to the background flow creating a radial shear, the gas flow starts deviating from its initial constant velocity, and for this reason we only evolve this test until $t = 2.0$, before boundary-induced waves can affect the Gaussian bump.

Figure 37 shows the concentration of the species with $\mathrm{St} = 0.1$ similar to Fig. 36. Errors remain less than 1%, indicating that the bump evolves as expected.

5.6.6 *2D axisymmetric protoplanetary disk in spherical coordinates: settling–diffusion equilibrium*

To test the dust fluid module in spherical coordinates, we ran a full protoplanetary disk setup in an axisymmetric 2D $\{r, \theta\}$ configuration until a vertical settling–diffusion equilibrium was achieved. Similar to the settling test in Section 5.6.2, we set a constant $\alpha = 10^{-2}$ and $\mathrm{St} = \{0.001, 0.01, 0.1, 1\}$, expecting the same solution as in Eq. (60). The vertical velocity should further follow the profile $v_{z,n} = -\mathrm{St}_n \Omega_K z$ from the terminal velocity approximation.

We set up a disk with $r \in [0.2, 2]$ R_0 and $\theta \in [\pi/2 - 0.4, \pi/2]$. We use $N_r = 512$ logarithmically spaced cells in the radial direction, and a higher resolution towards the disk midplane in the polar direction with a combination of a stretched grid with 256 cells for $\theta \in [\pi/2 - 0.4, \pi/2 - 0.1]$ and uniform with 512 cells for $\theta \in [\pi/2 - 0.1, \pi/2]$. We use the Roe Riemann solver for the gas, super-timestepping (STS) for gas viscosity and dust diffusion, a linear reconstruction, and RK2 time-stepping. The STS parameter STS_NU is set to the default value of 0.01, and the CFL criterion to 0.2.

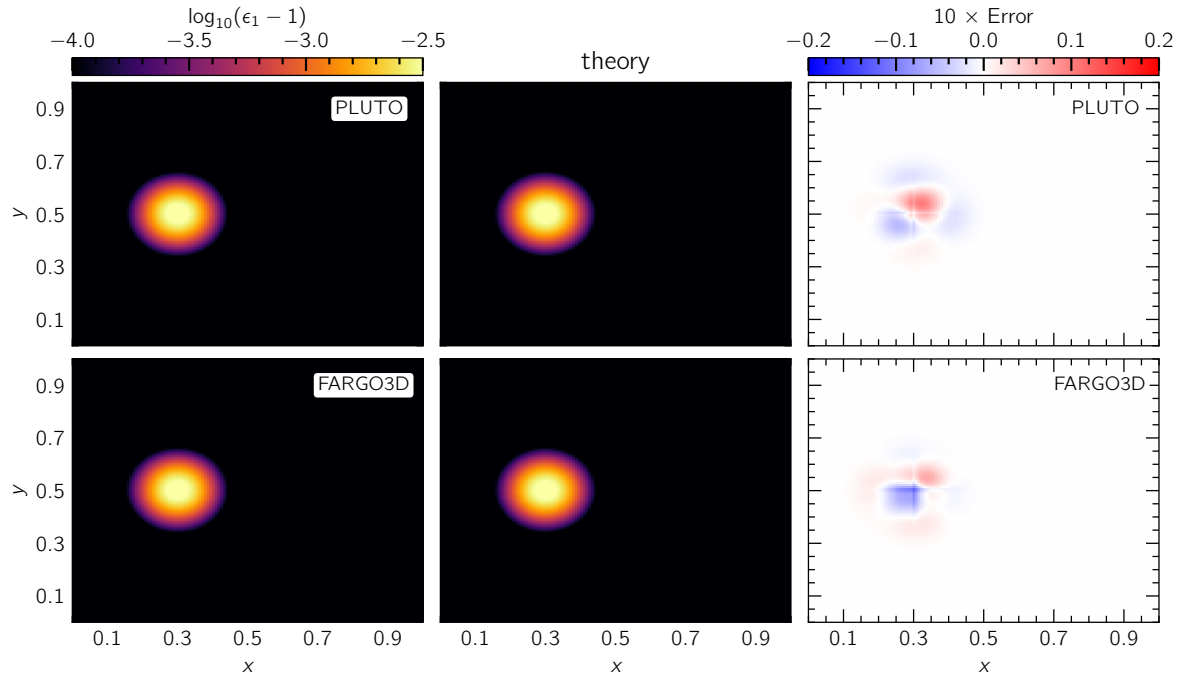


Figure 36: 2D profiles of the relative dust density normalized to the gas density on a logarithmic scale for the moving and diffusing Gaussian bump problem in Cartesian coordinates for two codes PLUTO (top left panel) and FARGO3D (bottom left panel). The second column plots the expected analytical profile following Eq. (64). The far right columns correspond to the relative error profiles for the two quantities normalized to the maximum value for the dust-to-gas ratio from the analytical profile.

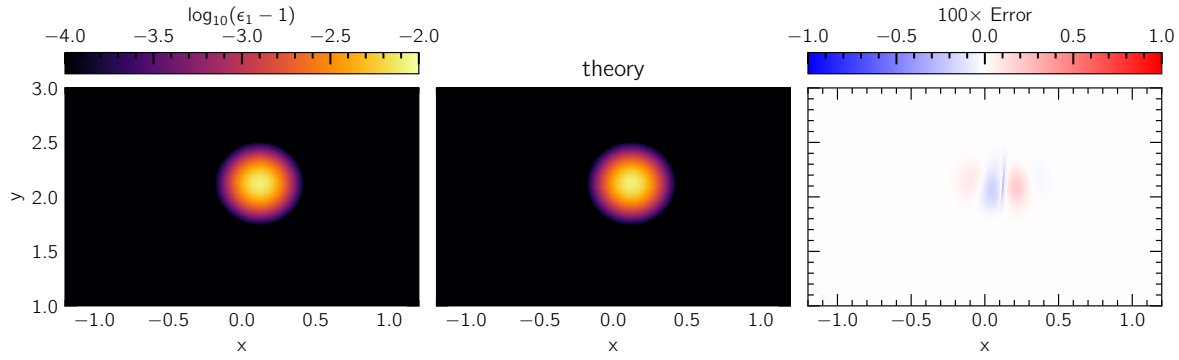


Figure 37: 2D plot of the relative dust density normalized to the gas density (left) for the polar Gaussian test compared to the analytical profile (middle), similar to the Cartesian test in the figure above. Relative error is plotted on the far right, and overall, the errors are below 1%.

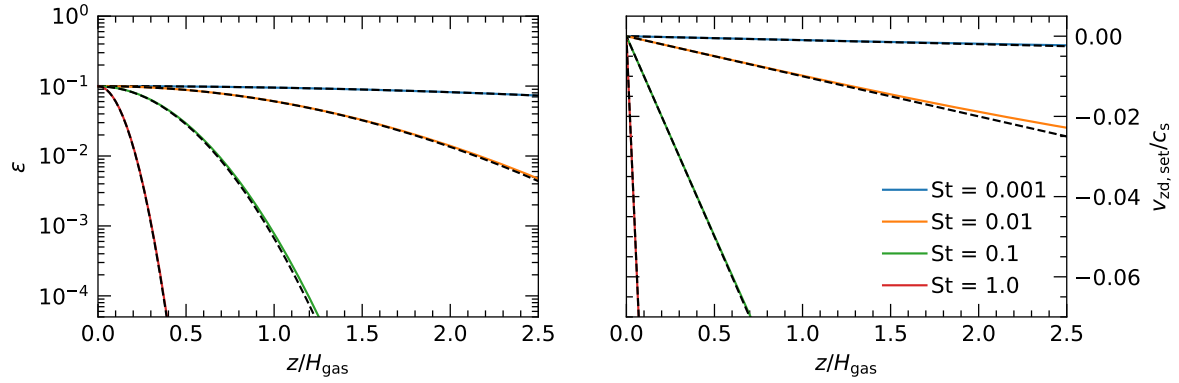


Figure 38: Dust-to-gas ratios ϵ_n and the vertical velocity $v_{z,i}$ for each species are plotted against the vertical coordinate normalized to the scale height corresponding to the profile in Eq. (60). The dashed black lines correspond to the respective analytical solutions, and the data show a good match to these values for up to 6 dust-to-gas scale heights.

The disk is initialized with the density and velocity for the gas profiles in equilibrium given by Nelson et al. (2013)

$$\rho(r,\theta) = \rho_0 \left(\frac{r\sin\theta}{R_0}\right)^p \exp\left[h^{-2}\left(\sin\theta - 1\right)\right], \tag{65a}$$

$$v_\phi(r,\theta) = v_K \left[(p+q)h^2 + (1+q) - q\sin\theta\right]^{1/2}, \tag{65b}$$

with $h = 0.1$, $p = -1.5$, and $q = -1$. We initialize four dust species with Stokes numbers $St = \{0.001, 0.01, 0.1, 1\}$ with their concentrations set to

$$\epsilon_n = 0.1 \exp\left(-\frac{St_n}{\alpha}\frac{z^2}{2H_g^2}\right), \tag{66}$$

and purely Keplerian velocities. Given that we use a constant St for all dust species and disable dust feedback, the choices of ρ_0 and R_0 are arbitrary.

Regarding boundary conditions, we impose a reflective disk midplane (at $\theta = \pi/2$), exponentially extrapolate all densities at the top polar boundary, and fix all other quantities to their initial values at the remaining boundaries. We further damp all quantities except for the azimuthal velocity to their initial profiles similar to de Val-Borro et al. (2006) at all boundaries except for the midplane. Finally, we floor the density to $\rho_{min} = 10^{-14}$ g/cm^3.

We then integrate the disk for 10 inner orbits. Figure 38 shows 1D vertical profiles of ϵ_n and the vertical velocity v_z for the different species at $R = R_0$ at steady state, both matching their analytical profiles very well. We therefore conclude that the implementation of the dust diffusion module in spherical coordinates is correct.

5.7 DISCUSSION AND CONCLUSIONS

In this section, we discuss some underlying assumptions and caveats of our formalism, suggest future directions, and outline the summary for this project.

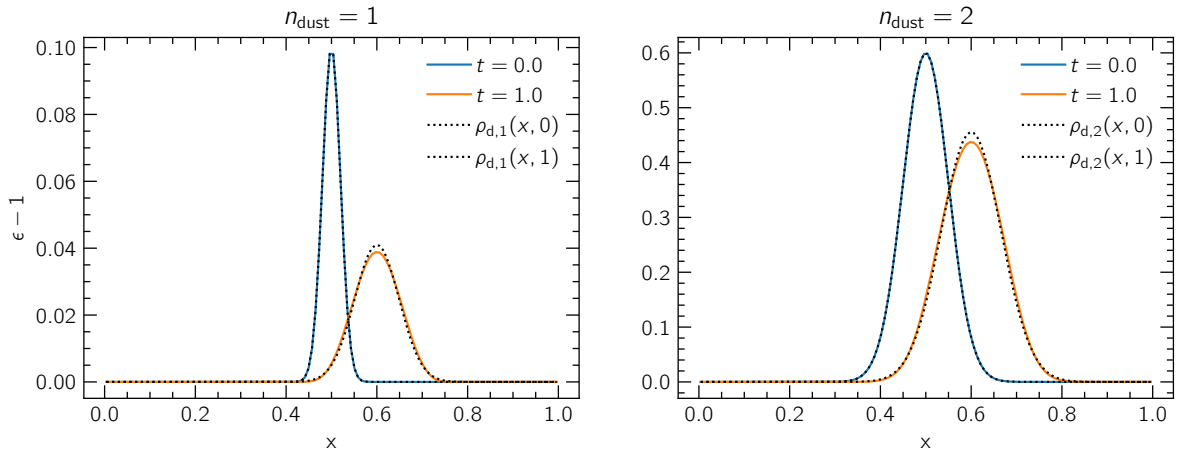


Figure 39: A replica of the 1D Gaussian diffusion test (described in Section 5.6.3, and presented in Figure 34), but with the additional terms from Huang and Bai, 2022. Compared to the fiducial formulation, the Gaussian profile appears to have experienced additional numerical diffusion.

5.7.1 *Comparison to the Huang and Bai (2022) diffusion formalism*

As seen in Eq. (58), the dust-diffusion treatment of Huang and Bai (2022) includes additional terms: i) they advect the "total velocity" $\boldsymbol{v}_{d,n} + \boldsymbol{v}_{d,dif,n}$, and ii) they include an additional term in the diffusion flux, $\boldsymbol{F}_{dif,n} \otimes \boldsymbol{v}_{d,n}$. It is argued that this is important for Galilean invariance, but their work does not include testing against analytical solutions, linear theory, or the more standard equations used in the literature. We nevertheless carried out a variant of the 1D Gaussian diffusion test outlined in Section 5.6.3, where we implemented Eq. (58) instead according to Huang and Bai (2022) (including the adjustment to the new conserved variable $\rho_{d,n}\boldsymbol{v}_{d,n} \rightarrow \rho_{d,n}\boldsymbol{v}_{d,n} + \boldsymbol{F}_{dif,n}$). The result is shown in Fig. 39.

We found that the behavior of the Gaussian bump is very similar between the two methods in terms of the diffusion rate and the symmetry of the bump about its peak, suggesting that both concentration and momentum are diffused appropriately in general. However, the solution differs slightly from the analytical expectation in Eq. (62) around the peak of the Gaussian. This does not necessarily imply that their method is wrong (in fact, our implementation of their method has technically not been rigorously tested against a problem with an analytical solution that is compatible with their approach). Rather, it highlights that the differences in the assumptions made in the derivation of the dust diffusion framework can manifest as additional effects in existing problem setups, with possibly more striking differences for more realistic non-linear problems. This remains largely unconstrained, and should be recognized for cross-code comparisons. Further testing between these methods with analytical solutions would be required to better understand the differences between implementations.

5.7.2 *The energy equation*

In this article, we only focus on momentum transport and ignore any changes in the energy equation due to the dust. Huang and Bai (2022) add terms to the energy equation of the gas to include frictional heating and work done by the drag, but no formalisms for a dust energy

equation. More recent work by Muley et al. (submitted) investigates the interaction of gas, radiation, and multiple species of dust in detail, with separate temperatures for each of these species. We refer the reader to that work and to Muley et al. (2023) for more information on this topic.

5.7.3 *Caveats of the pressureless dust-fluid method*

Lynch and Laibe (2024) discuss that within the fluid framework, the pressureless approximation neglects the turbulent pressure term, and by extension the anisotropic stress tensor that emerges from the kinetic equations of dust transport. They derive a moment-based method starting from the Fokker–Planck equations, providing a higher-order *ansatz* with a "6-dimensional Maxwell fluid". They argue that this could be a better description for dust diffusion and is dynamically important for dusty atmospheres where there is an equilibrium among dust stress, gravity, and the gas Reynolds stress. Beyond an improved diffusion closure, moment-based methods could become important to capture crossing trajectories for dust as they include higher-order velocity terms. These are absent in the single-stream approach of the pressureless dust-fluid formalism used in this article and in other standard codes.

With a more generalized Reynolds averaging framework, Binkert (2023) argue that gradient diffusion might be limited, and provide a mean-field theory-based dust transport model. Their method includes an additional set of momentum equations (one for each vector component) for the turbulent velocity of each dust species, in addition to the standard dust and gas continuity and (bulk-velocity) momentum equations. It remains to be seen how important these different effects and terms are, and whether including them yields meaningfully different results to the pressureless dust-fluid approximation. At the other end of the spectrum, the Lagrangian super-particle approach also suffers from grid averaging methods, and stiff regimes of high particle concentration. This being said, the pressureless dust module is still remarkably useful and a computationally efficient method that is generalizable to include complex physics.

5.7.4 *Summary*

We summarize the key results of this manuscript as follows:

1. We outline the method and the numerics of the multispecies drag and diffusion solvers for the dust fluid module in the `PLUTO` code.

2. We tested the drag solver with the multispecies radial drift and `DUSTYWAVE` tests to show that the implementation is robust for both steady-state and time-dependent problems, and we also show that both implicit and explicit drag solvers exhibit second-order convergence.

3. The non-linear streaming instability problems in the shearing-box approach `BA` and `AB` show a good match to the fluid implementations in Benítez-Llambay et al. (2019) and Huang and Bai (2022). The lack of convergence for the `AB` problem persists, presenting a need for a comparison between particle and fluid implementations similar to the work by Baronett et al. (2026).

4. We discussed the different dust diffusion implementations in the literature, pointing out key differences between codes FARGO3D and Athena++. We implemented a mass- and momentum-conserving concentration diffusion prescription with the associated terms in the momentum equation following a similar implementation used in cuDisc (Robinson et al., 2024), compatible with the super-timestepping scheme.

5. We implemented tests for both steady-state and time-dependent scenarios for the drag and diffusion solvers in the form of vertical settling and advection–diffusion tests. We showed the differences between implementing concentration diffusion with or without momentum diffusion and compared it directly to FARGO3D. We found that the momentum diffusion implemented in FARGO3D with the operator splitting scheme, although different from our finite-volume implementation, provides similar solutions but with an asymmetric error profile which could arise from the order in which the fluid variables are updated.

6. We also discuss the implementation of Huang and Bai (2022) and show that there is a fundamental difference in the equations FARGO3D and PLUTO implement, and as a result the former yields a different solution for the Gaussian test. We note that this could look different for non-linear problems and remains unconstrained.

7. Our implementation is tested for the three geometries available to PLUTO (Cartesian, cylindrical, spherical) and behaves as expected in all cases.

While noting the caveats of the pressureless dust fluid method, we show that this tool is still extremely useful for protoplanetary disk and other applications.

Beyond the results presented in this work, the PLUTO dust fluid module has been already adopted in several studies: streaming instability code comparisons such as Baronett et al. (2026) and Ostertag et al. (submitted), vertical shear instability and other dust drag instabilites (Bi et al., 2026; Fukuhara et al., 2025), and synthetic observations of dynamical signatures in protoplanetary disks (Fasano et al., 2026; Penzlin et al., 2025). It has also been coupled with radiation hydrodynamics (Ziampras et al., 2025a,b) and has recently been coupled to the dust coagulation method TriPoD (Ziampras and Birnstiel, 2026; Ziampras et al., 2026a), highlighting the flexibility and wide applicability of the dust module. Future upgrades will include the Runge–Kutta–Gegenbauer (RKG) time-stepping scheme for parabolic terms (Mattia et al., 2026) and porting to the GPU gPLUTO version.

6

Outlook and conclusions

This chapter contains results from **Sudarshan, P.**, Flock, M., et al. 2026, A&A 706, A198 [ADS] (see Chapter 4 for the author contributions)

and

Baronett, S., et al. (incl. **Sudarshan, P.**), under review in ApJ, "A Comparative Study of the Streaming Instability: Unstratified Models with Marginally Coupled Grains" [ADS]
Authors: Stanley A. Baronett, Wladimir Lyra, Hossam Aly, Olivia Brouillette, Daniel Carrera, Victoria I. De Cun, Linn E. J. Eriksson, Mario Flock, Pinghui Huang, Leonardo Krapp, Geoffroy Lesur, Rixin Li, Shengtai Li, Jeonghoon Lim, Sijme-Jan Paardekooper, David G. Rea, Debanjan Sengupta, Jacob B. Simon, Prakruti Sudarshan, Orkan M. Umurhan, Chao-Chin Yang, Andrew N. Youdin
Note: Authors listed alphabetically (non-contributory order) as part of the code comparison project; significant personal contribution through the PLUTO code implementation, numerical simulations, and manuscript writing; a full author contribution list is provided within the manuscript.

In this section, we describe the outlook and future work for this thesis, as well as our conclusions. We start by discussing the outlook and future work for the radiation hydrodynamic models of disks, and then move on to the dust. Later, we bridge the two topics, and provide the conclusions for this thesis.

6.1 Synthetic observations of shadows

In addition to the implications for observations suggested in Sect. 4.7.1, we also post-processed the final temperature state from our fiducial radiation hydrodynamic model `MFID` using the radiation transfer code RADMC-3D (Dullemond et al., 2012) to produce synthetic scattered-light images that could be observed using VLT-SPHERE in the H-Band (1.65 μm). We do not recompute the temperatures with `radmc mctherm`, but we recalculated the dust opacities with `optool` (Dominik et al., 2021) with the same parameters as the ones used in Sect. 4 (Krieger and Wolf, 2020) to also include the scattering components. We then convolve the image using a Gaussian filter with a beam size of $40'' \times 40''$, and add a coronograph of the same size around the star. Figure 40 shows the scattered-light image with the two starlight-driven bump and shadow pairs in our model having an imprint in the observables.

With future telescopes like the Extremely Large Telescope (ELT), high-resolution observations of dust thermal emission with the *METIS* instrument in the near-infrared (e.g., L-Band at ~ 3.8 μm) can provide further constraints on the temperature of the disk surface. Although thermal emission probes regions deeper than the scattering surface, provided the excellent angular resolution of the ELT, shadow-driven ring and gap pairs could be imaged. Synthetic

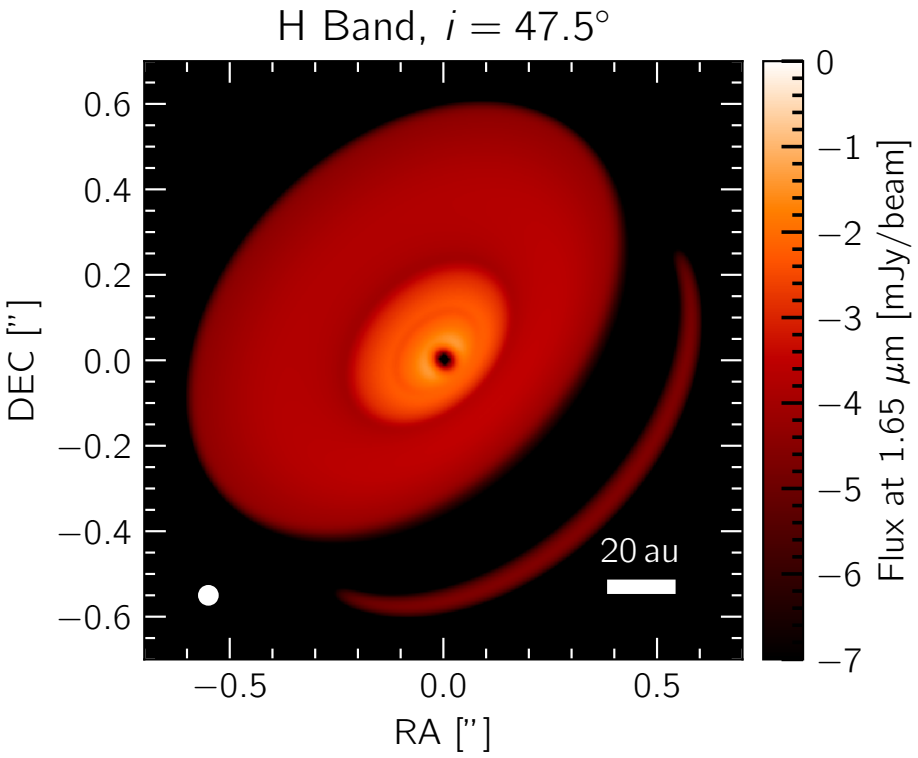


Figure 40: Synthetic scattered-light images in the H band of the final disk state in the fiducial model `MFID`. We chose a disk at a distance of 158 pc and an inclination of 47.5 deg similar to IM Lup (Avenhaus et al., 2018). The square in the bottom left denotes our beam size with a FWHM of $40'' \times 40''$.

scattered-light and thermal emission of shadows will prove to be a useful comparison not just in the context of the mechanism discussed in Chapter 4 but also in other shadow-driven substructures, such as Zhang and Zhu (2024) and Ziampras et al. (2025a). Comparisons to synthetic observations of optically thick molecular tracers such as ^{12}CO with ALMA will further give us insight into the correlation between shadow-driven structures at the surface in the gas and the dust.

6.2 TOWARDS COMPARING RADIATION METHODS

In our work following Chapter 4, we also compared the final temperature structures to the hydrostatic models of Melon Fuksman and Klahr (2022), who use the M1 moment-transfer method, with the flux-limited diffusion approach we use in this work, in Fig. 41. The parameters used for the model correspond to the `ML1` run (see Table 1). The temperature profiles in the optically thin atmosphere match between the models very well, and the analytical solution by Chiang and Goldreich (1997) and the corresponding RADMC-3D calculations (green line in Fig. 41). The background power-law of the midplane temperatures is also in good agreement, with some minor deviations given the different radiation treatments. Both radiation methods show the formation of thermal waves. However, with the M1 method, we see that the observed bumps are of higher amplitude than the ones we see with FLD.

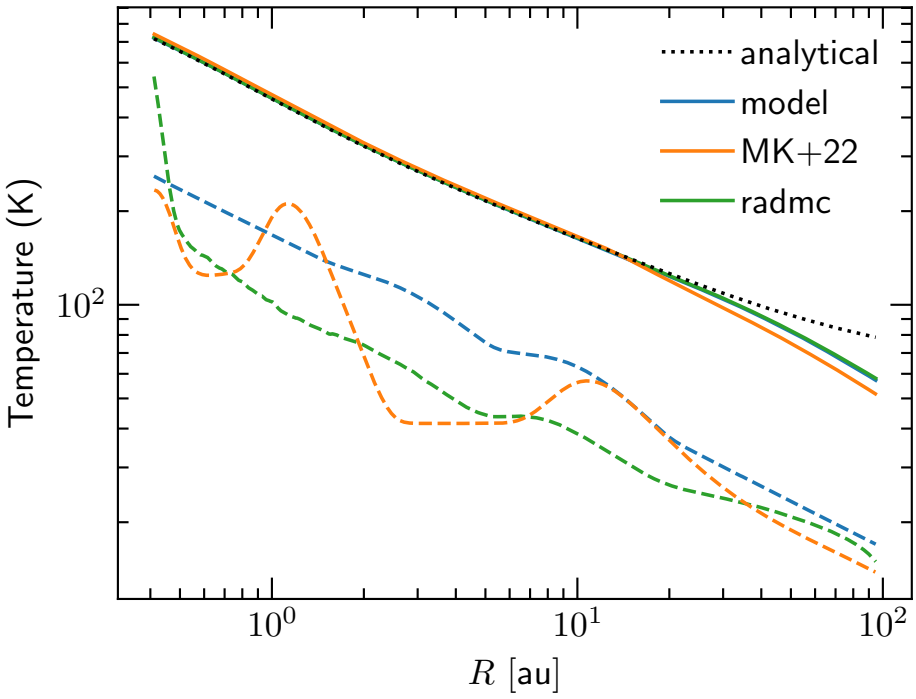


Figure 41: Comparison of 1D temperature profiles of the hydrostatic models at the end of the simulation for two different radiation methods — flux-limited diffusion (FLD, used in this work) and moment transfer (M1, Melon Fuksman and Klahr, 2022). Overall, we see very good agreement with temperatures with FLD and M1, with the optically thin atmosphere being captured quite well. The dotted black line shows the analytical solution for the temperature profile in the optically thin atmosphere (Chiang and Goldreich, 1997). The midplane temperatures are also in good agreement, with some minor deviations. The M1 method shows higher amplitude bumps than the FLD method. We also include the RADMC-3D calculations corresponding to our FLD model; these correspond to the green lines.

Both the M1 and the FLD methods overestimate the temperature at the midplane, which is expected to be lower using Monte Carlo methods (Dullemond et al., 2002; Krieger et al., 2025; Melon Fuksman et al., 2025; Pavlyuchenkov and Akimkin, 2025). This is also clear in Fig. 41 by the comparison of the FLD model (dotted blue line) to the corresponding RADMC-3D calculation (dotted green line). Kutra et al. (2024) argued that this casts doubt on the validity of the results in Melon Fuksman and Klahr (2022), suggesting that this effect may preclude the formation of a temperature staircase. However, despite this overestimation, both FLD and M1 models reproduce shadowed structures in hydrostatic simulations, and we find a parameter space where shadows persist in hydrodynamic models with FLD. Thus, we find the argument presented in Kutra et al. (2024, see Sect. 4.7.4) to lack strong physical motivation. Instead, the presence or absence of a staircase is sensitive to the assumed dust content, the production of initial bumps, and the correct treatment of diffusive cooling timescales resulting from the balance of stellar irradiation and thermal cooling.

Propagation of thermal waves requires careful treatment of radiation, and with the many methods of radiation transport available (Jiang, 2021; Melon Fuksman et al., 2021; Robinson et al., 2024), it is not always clear what the exact differences between them are. Comparisons between methods exist for standardized test problems such as the shadow or the beam-crossing tests (Jiang, 2021; Menon et al., 2022), but in the context of problems relevant to protoplanetary disks, such a comparison has not been made. Recently, Melon Fuksman et al. (2025) compared the M1 method to the half-moment transfer method, showing the differences that can arise when benchmarking them to Monte Carlo approaches like RADMC-3D.

Currently, we are working on such a comparison, compiling the differences between FLD, M1, half-moment, and discrete ordinates, including frequency-averaged and multigroup methods (Sudarshan et al., work in progress). We use the most popular problem for protoplanetary disks: measuring temperatures in the gap carved by a planet in the hydrostatic limit. Early results suggest that while moment methods (both FLD and M1) struggle to capture the warm gap, heated by re-emission from the planet-gap edge, multi-group schemes seem to help bridge the gap, with differences compared to Monte Carlo schemes. More comparisons between the methods will prove useful to understanding the underlying setup versus method-dependent differences, further informing us on problems where we can get away with using simpler approximations to gain computational efficiency.

6.3 CODE COMPARISON OF DUST MODELS FOR STREAMING INSTABILITY

With our implementation of the multifluid dust module in the `PLUTO` code, we can carry out various studies. Similar to the radiation method, there are many approaches to modeling dust in the literature, from describing them as a fluid or as Lagrangian super-particles to numerical implementations in finite-difference and finite-volume codes. We were involved in a code comparison study that studied the `BA` problem of the streaming instability (see Sect. 5.5.4 for a full description of this test), where the goal was to study whether, despite the differences in the methods and implementations, seven different codes (including 9 different approaches, both particle and fluid) yielded similar results.

Figure 42 shows the snapshots of the dust-to-gas mass ratios (similar to the top Fig. 30) at different times $T = \Omega_0^{-1}$, showing the time evolution to the saturated state of the `BA` problem. While the initial evolution for the different codes looks different, at this resolution of 1024^2, the final saturated states of the `BA` problem show good convergence between both particle

and fluid approaches. This is further clear in the maximum dust density profiles normalized to the gas density in the top panel of Fig. 43, where the bottom panel shows moving averages over time windows of 10T. However, overplotted in the bottom panel are the results for the lower-resolution model 512^2, where we see that the simulations using the fluid approach reach lower maximum dust densities than the particle codes, with an overall scatter of about an order of magnitude. Hence, the interpretation of planetesimal populations from streaming instability studies should be carried out with caution, given the resolutions used and the method. Future code comparisons in stratified problems and realistic conditions will provide better constraints on this. However, the result that the problem converges very well at high resolutions across codes is important, and it confirms that our implementation of the dust fluid module in Chapter 5 is accurate.

6.4 TOWARDS MORE COMPLEX PHYSICS, AND COUPLING DUST AND RADIATION

Within our current framework of modules available in PLUTO, the dust fluid module can be further coupled with non-ideal MHD (Flock et al., 2013; Mattia et al., 2026) to study magnetized winds and the properties of dust entrained in them. This has implications for interpreting JWST observations of winds (Pascucci et al., 2025). The dust fluid module is also coupled to the dust coagulation method TriPod (Pfeil et al., 2024) and to interpret observations (Fasano et al., 2026). Finally, the dust module can be coupled with the radiation modules (Flock et al., 2013; Melon Fuksman et al., 2021) to study the influence of dust settling on the propagation of shadows (as discussed in Sec. 4.6). Currently, studies by Ziampras et al. (2025a,b) have already used FLD for the dust module; however, these are vertically integrated simulations. With the dust diffusion module providing additional stability, axisymmetric 2D simulations of dust and radiation hydrodynamics are planned for future work.

6.5 CONCLUSIONS

To circle back to the two initial questions we posed in the introduction of this thesis:

Q1. Can stellar irradiation impact the thermal structure of the disk, generating shadows and substructures?

and

Q2. How can we accurately model dust dynamics in modern hydrodynamic codes?

We summarize the following conclusions.

Answering Q1, following Chapter 4,

1. Our radiation hydrodynamic models of protoplanetary disks showed that starlight-driven shadowing in disks can exist in optically thick, slowly cooling disks, as suggested by our models of high dust-to-gas ratios and surface densities.

2. In the hydrostatic limit, thermal waves form and propagate inwards, recovering the thermal wave instability. However, with 2D radiation hydrodynamics, instead of an instability, we see temperature bumps at intermediate disk radii form and settle into stable states, with a long-term staircase geometry in the disk.

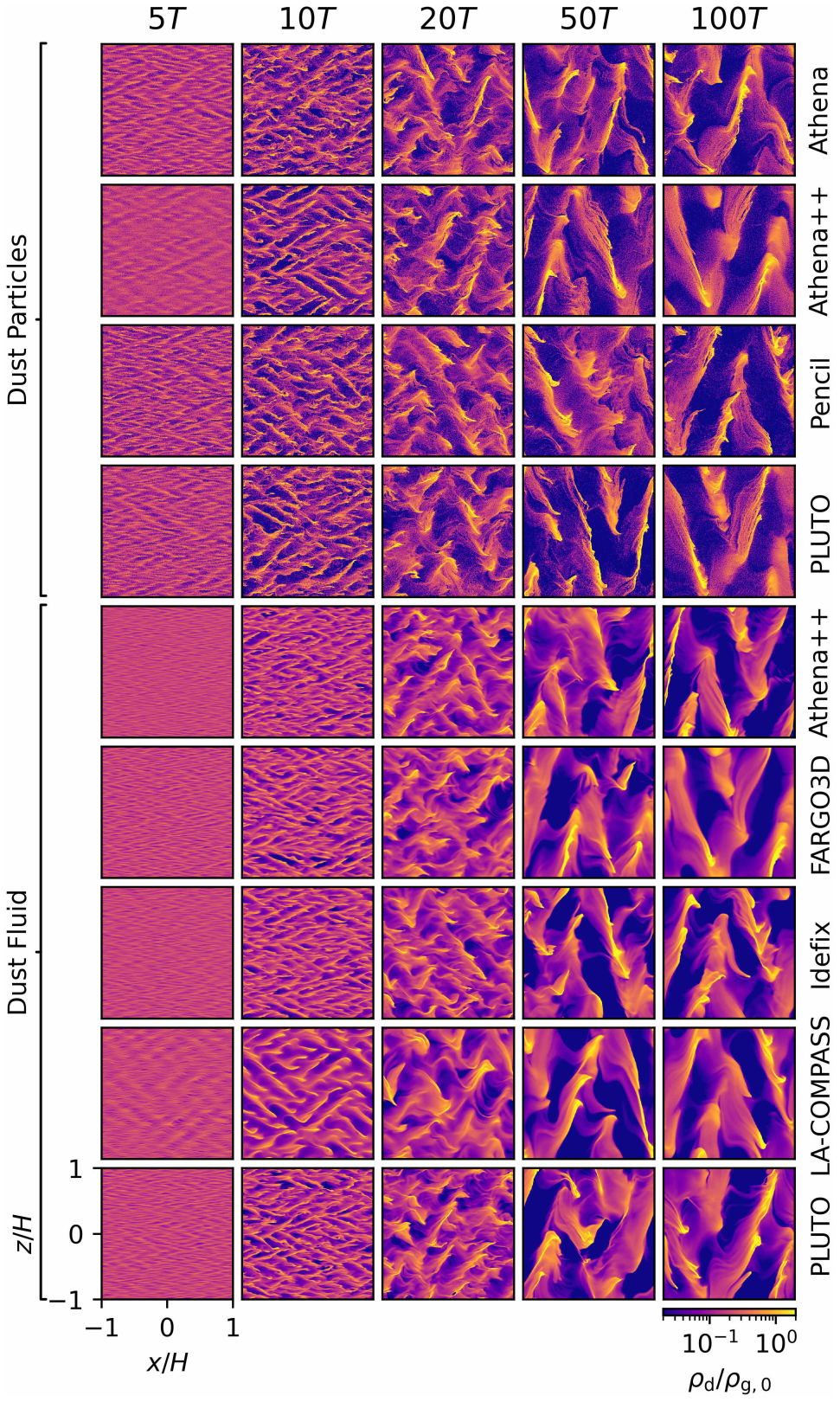


Figure 42: 2D x-z profiles (normalized to the gas scale height H) of dust densities with respect to the initial gas density $\rho_d/\rho_{g,o}$ on a logarithmic scale for different codes, similar to Fig. 47 for i) particle codes: `Athena` (Bai and Stone, 2010a), `Athena++`, `Pencil`(Youdin and Johansen, 2007), `PLUTO` (Mignone et al., 2019) and ii) fluid codes: `Athena++` (Huang and Bai, 2022), `FARGO3D` (Benítez-Llambay et al., 2019), `IDEFIX` (Lesur et al., 2023a), `LA-COMPASS` (Fu et al., 2014) and `PLUTO` (this work, Chapter 5). The resolution used for these simulations is 1024^2.

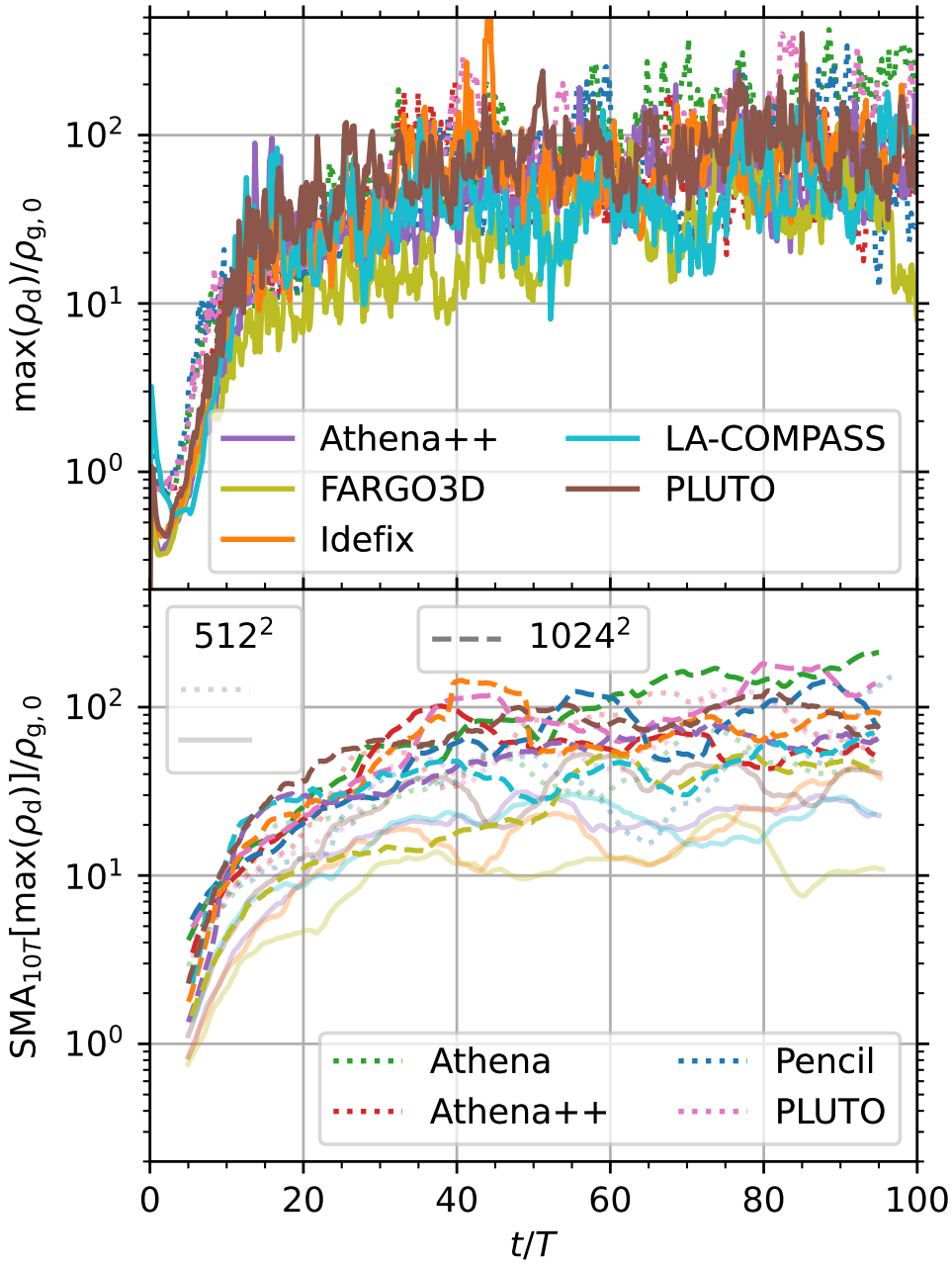


Figure 43: Maximum dust densities normalized to the initial gas density $\max(\rho_d/\rho_{g,o})$ for the different codes with particle approaches denoted in dotted colors and the fluid approaches in solid colors (top). The bottom panel includes the moving averages of the $\max(\rho_d/\rho_{g,o})$ over 10T for simulations with resolutions 1024^2 and 512^2.

3. We found that the innermost bump corresponds to an opacity transition in the disk, suggesting that structures in the disk can trigger additional shadows downstream, as a natural consequence of starlight-driven shadowing.

4. The amplitudes of the shadows are less pronounced than previous work that uses simpler prescriptions, indicating the importance of accurately accounting for the vertical and in-plane diffusion of radiation to avoid overestimating features.

5. Our hydrostatic models with dust settling show sharp irradiation absorption surfaces in the disk, with superheated dust layers but no shadowing. The $\tau = 1$ lines move substantially towards the midplane. Further, we highlight the importance of accurate modeling of dust dynamics in the disk for interpreting these shadows.

6. We discuss the requirement of better initial conditions for the structure and the optical depth of the inner disk < 1 au and how these can significantly impact the rest of the temperature geometry.

7. Finally, we show that these shadows could be observable in scattered light and thermal observations, and show that they affect the midplane pressure gradient, indicating that they could be signs of dust traffic jams.

Answering Q2, following Chapter 5,

1. We introduce and outline the method and implementation of the one-fluid and multi-fluid dust module in the `PLUTO` code.

2. Focusing on momentum transport, we implemented dust–gas drag and diffusion prescriptions. For the drag, we provide explicit and implicit solvers, and the diffusion equations are written explicitly with an option for super-timestepping.

3. We test the drag solvers and their convergence with the radial drift and linear `DUSTYWAVE` tests, comparing them to their analytical solutions and showing that the solvers are accurate and show second-order convergence.

4. We benchmark the drag solver with the non-linear streaming instability problem in the shearing box approximation and compare it to the results found in the literature.

5. We describe the differences in the implementation of the dust diffusion equations across codes, and provide a gradient-based concentration diffusion formalism for `PLUTO`.

6. We provide tests to compare against analytical solutions in all geometries and show that the diffusion solver has been implemented correctly and shows second-order convergence.

7. This module is ready to use and will be made public very soon for access by PLUTO users in the community. It is generalizable to include additional physics such as radiation, dust growth, magnetic fields, and more.

Part III

APPENDICES

A

Appendix for Chapter 4

A.1 Thermal relaxation timescale

Previous work has supported the idea that the ratio between the cooling timescale t_{cool} and the dynamical timescale ($\approx \Omega_K^{-1}$) is important in determining the formation and propagation of thermal waves (e.g., Dullemond, 2000; Melon Fuksman and Klahr, 2022). We therefore compute the cooling timescale to both assess its effect on the dynamics of the TWI as well as to estimate the radiative diffusion timescale between hydrostatic iterations in our models. For well-coupled dust and gas, the cooling parameter $\beta = t_{cool}\Omega_K$ can be written as a sum of optically thick and thin cooling components in the flux-limited diffusion approach (following Flock et al., 2017b; Ziampras et al., 2024),

$$\beta = \frac{\Omega_K}{\eta}\left(H^2 + \frac{l^2}{3}\right), \quad \eta = \frac{16\sigma_{SB}T^3}{3\kappa_P\rho^2 c_v}, \quad l = \frac{1}{\kappa_P\rho} \tag{67}$$

where σ_{SB} is the Stefan-Boltzmann constant. We plot β radially and at a vertical slice corresponding to $R = 1\,au$ in Fig. 44. Most of our radial computational domain is dominated by the optically thick component ($\propto H^2$) until up to $R \sim 50\,au$. We find starlight-driven staircases between 1–30 au, in regions corresponding to long-to-intermediate cooling timescales radially. Cooling is inefficient in the inner parts where the cooling parameter is as high as $\beta = 10^3$ and then it decreases reaching up to $\beta = 0.1$ at $\sim 15\,au$. This indicates that the outer part of the disk is susceptible to the vertical shear instability (Flock et al., 2020; Nelson et al., 2013; Pfeil and Klahr, 2019), as also supported by our study (see Fig.22). In the inner disk and along the vertical direction, the region around the midplane remains optically thick up to about four scale heights ($|z|/r \lesssim 0.12$), with a rapidly cooling optically thin atmosphere where $\beta \lesssim 10^{-4}$. The spikes in the vertical profiles of the cooling timescale correspond to the numerical artifact in the temperature (see Fig. 16) and the flat profile in the upper disk boundaries corresponds to the density floor.

A.2 The effect of $\tau_{\nu,0}$ on the irradiation source term

In Sect. 4.7.6 we discussed the effect of the optical depth profile $\tau_{\nu,0}$ at the inner radial end of our domain, given that it is an important initial condition that sets our disk model. We do not fully know the exact structure of the inner disk in reality. Starting out with shadows in the inner disk suggests that the disk outside of it could be subject to nonuniform heating. This could further help starlight-driven shadowing. In this section we compare the effect of two different profiles of $\tau_{\nu,0}$ on the irradiation source term.

We compare the profile used in this work (Eq. (24)) with the one used in Melon Fuksman and Klahr (2022). Figure 45 shows the irradiation source term sampled at the $\tau = 1$ surface for both cases. We find very good agreement between the two prescriptions through most of the disk outward of 1 au. In the innermost parts, we note that the orange line (used by

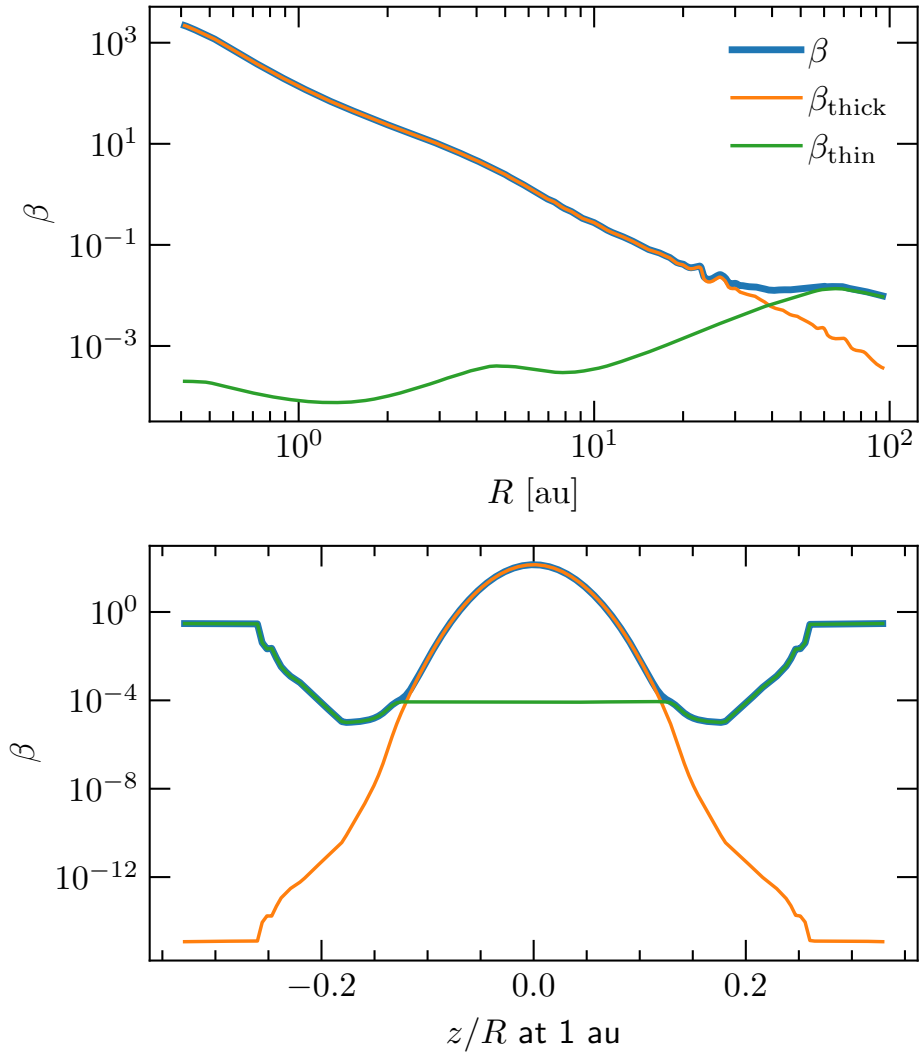


Figure 44: Top: Radial profile of the cooling timescale β for the disk. The regions of the disk where we find starlight-driven shadows lie in the intermediate cooling timescale regime ($1 - 30$ au). Bottom: Vertical profile of the cooling timescale β at R = 1 au. We see that for most of the disk (except for the atmosphere, and outermost regions), optically thick cooling is more dominant.

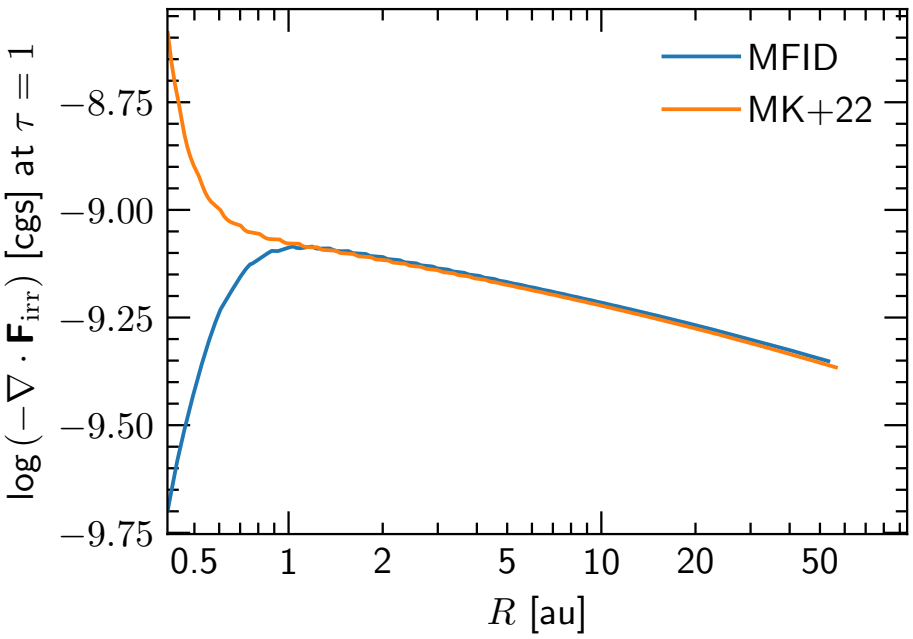


Figure 45: Comparison of the irradiation source term sampled at the $\tau = 1$ surface for two different profiles for the initial optical depth $\tau_{\nu,0}$. The blue line shows the profile used in this work while the orange line shows the profile used in Melon Fuksman and Klahr (2022). We see that they match very well outside 1 au but differ inward. While Melon Fuksman and Klahr (2022) use a radially decreasing, smooth profile, we see that our prescription can create little nonuniformities in heating that might affect the innermost bump.

Melon Fuksman and Klahr, 2022) is a smooth, radially decreasing quantity, while there is a little hotspot in the profile we use. This could influence the growth of the innermost bump in our model. Better models that motivate the inner disk structure from self-consistent radiation hydrodynamic models will help constrain this initial condition better. We nevertheless stress that the presence of a second bump (e.g., at 10 au in Fig. 16) is unaffected by this.

B

Appendix for Chapter 5

B.1 Code structure

In Fig. 46 we present three indicative flowcharts of the code structure, for completeness and to showcase the different implementations of the dust fluid approach. Panel *a* shows the explicit scheme, where dust transport and source terms (e.g., gravity), dust–gas drag, and dust diffusion are all implemented in a fully unsplit manner. Additional, gas-related operator-split source terms (e.g., cooling) are handled separately from the main hydrodynamics block using Strang splitting (i.e., before/after the main block on even/odd steps). In panel *b* we show the "IMEX" scheme, where dust–gas drag is treated implicitly instead. Finally, in panel *c* we show the implementation of the combined IMEX–super-timestepping (STS) scheme, which circumvents the strict timestep limitations of both very small stopping times and very high dust diffusivities for drag and dust diffusion respectively.

B.2 Shearing-box equations

For N dust species, the shearing box equations used for the streaming instability problem can be studied by adding an additional external force that translates as a constant pressure gradient in a box, like in formalisms used by Bai and Stone (2010b) and Benítez-Llambay et al. (2019). Using this, the equations for the gas and the nth dust species, where $n = 1, .., N$ can be written as:

$$\frac{\partial \rho_g}{\partial t} + \nabla \cdot (\rho_g \boldsymbol{v_g}) = 0, \tag{68a}$$

$$\frac{\partial \rho_d}{\partial t} + \nabla \cdot (\rho_d \boldsymbol{v_d}) = 0, \tag{68b}$$

$$\begin{aligned}\frac{\partial (\rho_g \boldsymbol{v_g})}{\partial t} + \nabla \cdot (\rho_g \boldsymbol{v_g} \otimes \boldsymbol{v_g} + P\mathbf{I}) = \rho_g \chi_0 \Omega_0 \boldsymbol{e}_x - 2q\rho_g \Omega_0^2 x \boldsymbol{e}_x \\ -2\Omega_0 \times \left(\rho_g \boldsymbol{v}_g\right) + \sum_{n=1}^{N} \frac{\rho_{d,n}(\boldsymbol{v}_{d,n} - \boldsymbol{v}_g)}{t_{s,n}},\end{aligned} \tag{68c}$$

$$\begin{aligned}\frac{\partial (\rho_{d,n} \boldsymbol{v_{d,n}})}{\partial t} + \nabla \cdot (\rho_{d,n} \boldsymbol{v_{d,n}} \otimes \boldsymbol{v_{d,n}}) = -2q\rho_{d,n} \Omega_0^2 x \boldsymbol{e}_x \\ -2\Omega_0 \times \left(\rho_{d,n} \boldsymbol{v}_{d,n}\right) + \frac{\rho_{d,n}(\boldsymbol{v}_g - \boldsymbol{v}_{d,n})}{t_{s,n}}.\end{aligned} \tag{68d}$$

The term added to mimic the constant pressure gradient in the gas momentum equation is defined by $\chi_0 \Omega_0$ where χ_0 is a constant velocity, chosen as $\chi_0 = 2h_0 v_{K,0}$ with aspect ratio

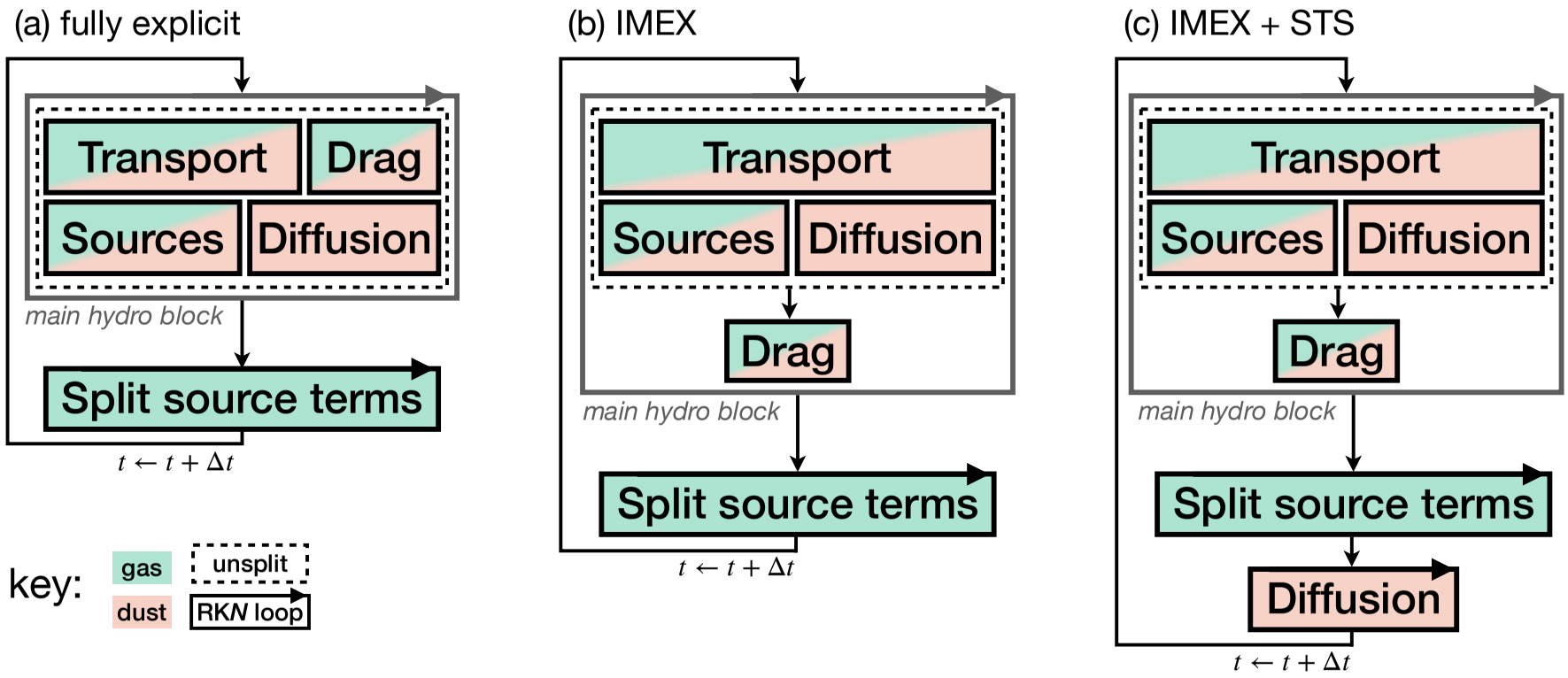


Figure 46: Flowcharts illustrating the structure of the code with the relevant gas and dust terms important for the dust fluid module. Dust–gas drag is treated implicitly in panels *b* and *c*, while dust diffusion is integrated using super-timestepping in panel *c*. Gas-and dust-related terms are denoted in green and orange, respectively. Dashed boxes indicate the terms included within an "unsplit" (sub)step. Solid boxes indicate a Runge–Kutta loop of order N.

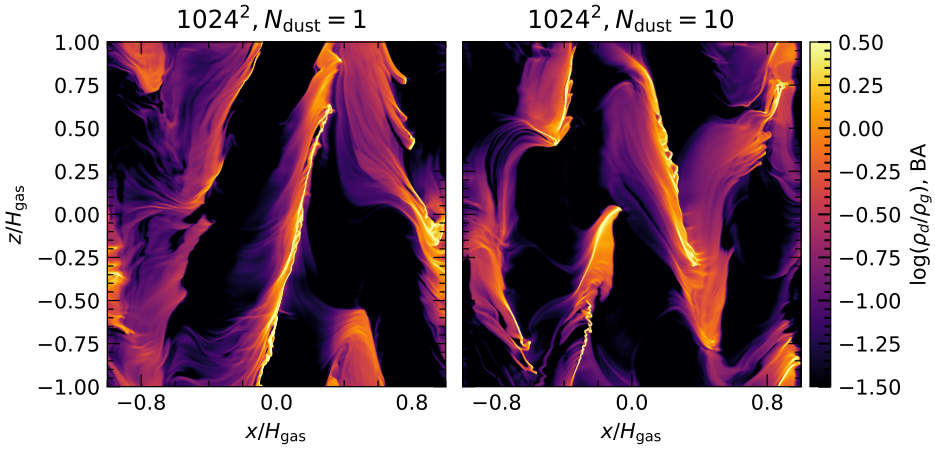


Figure 47: 2D snapshots of the dust-to-gas ratio in logarithmic scale for the BA streaming instability problem, with the panel on the left indicating the simulation with resolution 1024^2 also presented in Sect. 5.5.4 and the panel on the right indicating the simulation at the same resolution but with the mass divided between 10 dust species at time $100\Omega_0^{-1}$.

$h_0 = c_{s,0}/v_{K,0}$ and sound speed $c_{s,0}$. We use a value for $h_0 = 0.05$, and $v_{K,0} = 1$. The shear parameter is given by q, and for Keplerian shear, $q = 3/2$. The gas pressure is defined as, $P = c_{s,0}^2 \rho_g$, and e_x is the unit vector in the x direction. The Stokes number of the nth dust species is given by $t_{s,n}$. We use the `FARGO` module, and hence the background flow is kept fixed.

B.3 BA STREAMING INSTABILITY PROBLEM WITH 10 SPECIES

We ran the BA problem introduced in Section 5.5.4 at resolution 1024^2 with an additional setup '$1024^2 - 10$' where the dust-to-gas mass ratio was divided between 10 dust species, to see how the time evolution looks. The simulations are initialized with random noise, like in the main text, and hence they formally go through a different evolution. Figure 47 shows the 2D profiles for the two setups with one and 10 dust species after $100\Omega_0^{-1}$, and we can see that they reach very similar saturated states, barring the differences due to the initial noise.

The profiles of the maximum dust densities normalized to the reference gas density values in Fig. 48 also show that they reach the same average maximum densities for both simulations. This is a feature of the dust fluid module not being able to capture crossing trajectories, as the dust species do not interact with each other.

B.4 CONCENTRATION DIFFUSION WITH NO MOMENTUM CORRECTION

In this section, we perform the same 1D Gaussian bump test as outlined in Sect. 5.6.3, but here we only solve the continuity equation for dust diffusion (Eq. (56a)) with no corresponding momentum terms. Figure 49 corresponds to the relative dust density with respect to the gas density for the two dust species. We show that after time $T = 1$, the simulation data, while being close to the analytical solution, clearly shows a kink. This detail is missed in a steady-state test, showing that it is indeed important to include the corresponding momentum corrections to the concentration diffusion prescription being used.

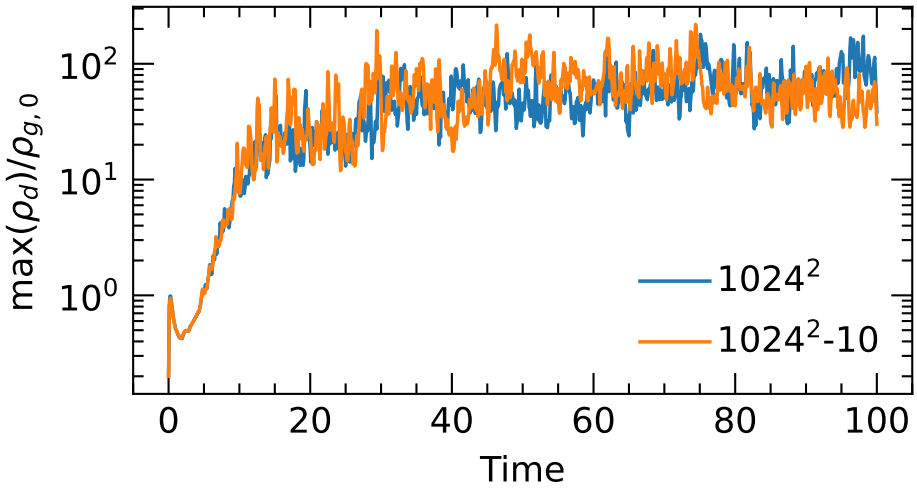


Figure 48: Maximum dust densities normalized to the initial gas density for the two setups with one and ten dust species, respectively, at resolution 1024^2.

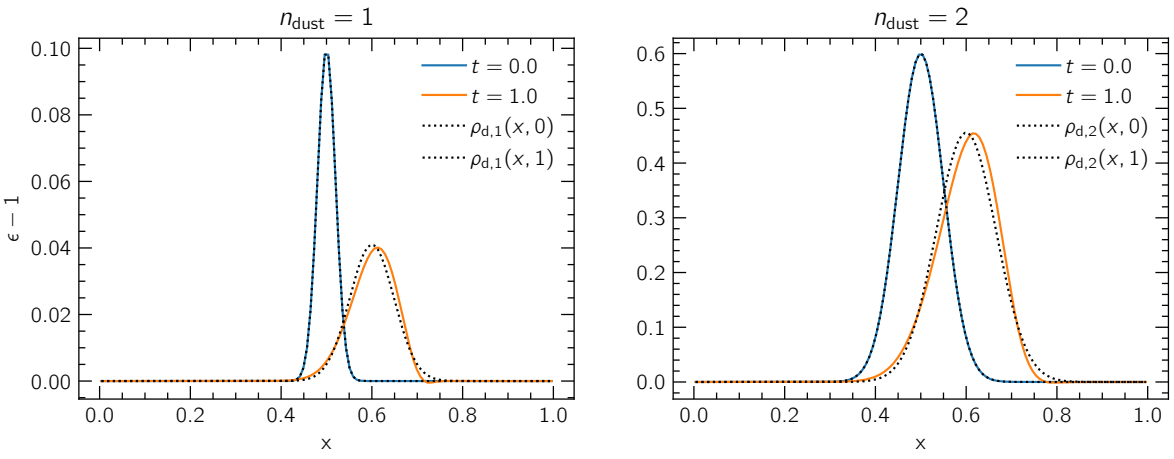


Figure 49: 1D profiles for the moving Gaussian test for the two dust species (left and right) with only concentration diffusion. Relative dust densities are plotted normalized to the gas densities with respect to the x-coordinate; the blue line corresponds to the initial state of the Gaussian, and the orange line after time t=1. The dashed black lines are the correct analytical solutions.

C

Personal Bibliography

This dissertation contains work that is part of the following articles:
Note that, the papers where I am the lead author make up most of the results sections.

1. **Sudarshan, P.**, Ziampras, A., et al., submitted to ApJS, "Multifluid dust dynamics in `PLUTO`: Benchmarking dust–gas drag and diffusion against existing implementations"
2. Baronett, S., et al. (incl. **Sudarshan, P.**)[1], under review in ApJ, "A Comparative Study of the Streaming Instability: Unstratified Models with Marginally Coupled Grains" [ADS]
   Note: Authors listed alphabetically (non-contributory order) as part of the code comparison project; significant personal contribution through the PLUTO code implementation, numerical simulations, and manuscript writing
3. **Sudarshan, P.**, Flock, M., et al. 2026, "Starlight-driven flared-staircase geometry in radiation hydrodynamics simulations of protoplanetary disks" [ADS]
4. Ziampras, A., **Sudarshan, P.**, et al., 2025, MNRAS, 536, 4, pp.3322-3337 "Dusty substructures induced by planets in ALMA disks: how dust growth and dynamics changes the picture" [ADS]

Other works

1. Sanchez-Bermudez, J., **Sudarshan, P.**, et al., to be submitted to MNRAS Letters, "The GRAVITY young stellar object survey: multiplicity of FS CMa revealed with infrared interferometry", part of the GRAVITY YSO Collaboration.
2. Muley, D., Melon Fuksman, D., **Sudarshan, P.** et al., under review in A&A, "Simulating the thermodynamics of gas, radiation, and multispecies dust. Method and applications to protoplanetary disks"
3. Cecil, M., et al. 2026 (incl. **Sudarshan, P.**), A&A, "The role of detailed gas and dust opacities in shaping the evolution of the inner disc edge subject to episodic accretion" [ADS]
4. Ueda, T., et al. 2024 (incl. **Sudarshan, P.**), Nature Astronomy 8, 1148-1158 (2024) "Support for fragile porous dust in a gravitationally self-regulated disk around IMLup" [ADS]
5. **Sudarshan, P.**, Penzlin, A. B. T., Ziampras, A., Kley, W., & Nelson, R. P., 2022, A&A, 664, A157 "How cooling influences circumbinary disks" [ADS]

Note: research works in preparation are excluded.

1 This work is also a part of the doctoral thesis work of Stanley Baronett, University of Nevada, Las Vegas (2026), and is only included in the outlook chapter 6.

D

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